OSCILLATORS AND TIMERS

CHAPTER 7

In this chapter we meet oscillators and timers, the circuits that provide the essential "heartbeats" and timing of electronics. As we'll see, many of the important devices and techniques involve a blend of analog and digital electronics. We'll introduce the digital know-how you need for a first look, and point to later sections of the book where you can get a deeper knowledge of digital techniques.

7.1 Oscillators

7.1.1 Introduction to oscillators

Within nearly every electronic instrument it is essential to have an oscillator or waveform generator of some sort. Apart from the obvious case of signal generators, function generators, and pulse generators themselves, a source of regular oscillations is necessary in any cyclical measuring instrument, in any instrument that initiates measurements or processes, and in any instrument whose function involves periodic states or periodic waveforms. That includes just about everything. For example, oscillators or waveform generators are used in digital multimeters, oscilloscopes, radiofrequency receivers, computers, every computer peripheral (tape, disk, printer, terminal), nearly every digital instrument (counters, timers, calculators, and anything with a "multiplexed display"), every consumer electronic device (cell phone, camera, music or video player or recorder), and a host of other devices too numerous to mention. A device without an oscillator either doesn't do anything or expects to be driven by something else (which probably contains an oscillator). It is not an exaggeration to say that an oscillator of some sort is as essential an ingredient in electronics as is a regulated supply of dc power.

Depending on the application, an oscillator may be used simply as a source of regularly spaced pulses (e.g., a "clock" for a digital system), or demands may be made on its stability and accuracy (e.g., the time base for a frequency counter), its adjustability (e.g., the local oscillator in a transmitter or receiver), or its ability to produce accurate waveforms (e.g., the ramp generator in a dual-slope analog-to-digital converter).

In the following sections we treat briefly the most popular oscillators, from the simple *RC* relaxation oscillators to the stable quartz-crystal oscillators. Our aim is not to survey everything in exhaustive detail, but simply to make you acquainted with what is available and what sorts of oscillators are suitable in various situations.

7.1.2 Relaxation oscillators

A very simple kind of oscillator can be made by charging a capacitor through a resistor (or a current source), then discharging it rapidly when the voltage reaches some threshold, beginning the cycle anew. Alternatively, the external circuit may be arranged to reverse the polarity of the charging current when the threshold is reached, thus generating a triangle wave rather than a sawtooth. Oscillators based on this principle are known as *relaxation oscillators*. They are inexpensive and simple, and with careful design they can be made reasonably stable (better than 1%) in frequency.

A. Basic op-amp – comparator relaxation oscillator

In the past, negative-resistance devices such as unijunction transistors and neon bulbs were used to make relaxation oscillators, but current practice favors op-amps, comparators, or special timer ICs. Figure 7.1A shows a classic *RC* relaxation oscillator. The operation is simple: assume that when power is first applied, the comparator output goes to positive saturation (it's actually a toss-up which way it will go, but it doesn't matter). The capacitor begins charging up toward +5 V, with time constant *RC*. When it reaches one-half the supply voltage, the op-amp switches into negative saturation (it's a Schmitt trigger), and the capacitor begins discharging toward −5 V with the same time constant. The cycle repeats indefinitely, with period 2.2*RC*, independent of supply voltage.

Exercise 7.1. Show that the period is as stated.

Rail-to-rail CMOS output-stage comparators[1] (see §§4.3.2A. 12.1.7 and Table 12.2) were chosen because

[1] In this case the TLC3702 or LMC6762.

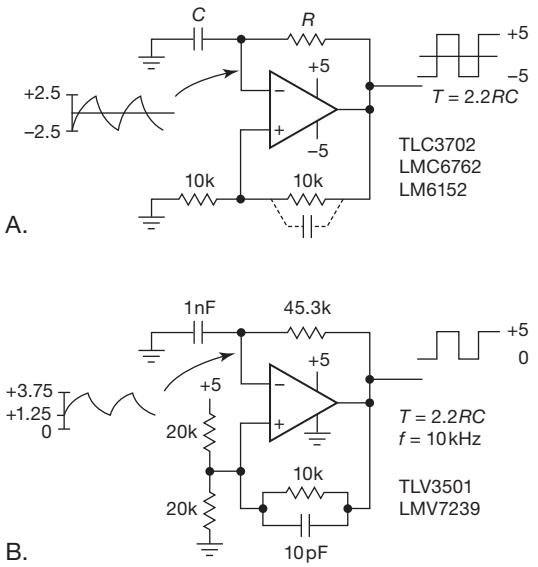


Figure 7.1. Classic op-amp (or comparator) relaxation oscillator, using comparators with rail-to-rail output stages. A. Dual supply with symmetrical bipolarity square-wave output. B. Single-supply variant, with parts values (for 10 kHz), including speed-up capacitor. See also Figure 4.39 in §4.3.3.

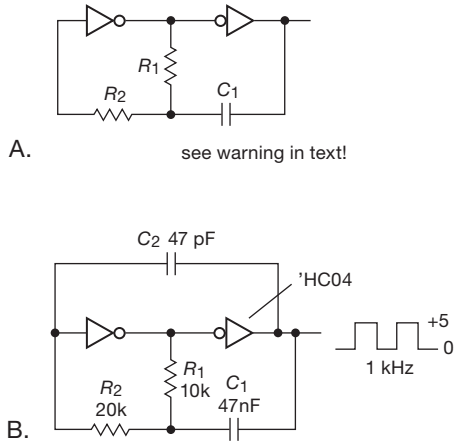


Figure 7.2. Relaxation oscillator with CMOS digital logic inverters. A. Beware conventional wisdom! B. Taming parasitic instabilities with a small speed-up capacitor C_2.

their outputs saturate cleanly at the supply voltages. Comparators like the TLC3702 are much faster (its propagation time is ∼5 ns) than op-amps of similar technology, because they do not need compensation for stable operation with negative feedback (see §4.9), so they are a good choice if you want to go above a few kilohertz. The bipolar LM6132–54 series of op-amps also swings rail-to-rail and, unlike their CMOS cousins, allows operation at a full ±15 V. However, if op-amps (rather than comparators) are used, this circuit makes considerable demands on the op-amp speed, because the output square wave goes rail-to-rail; even something like the LM6152, with its f_T of 75 MHz and large-signal slew rate of 45 V/μs, has enough speed for operation only up to 100 kHz or so. Note that this circuit does not operate the op-amp in the linear region, with the customary negative feedback;[2] so you could use an uncompensated op-amp (see §4.9) for better speed.

You can run this kind of circuit from a single supply voltage, as shown in Figure 7.1B, if you add one resistor. Here we've taken advantage of the faster comparators that are available for lower supply voltages only. The TLV3501 (supply voltage range of 2.7 V to 5.5 V only) has a propagation time of just 3 ns in this circuit, allowing operation up to tens of megahertz; here it's loafing along at a mere 10 kHz.[3]

By using current sources (instead of resistors) to charge the capacitor, a good triangle wave can be generated. A clever circuit using that principle was shown in §4.3.3.

B. CMOS logic relaxation oscillators

You can build simpler *RC* relaxation oscillators by using CMOS digital logic inverters (Chapters 10–12) instead of an op-amp or comparator. Figure 7.2A shows a circuit often seen in the literature.[4] The good news is that it's simple; the bad news is that it doesn't work! Specifically, its output waveform has ragged edges, plagued with fast (∼100 MHz) parasitic oscillations at each transition; this is due to the relatively slow rise time at the input to the first inverter (caused by capacitive loading). The final (good) news is that there's a simple fix, namely the inclusion of a small speed-up capacitor (C_2 in Figure 7.2B). Figure 7.3 shows measured waveforms, both for the conventional circuit (Figure 7.2A), and for the improved circuit.

One slightly worrisome aspect of this circuit is the fact that the input protection diodes are forced into conduction each cycle by the charged capacitor C_1; it's not really a

[2] The fast feedback is positive, and the output swing alternates between positive and negative saturation.

[3] If, instead, you're interested in very low operating current, you could use the impressive LPV7215 comparator: it runs at less than a microamp supply current, with propagation time ∼10 μs. To exploit its very low power, of course, you must use high resistor values, say ∼10 MΩ, which are OK, given the comparator's very low input current (<1 pA).

[4] For example, in the datasheet for the 74HC4060, and in our previous book edition.

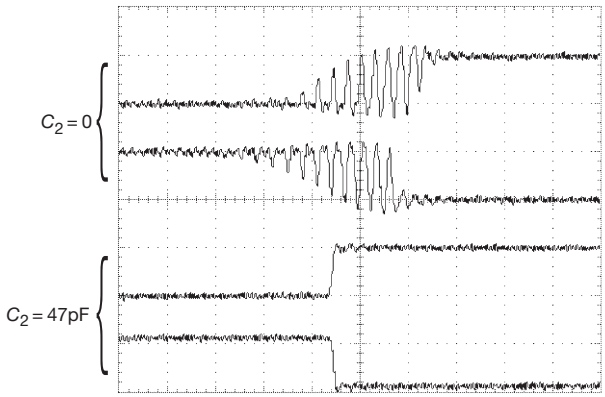


Figure 7.3. Parasitic oscillations plague the simple oscillator circuit of Figure 7.2A. The upper pair of waveforms show an ~90 MHz instability on the rising and falling edges of a 1 kHz oscillator (made with commonplace 74HC04 logic running at 5 V). Adding a 47 pF speed-up capacitor (Figure 7.2B) cleans it up nicely. Horizontal: 40 ns/div; vertical: 5 V/div

problem, however, because the current is safely limited by R_2. But if it bothers you, you'll like the circuit of Figure 7.4A, where a 2:1 voltage divider reduces the swing applied to the capacitor, preventing input clamping. Figure 7.4B is another variation on the logic oscillator theme, taking aim at the parasitic oscillation problem. When we breadboarded these two oscillators on the test bench, however, we found it still necessary to include the 47 pF speed-up capacitors to eliminate parasitics.

You can make an even simpler CMOS oscillator by just hooking *RC* feedback around a CMOS logic inverter with Schmitt trigger input (Figure 7.5). It is guaranteed to oscillate, with clean output transitions and a full logic swing. However, its frequency is not particularly well determined, because hysteresis is not a well-controlled parameter in logic ICs – it's provided to clean up slow inputs, not to do anything precise (you're lucky to get it at all!). The 74HC14, for example, specifies only that the hysteresis amplitude (that is, the difference between rising and falling thresholds) is somewhere between 0.5 V and 1.5 V![5] That means you can expect a frequency spread of 50% or more among oscillators with the same *R* and *C* values. The frequency will also vary with supply voltage; we found frequency approximately proportional to supply voltage for the oscillator of Figure 7.5. Finally, this oscillator generates a somewhat unstable output, with as much as a few percent "jitter" in the timing of successive edges (enough

[5] They specify the individual thresholds, also, with comparable imprecision: the rising threshold is somewhere between 1.8 V and 3.5 V, and the falling threshold is somewhere between 1.0 V and 2.5 V.

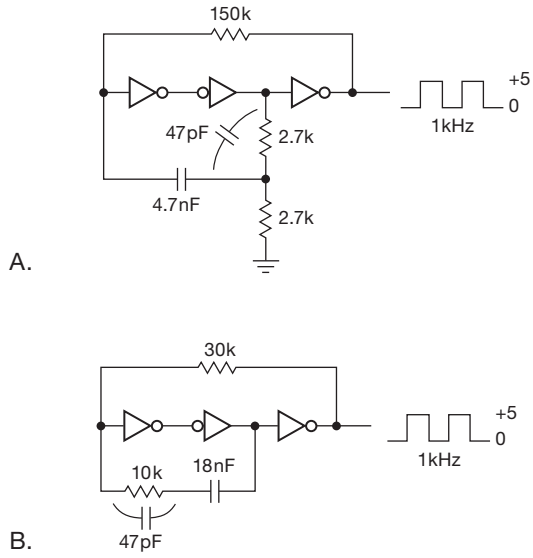


Figure 7.4. CMOS relaxation oscillator variants. A. Half-scale swing prevents input diode clamping (J. Thompson's design). B. Improved stability with one inverter and one buffer (E. Wielandt's design). In our test setup the 47 pF capacitor was needed to prevent parasitics.

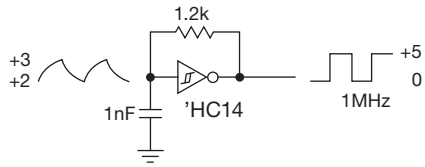


Figure 7.5. Simplest CMOS oscillator.

to see on an oscilloscope), with sensitivity to digital noise on the power supply.

C. Unijunction transistor relaxation oscillator

There are several ways to make a relaxation oscillator that exploit the "negative-resistance" characteristic of devices such as tunnel diodes, gas-filled discharge tubes, diacs, and unijunction transistors. In the circuit of Figure 7.6, for example, the programmable unijunction transistor (PUJT) is a 3-terminal, 4-layer (*pnpn*) device; it looks like an open circuit to the charging *RC* until the capacitor voltage reaches a diode drop above the gate (G) voltage (set by the R_2R_3 divider), at which point the PUJT conducts heavily from anode (A) to cathode (K), discharging the capacitor and beginning a new cycle. The discharge current turns on the output transistor switch, generating a saturated output pulse to ground. With the values shown, the oscillator

generates 10 μs negative-going output pulses at 10 Hz, with a total supply current of just 1 μA.

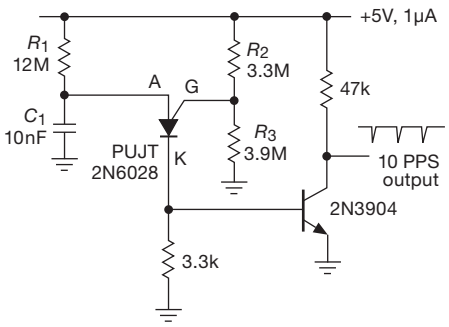


Figure 7.6. Programmable unijunction transistor (PUJT) relaxation oscillator.

Continuing this theme, Figure 7.7 shows a couple of funky oscillators that we couldn't resist hooking up, just to remind us of the old days of electronics. They also exploit "snapback" negative resistance, at somewhat higher voltages than the PUJT, in this case of a neon lamp and a 4-layer diac; the latter is widely used as a triac trigger in phase-control ac powerline dimmers (of the sort that are common in residential wall switches).

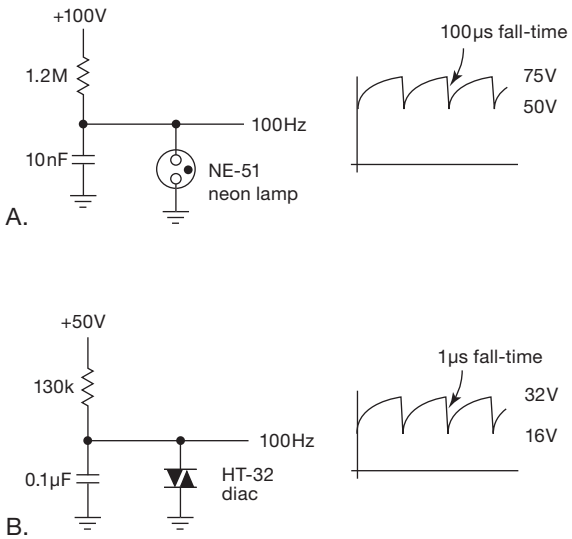


Figure 7.7. Two unusual relaxation oscillators that exploit devices with a negative-resistance "snapback" VI characteristic. The Littelfuse HT32 and ST32 diacs were discontinued in 2009, but you can get the comparable DB3 from at least three manufacturers.

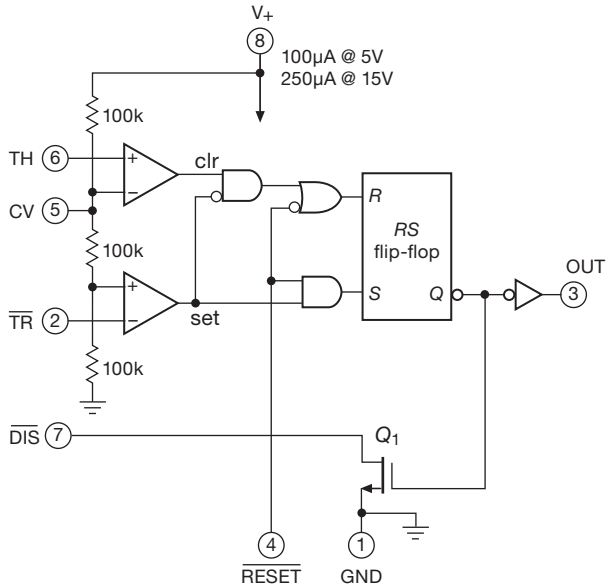


Figure 7.8. Block diagram of the legendary 555 in its contemporary CMOS implementation.

7.1.3 The classic oscillator–timer chip: the 555

The next level of sophistication involves the use of timer or waveform-generator ICs as relaxation oscillators. The most popular chip around is the legendary 555 (and its many successors), designed originally in 1970 by Hans Camenzind at Signetics. It is also a misunderstood chip, and we intend to set the record straight with the equivalent circuit shown in Figure 7.8. Some of the symbols belong to the digital world (Chapter 10 and following), so you won't become a 555 expert for a while yet. But the operation is simple enough: the output goes HIGH (near V_{CC}) when the 555 receives a $\overline{\text{TRIGGER}}$ input, and it stays there until the THRESHOLD input is driven, at which time the output goes LOW (near ground) and the DISCHARGE transistor is turned on. The $\overline{\text{TRIGGER}}$ input is activated by an input level below $\frac{1}{3}V_{CC}$, and the THRESHOLD is activated by an input level above $\frac{2}{3}V_{CC}$.

The easiest way to understand the workings of the 555 is to look at an example (Figure 7.9). Before power is applied, the capacitor is discharged; so when power is applied the 555 is triggered, causing the output to go HIGH, the discharge transistor Q_1 to turn off, and the capacitor to begin charging toward 15 V through R_A+R_B. When it has reached $\frac{2}{3}V_{CC}$ (+10 V), the THRESHOLD input is triggered, causing the output to go LOW and Q_1 to turn on, discharging C toward ground through R_B. Operation is now cyclic, with C's voltage going between $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$, with period

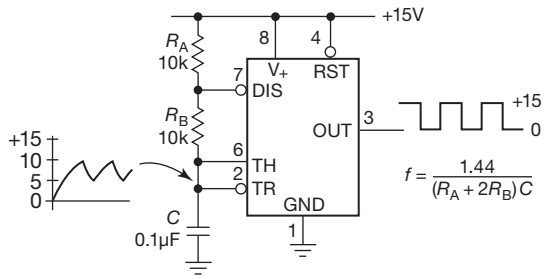


Figure 7.9. The 555 connected as an oscillator.

$T = 0.693(R_A + 2R_B)C$. The output you generally use is the square wave[6] at the output.

Exercise 7.2. Show that the period is as advertised, independent of supply voltage.

The original (bipolar-transistor version) 555 makes a respectable oscillator, with stability approaching 1%. It can run from a single positive supply of 4.5 to 16 V,[7] maintaining good frequency stability with supply-voltage variations because the thresholds track the supply fluctuations. The 555 can also be used as a timer to generate single pulses of arbitrary width (see §7.2.1E), as well as a bunch of other things. It is really a small kit, containing comparators, gates, and flip-flops. It has become a game in the electronics industry to try to think of new uses for the 555 – you too can become an alpha-nerd!

A caution about the bipolar 555: many versions of this IC (in common with some other timer chips) generate a big supply-current glitch (as much as 150 mA) during each output transition.[8] Be sure to use a hefty bypass capacitor near the chip. Even so, the 555 may have a tendency to generate double output transitions; the CMOS versions (discussed next) are better in this regard, but still not cured of all bad behavior.

A. CMOS 555s

Some of the less-desirable properties of the original bipolar 555 (high supply current, high trigger current, double output transitions, and inability to run with very low supply voltage) have been remedied in a collection of CMOS successors. You can recognize these by the tell-tale "555" somewhere in the part number. Table 7.1 on the next page lists most of these that we could find, along with their important specifications. Note particularly the ability to operate at very low supply voltage (down to 1 V) and the generally low supply current. These chips also can run at higher frequency than the original 555. The CMOS output stages give rail-to-rail swing, at least at low load currents (but note that these chips don't have the output-current muscle of the standard 555). All chips listed are CMOS except for the original 555 and the ZSCT1555.

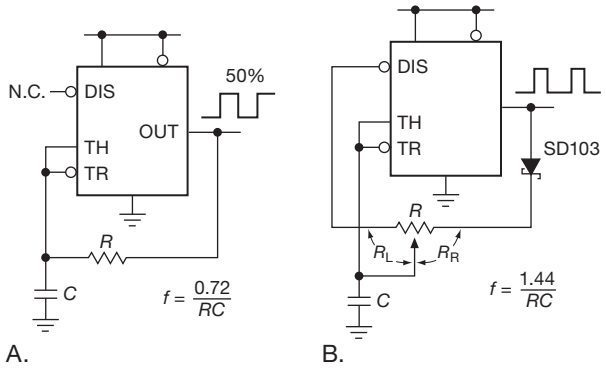


Figure 7.10. More CMOS 555 oscillator circuits. A. 50% duty cycle (square wave). B. Constant frequency, with fully variable duty cycle.

B. 50% duty cycle

The 555 oscillator of Figure 7.9 generates a rectangular-wave output whose duty cycle (fraction of time the output is HIGH) is always greater than 50%. That is because the timing capacitor is charged through the series pair R_A+R_B, but discharged (more rapidly) through R_B alone. But you can make a *CMOS* 555 (with its rail-to-rail output swing) give you exactly 50% duty cycle (a true square wave) with the circuit of Figure 7.10A. The trick is to use a single charge–discharge resistor, connected to the output; so the capacitor is either charging toward $+V_{CC}$ (with $\frac{2}{3}V_{CC}$ threshold) or discharging toward ground (with $\frac{1}{3}V_{CC}$ threshold). It's like two people throwing a ball back and forth, with a dog tirelessly zigzagging back and forth, trying to catch it.[9] You should be able to show that $f_{osc} = 0.72/RC$.

Exercise 7.3. Show that this result is correct.

[6] A *rectangular wave*, to be precise, because it spends 2/3 of the time HIGH and 1/3 of the time LOW. But it's conventional to use the term "square wave," as we do here, to distinguish a 2-level waveform (whatever its symmetry) from a continuous waveform like the exponentials at the capacitor.

[7] Variants like the bipolar ZSCT1555, as well as the CMOS versions, can run on lower voltages, down to 0.9 V for some types; see Table 7.1.

[8] Better-behaved versions will usually brag about it on the datasheet, for example "No output cross-conduction current spikes" (Micrel MC1555 IttyBitty™ RC Timer/Oscillator).

[9] Hope springs eternal in a charging capacitor.

Table 7.1 555-type Oscillators[a]

Part #	Mfg	Qty per package 1	2	4	Supply voltage min (V)	max (V)	Supply current/osc typ @ 5V (μA)	Trig & thresh current max (nA)	Max freq typ @ 5V (MHz)	R_{out} typ @ 5V sink (Ω)	source (Ω)
555		•	•	–	4.5	18	3000	2000	0.5	12[b]	100[c]
ZSCT1555	ZT	•	–	–	0.9	6	150	100	0.3[d]	35[e]	0.15[e]
ICM7555	IL	•	•	–	2	18	60	10	1	50	400
TLC551	TI	•	•	–	1	18	15[f]	0.01[t]	1.8	25	200
TLC555	TI	•	–	–	2	18	170	0.01[t]	2.1	25	200
LMC555	NS	•	•	–	1.5	15	100	0.01[t]	3	40	150
ALD555	AL	•	–	–	2	12	100	0.2	2	20	250
ALD1502	AL	•	•	•	2	12	50	0.4	2.5	20	200
MIC1555	MI	•	–	–	2.7	18	240	50	5[g]	25	100

Notes: (a) all are CMOS except first two (bipolar) entries. (b) I_O (mA) at V_O=0.3V. (c) I_O (mA) at V_{sat}=1.7V. (d) min, @ V_S=1.5V. (e) I_O (mA) at V_{sat}=±0.35V and V_S=1.5V. (f) at V_S=1V. (g) at V_S=8V. (t) typical.

C. Full duty cycle control

Figure 7.10B shows how to make a fixed-frequency output whose duty cycle can be varied from near 0% to near 100%. The frequency would be completely constant, and independent of the duty-cycle setting, except for the effect of the diode drop during charging; that's why we used a low-drop ($V_F = 0.3$ V at 10 mA) SD103C Schottky diode.

Exercise 7.4. Show that $f_{osc} = 1.44/RC$.

D. Sawtooth oscillator

By using a current source to charge the timing capacitor, you can make a ramp (or "sawtooth-wave") generator. Figure 7.11A shows how, using a simple *pnp* current source. The ramp charges to $\frac{2}{3}V_{CC}$, then discharges rapidly (through the 555's *npn* discharge transistor, pin 7) to $\frac{1}{3}V_{CC}$, beginning the ramp cycle anew. Note that the ramp waveform appears on the capacitor terminal and must be buffered with an op-amp since it is at high impedance. In practice, this circuit exhibits a subtle flaw: when a small value capacitor is used, the discharge is so rapid that the bottom of the sawtooth drops below $\frac{1}{3}V_{CC}$ before the discharge transistor shuts off. This is remedied, as shown, by the inclusion of a small resistor in series with the $\overline{\text{DIS}}$ pin, chosen for a discharge time constant of $\sim 5\,\mu$s.[10]

Figure 7.11B shows an alternative, namely to delay the falling-edge signal to TR′ so that the discharge interval is extended long enough to ensure complete discharge; 1 μs was adequate, for the circuit values shown.

In Figure 7.11B we drew a current source symbol, because there are several alternatives to the discrete *pnp* current source. Figure 7.12 shows some simple 2-terminal favorites, namely a JFET "current regulator diode," and two integrated circuits. As discussed in §3.2.2, a JFET with

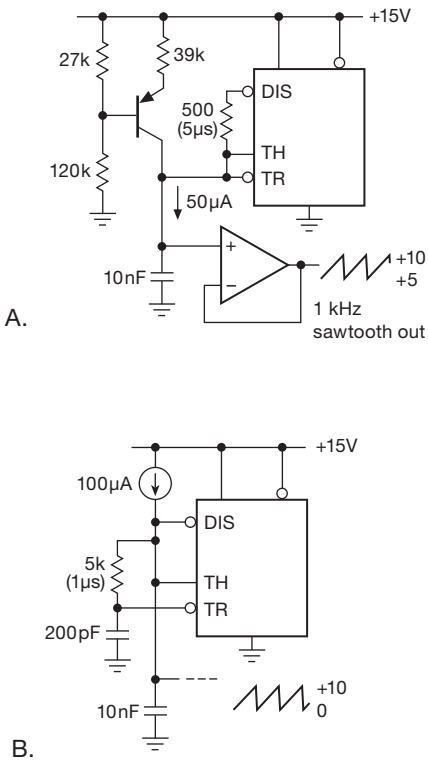


Figure 7.11. Sawtooth oscillators with the CMOS 555. A. Discrete *pnp* current source charges C, whose discharge is slowed to prevent $V_+/3$ undershoot. B. Delayed TR′ on falling edge causes full discharge to 0 V.

[10] We found, by measurement, that the minimum discharge times required to prevent undershoot were approximately 1 μs, 5 μs, and 10 μs, for samples of the LMC555, ICL7555, and bipolar 555, respectively.

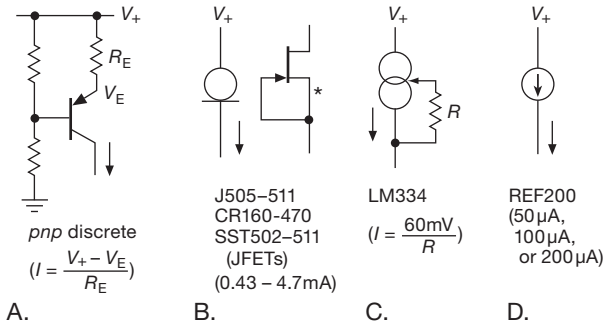


Figure 7.12. Current-source choices for the sawtooth oscillator.

gate tied to source runs at constant current when biased with $\gtrsim$1 V; these are available as packaged 2-terminal devices, in a limited range of currents (0.43 mA–4.7 mA), and with operation up to 100 V. The LM334 is also a 2-terminal (floating) current source, with a third pin that lets you program the current by connecting a resistor as shown; the current is approximately[11] 60 mV/R, and, like the JFET, works down to ∼1 V (its maximum is 40 V). The REF200 is a temperature compensated 2-terminal (floating) 100 μA current source (with several selectable multiples), operable from 2–40 V.

Ratiometric sawtooth oscillator

And now for an interesting variation. In many situations you *want* a stable current source like the REF200, designed to deliver a current that does not depend on the voltage across it (the REF200 is quite good in this regard, its current varying less than 0.1% for voltages from 2 V to 30 V; see Figure 9.37). And that would be fine here, delivering a constant frequency sawtooth in the circuit of Figure 7.11B – providing, of course, that the 555's thresholds stay constant. But if the supply voltage (here V_+=+15 V) were to change, the thresholds would follow proportionally (at $V_+/3$ and $2V_+/3$), and the frequency of oscillation would change. And this would happen if you were to use something like a 9 V battery for your V_+ supply.

There's an elegant way to finesse this problem, namely to make the current source's output proportional to supply voltage, which compensates exactly for the frequency change that would otherwise occur. This is the technique of *ratiometric* design, elegant and powerful. The simple *pnp* current source of Figure 7.12A is almost what you want: it would be exactly right except for the ∼0.6 V base–emitter drop. And you can fix that by rigging up a transistor current source with V_{BE} cancellation.

[11] It's actually "PTAT" — proportional to absolute temperature.

Look at Figure 7.13: in the first circuit a diode in the base divider adds a voltage drop that approximately matches the transistor's base–emitter drop. That's pretty good, but not ideal because (a) the V_{BE} match is imperfect, and (b) the diode's drop means that the voltage drop across R_1 is not quite proportional to V_+ (figure out why).

This is fixed in the second circuit (Figure 7.13B), where Q_1's base voltage tracks V_+ exactly, and its V_{BE} downward drop cancels Q_2's drop going back up. It's imperfect only insofar as the V_{BE}'s aren't equal, both because of transistor mismatch and because of I_C mismatch (à la Ebers–Moll). The third circuit fixes the V_{BE} mismatch (by using matched transistors operating at the same current), but it's got the not-quite-proportional current problem of Figure 7.12A; that is, the current through "programming resistor" R_p is proportional to $V_+ - V_{BE}$. Note, by the way, that this circuit is the classic current mirror that we saw back in §2.3.7.

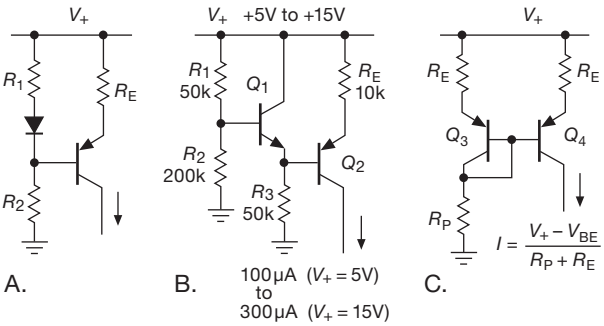


Figure 7.13. Current sources with output approximately proportional to the supply voltage ($I_{out} \propto V_+$), to make f_{out} in Figure 7.11 independent of supply voltage.

E. Triangle oscillator

Figure 7.14 shows a simple way to generate a *triangle* wave with a CMOS 555. The rail-to-rail output square wave is used to generate a current source–sink (of alternating polarity), producing a triangle waveform (going between the usual $\frac{1}{3}V_{CC}$ and $\frac{2}{3}V_{CC}$) at the capacitor. The configuration of diodes is the familiar *bridge rectifier* (§1.6.2), here used to trick the unipolar 2-terminal current-source device into thinking that the current is always flowing in the same (normal) direction, whereas to the outside world it's a bidirectional current source. (You can think of the bridge as presenting to the current source a rectified version of the alternating-polarity external current.) We used Schottky diodes here, to minimize the two-diode forward drop of the bridge. As with the sawtooth oscillator, you have to buffer the high-impedance waveform with an op-amp. This

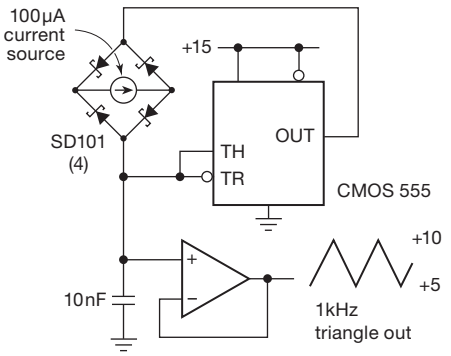


Figure 7.14. Triangle-wave oscillator with the 555. This circuit requires a floating (2-terminal) current source, as in Figure 7.12B–D.

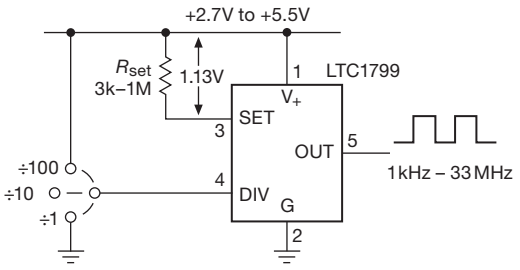


Figure 7.15. The LTC1799 (and its descendants) generates stable rail-to-rail square waves with single-resistor programming.

circuit is simple (taking advantage of the oscillator-oriented innards of the 555), but its performance is not nearly as good as the more elaborate op-amp-based circuit of Figure 4.39 on page 239 or Figure 4.83 on page 267.

Exercise 7.5. Demonstrate that you understand the circuits of Figures 7.11 and 7.14 by calculating the frequency of oscillation in each case.

7.1.4 Other relaxation-oscillator ICs

The classic 555 spawned compatible and improved CMOS successors, as we've seen; it's still alive and well. And it is flexible enough to do lots of good tasks, including timing and pulse generation, which we treat later in the chapter. But there's been a lot of progress in semiconductor electronics since the 555's introduction in 1971, and you can get some very nice contemporary oscillator–timer chips that may well be your better choice.[12] Here are a few of our favorites.

A. LTC1799 and LTC6900 series

These elegant ICs come from the wizards at Linear Technology, who call them "silicon oscillators." The LTC1799 runs from a single positive 2.7 V to 5.5 V supply (drawing about a milliamp), it generates a 50% duty-cycle rail-to-rail square-wave output, and its output frequency is set by a single external resistor (or current source). It operates from 1 kHz to 33 MHz (it has an internal ÷1, ÷10, or ÷100 frequency divider, N, selected by tying the DIV input LOW, open, or HIGH, to produce its 33,000:1 range). That's it – one external component, and off you go! It has very good accuracy (±0.5%, typ), temperature stability (±0.004%/°C, typ), and voltage coefficient (±0.05%/V) Figure 7.15 shows how to use it. The similar LTC6900 series includes the LTC6900 (somewhat lower power consumption), the LTC6905 (17 MHz–170 MHz), and the LTC6903/4 (1 kHz–68 MHz, programmed via a 3-wire serial digital connection). The latter part would be particularly well suited to a system with a resident microcontroller (Chapter 15), which can effortlessly send digital commands hither and yon.

Figure 7.16 shows the dependence of output frequency on resistor value. If you like equations, you may be happier with this:

$$f_{\text{osc}} = \frac{1}{N} \cdot \frac{100}{R(\text{k}\Omega)} \quad \text{MHz}.$$

Because the frequency is set by an input current at the SET input terminal, you can use instead an externally generated current to adjust the frequency (we'll call it an "ICO," for "current-controlled oscillator"), keeping in mind that the SET input sits approximately 1.13 V below the positive rail. For current programming the datasheet suggests currents in the range of 5 μA to 200 μA, with ×10 range switching as before (which could be done electrically, for example with 3-state logic or with a pair of transistor switches). The V_+-referenced voltage level at the SET input makes *voltage* programming a bit awkward: the datasheet suggests one method, namely to apply a control voltage through a second resistor to the SET input, to add or subtract a variable current from that supplied by R_{SET}; but this method has its problems, and you're probably best off just generating a current externally to drive SET.

B. LTC699x "TimerBlox"

A few years after their LTC1799/6900, Linear Technology introduced the "TimerBlox®" series of timing functions – oscillators, pulse-width modulators, one-shots

[12] Being of recent design, they are invariably available in small surface-mount (SMT) packages. This can be a disadvantage, though, if you are looking for easy-to-prototype traditional *through-hole* packaging, because new designs increasingly come *only* in SMT.

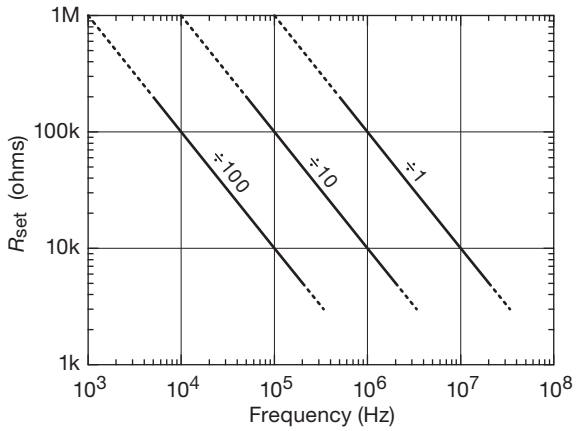


Figure 7.16. LTC1799 output-frequency programming. The manufacturer advises you to stay on the solid lines for best accuracy.

(monostables), and delay/debounce. Rather than trying to make a "one size fits all" chip (with lots of pins), they instead tailored separate chips, in the small 6-pin SOT23-6 package, or the even-smaller 2 mm × 3 mm DFN package. These chips share many properties, for example a single-supply (2.25–5.5 V), good out-of-the-box accuracy (∼2% worst-case), single resistor tuning, and a single analog pin for user programming of 16 choices of range and mode (compare with the LTC1799/6900, where one pin sets one of three ranges). LTC has pulled out the stops to create a set of simple, function-specific, and small parts[13] to handle our oscillator and timing problems.

Here is a listing of the TimerBlox parts available at the time of writing:

Part #	Function	Total Range	Comments
LTC6990	VCO	488 Hz–2 MHz	16:1 tuning range, as eight octave (×2) ranges
LTC6991	LF osc; timer	29 μHz–977 Hz	1 ms–9.5 hr, as eight ×8 ranges
LTC6992-x	PWM	3.8 Hz–1 MHz	0%–100%, 5%–100%, 5%–95%, or 0%–95%
LTC6993-x	One-shot	1 μs–34 s	normal or retrig; rising or falling edge
LTC6994-x	delay/ debounce	1 μs–34 s	delay single or all edges; reject narrow pulse

The LTC6991 takes aim at the problem of very low-frequency square-wave oscillators, with a total output frequency range of approximately 30 *micro*hertz to 1 kHz;

[13] The SOT23 is the experimenter's ideal surface-mount part, large enough for ordinary soldering-iron assembly; LTC could have chosen a smaller machine-assembly part for the "larger" package.

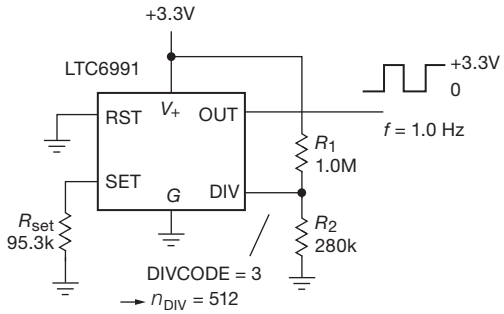


Figure 7.17. Resistor-programmed 1 Hz oscillator.

thinking of this as a timer, that's a range of periods from 1 millisecond to 9 hours. You select one of eight ranges, whose "center" frequencies (corresponding to R_{set}=200 kΩ) are spaced apart by factors of eight: 0.00012 Hz, 0.001 Hz, 0.008 Hz, 0.064 Hz, 0.5 Hz, 4 Hz, 32 Hz, 250 Hz. Within a chosen range, the external resistor R_{set} (50–800 kΩ) tunes the oscillation frequency over a 16:1 range. For example, in the 4 Hz range you can tune continuously from 1 Hz (R_{set}=800k) to 16 Hz (R_{set}=50k).

Now for the mystery: how can you select one of 16 operating modes (eight ranges, two polarities) with a single pin, without the need for any kind of serial programming? Easy: these chips contain an internal 16-level (4-bit) ADC, with the power supply voltage V_+ as full-scale reference, to determine the DIVCODE value (an integer from 0 to 15). So you simply apply a dc voltage in the range 0 to V_+ to the DIV programming pin, most easily done via a pair of resistors from V_+ to ground.[14] That voltage divider's output (call it V_k) should aim for the midpoint of one of the sixteen subdivisions from 0 V to V_+. Stated precisely, you would use a pair of 1% resistors that produces an output voltage $V_k = V_+(2k+1)/32$ in order to select[15] DIVCODE=k. That DIVCODE then determines the frequency range and (in this case) the polarity of output and reset.

As an example, Figure 7.17 shows how to generate a

[14] You could instead use the voltage output of a DAC, if you want digital control.

[15] This is an, uh, *interesting* scheme. Why not use two more pins, for a total pincount of eight, with programming like the LTC1799 (each one tied to GND, V_+, or open)? Evidently, in their zeal to minimize pincount and package size, and simultaneously to avoid the need for any kind of serial programming, the designers chose this analog scheme, with internal ADC. We'd like to see them offer an 8-pin SOIC (and DFN) option, with 3-level selection on three pins; that way you'd get up to 27 mode choices, without the need for any external resistors. And, hey, a DFN-8 takes up less space than a DFN-6 plus two resistors! OK, you clever folks at LTC, what do you think?

1 Hz square wave with 3.3 V logic level output (i.e., 0 V and +3.3 V). We selected the 0.5 Hz range (k=3), for which the voltage divider ratio is $R_2/(R_1+R_2)$=7/32; then we selected R_{set}=95.3 kΩ for an f_{out} of 1.0 Hz according to the datasheet's exact formula for oscillator period

$$T = \frac{1}{f_{out}} = \frac{R_{set}}{50\text{k}\Omega} \cdot n_{div} \cdot 1.024 \times 10^{-3} \text{ (seconds)}$$

where the chip's internal divider ratio n_{div} is determined[16] from the DIVCODE according to $n_{div} = 2^{3k}$, thus n_{div}=512 here. These chips run on reasonably low power; this circuit draws about 0.1 mA when unloaded.

Later in the chapter, in connection with timers, we'll see another example in which the LTC6991 is used to generate a switched "hour of power" (Figure 7.65). And we'll see also how to use other members of this family, in VCO and timer applications (§7.2.4).

C. Oscillator + divider

Another class of oscillator–timers uses an oscillator (relaxation or otherwise) followed by a digital counter, to generate long delay times without resorting to large resistor and capacitor values. Examples of this are the 74HC4060 and the Maxim ICM7240/50/60. These CMOS parts generate an output pulse for some number N of oscillator cycles,[17] and they run on a fraction of a milliamp. These timers (and their near relatives) are great for generating delays from a few seconds to a few minutes.

Some more recent additions to this class of oscillator include the LTC6903/4 (described earlier), and the Maxim DS1070/80 series. The LTC ICs operate from the same supply voltages as the LTC1799/6900 and generate rail-to-rail square-wave outputs over the larger range of 1 kHz to 68 MHz; but they require *no* external components! The trick is that the frequency is set by sending a pair of numbers (a 4-bit scaling factor and a 10-bit frequency coefficient) via a bit-serial digital-input line. That might sound complicated; but in fact it is remarkably easy to do in any system where you have an embedded microcontroller (a small computer-on-a-chip), which in contemporary electronics includes just about any electronic circuit. We'll see much more of that in Chapter 15 ("Microcontrollers").

The DS1070/80 series "EconOscillators" is similar, with serial programming to set the frequency; that is, held in internal nonvolatile memory, so you need to program it only once, or when you want to change it. The DS1085 is currently the fanciest member of the family, with four programmable settings that let you choose an output frequency (in the range of 8.1 kHz to 133 MHz) to better than one part in a thousand. Beware, though, that the initial accuracy and stability of this "silicon oscillator" is only at the ±1% level; that is, its resolution (step size) greatly exceeds its accuracy and stability. Think of it as a "1% oscillator" whose frequency you can set in the field.[18]

There's a nice generalization of these "divide-by-N" methods, namely the *phase-locked loop* (PLL). We treat them later in this chapter (§7.1.8B) and in Chapter 13.

D. Voltage-controlled oscillators

Other IC oscillators are available as voltage-controlled oscillators (VCOs), with the output rate variable over some range according to an input control voltage. We saw the germ of this idea when we used a current source to charge the capacitor in the 555; with little additional effort we could have made the current proportional to a controlling input voltage. There are many uses for a VCO, and so there are many offerings from the chip manufacturers. Some of these have frequency ranges exceeding 1000:1. Examples are the original NE566 and later designs like the ICL8038, MAX038, XR2206/7, and 74LS624–9 series.

The 74LS624 series, for example, generates digital-logic-level outputs up to 20 MHz and uses external *RC*'s to set the nominal frequency. Faster VCOs like the 1648 can produce outputs to 200 MHz, and there are much higher-frequency VCO techniques (like Gunn-diode oscillators and YIG oscillators) that operate in the many *giga*hertz range.

Where linearity is important, a precision voltage-to-frequency (V/F) converter like the AD537, LM331, or AD650 really does the job, with worst-case linearity of 0.15%, 0.01%, or 0.005%, respectively. Most VCOs use internal current sources to charge and discharge a capacitor, and many therefore provide triangle-wave outputs. The classic Exar XR2206 goes further – it includes a set of "soft" clamps to convert the triangle wave to a not-too-great sine wave; they call this a *sine shaper*, and it produces a sine-looking wave with <1% distortion. Depending on external timing components, it goes from fractional hertz

[16] Valid for k between 0 and 7; see the datasheet for more detail.

[17] Specifically, 10 choices of $N = 2^k$ (going from k=4 to k=14) for the 74HC4060; any of N=1 to 255 for the ICM7240; N=128 for the ICM7242; any of N=1 to 99 for the ICM7250; and N=60 for the ICM7260.

[18] For better accuracy, you can use a crystal oscillator (§7.1.6) upstream of the divider. A nice example, sadly discontinued, was the Epson SPG series of "selectable-output crystal oscillators," in which you chose the output frequency with six programming pins that you connect to ground or to the +5 V supply.

(at the low end) to 1 MHz (at the high end),[19] with a sweep range of 1000:1 and a temperature stability of frequency of 0.002%/°C. You can also use it as a triangle-wave generator, in which mode it lets you adjust the duty cycle from 1% to 99%.

In Chapter 13 we'll see some additional VCO methods, there in the context of "voltage-to-frequency conversion." Those methods are *synchronous* and require a fixed-frequency clocking input; those clocking pulses are passed through to the output, or not, such that the average output frequency is accurately proportional to the input voltage; see §13.8.1 and §13.9.

VCO chips sometimes have an awkward reference for the control voltage (e.g., the positive supply) and complicated symmetrizing schemes for sinewave output. It is our opinion that the ideal VCO has yet to be developed. Many of these chips can be used with an external quartz crystal, as we discuss shortly, for much higher accuracy and stability; in such cases the crystal simply replaces the capacitor.

You can make VCOs with techniques other than *RC* (or current-driven) relaxation oscillators. For example, the frequency of an *LC* oscillator (§7.1.5D) can be electrically tuned with a voltage-variable capacitor (a *varactor*), though the tuning range is much less (typically 1–10%) than a tunable relaxation oscillator. Likewise it is possible to "pull" the frequency of a quartz crystal over a narrow range of perhaps 0.01%. Other oscillator technologies (Gunn oscillators, dielectric resonator oscillators, YIG-tuned oscillators, "current-starved" inverter chains, and others) permit electrical tuning by various means, an essential element of phase-locked-loop frequency synthesis (§13.13).

7.1.5 Sinewave oscillators

For many applications you need a real *sine* wave, rather than the rectangular, triangular, or other waveforms you get from relaxation oscillators. Examples are in audio-frequency test and measurement, in radio and video communications, and in medical and scientific research and applications. It is common in those applications to talk about *spectral purity* or *harmonic distortion*, which are measures of departure from an ideal sinewave.

The *RC* relaxation oscillators we've been talking about don't generate sinewaves – their native waveforms are ramps (linear or *RC* exponential) and rectangular waves. The XR2206 in §7.1.4D illustrates one method of making an approximate sinewave, namely by clipping a triangle with a succession of soft clamps; this method is used in some analog function generators.[20] However, there are oscillator configurations that generate sinewaves directly, and there are other tricks for making sinewaves from square waves.

Examples of native sinewave oscillators are the *Wien bridge* (which uses humble *R*'s and *C*'s), the *resonator oscillators* (which use resonators such as an *LC* circuit, quartz crystal, coaxial or cavity resonator, or even an atomic–molecular resonator), and the method of *direct digital synthesis* (DDS).

In this section we look at ways to make sinewaves, beginning with a few tricks for making sinewaves from square waves (or other non-sinusoidal waveforms), then moving to techniques that generate sinewaves directly. In a later section (§7.1.9) we discuss *quadrature oscillators*, which generate a pair of signals with a 90° phase relationship.

A. Sine from square

The simple trick here is to lowpass filter the square (or whatever) wave, to remove all but the fundamental frequency. An easy way to think about it is to remember that any[21] periodic waveform can be deconstructed into a set of sinusoidal components (the waveform's "Fourier series"), each with some fixed amplitude and phase; when you add those components together, you reconstruct the original waveform. The lowest frequency component (at the periodicity of the original waveform) is the *fundamental*, and all higher components ("harmonics") are integer multiples of the fundamental frequency (that is, at $2f_0, 3f_0, 4f_0, \ldots$). So you can create a sinusoidal wave from an arbitrary periodic wave by lowpass filtering at a frequency higher than the

[19] The MAX038 was similar, but faster – 20 MHz. We are saddened by its passing. Its obituary, posted on the Maxim website, reads "This product was manufactured for Maxim by an outside wafer foundry using a process that is no longer available."

[20] Such techniques of "corrupting" a triangle wave with diode clamps do not produce a high-quality sinewave: the resulting distortion can rarely be reduced below 1%. By comparison, most audiophiles insist on distortion levels well below 0.1% for their amplifiers. To test such low-distortion audio components, pure sinewave signal sources with residual distortion less than 0.01% or so are required.

[21] Well, *nearly* any: mathematicians are adept at finding pathological functions to rebut any loosely-worded proposition. We should probably say "any *well-behaved* periodic waveform"; but we can say, confidently, "any periodic waveform you can actually create with electronics."

fundamental but lower than the second harmonic,[22] thus extracting only the fundamental sinusoidal component.

As long as the filter doesn't pass the harmonic spectrum significantly, you'll get a good sinewave. As we discussed in Chapter 6, you can make lowpass filters in several ways: as "continuous-time" analog filters (with a network of discrete L's and C's; or, at lower frequencies, with active filters; or with discrete-time switched-capacitor filters; or with the numerical methods of digital signal processing).

To demonstrate this technique, we hooked up a 555 as a 1 kHz oscillator (Figure 7.10A, with R=75 k and C=10 nF), and ran the square-wave output through a lowpass active filter (a continuous-time 8-pole Butterworth) with breakpoint at 1.5 kHz. The output looked like a pretty good sinewave, and in fact it measured just 0.6% distortion[23] This technique requires a somewhat complex analog circuit, and it lacks frequency agility (i.e., once you've chosen the filter's breakpoint frequency, you can vary the oscillator frequency by only a small amount, say ±25%).

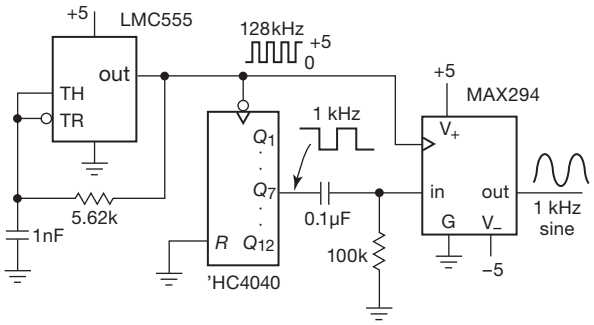


Figure 7.18. Sinewave generation with tracking lowpass filter. The MAX294 (or similar MAX293 or LTC1069-1) is an 8-pole elliptic lowpass switched-capacitor filter requiring no external components.

Switched-capacitor filters are simpler to use, being available as inexpensive clocked ICs; instead of requiring R's and C's to set the passband, the clock frequency tunes the filter – it determines the breakpoint frequency. Figure 7.18 shows a simple circuit in which a MAX294 switched-capacitor lowpass filter is used to convert an input square wave to an output sinewave. The MAX294's breakpoint is at $f_{CLK}/100$, so we've clocked it at 128 times the input square wave frequency. This puts the breakpoint at

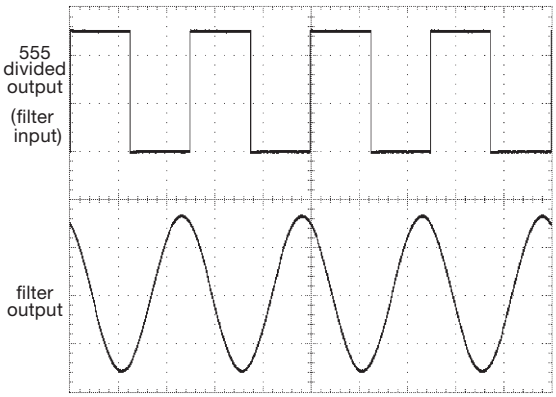


Figure 7.19. Waveforms from the circuit of Figure 7.18: the 1 kHz square-wave input ($f_{OSC}/128$), and the resulting sinewave after lowpass filtering. The output amplitude is equal to the fundamental component of the square wave input, which is a factor of $4/\pi$ larger. Horizontal: 400 μs/div. Vertical: 2 V/div.

$1.28 f_{IN}$, and results in the handsome sinewave shown in Figure 7.19; it measured a very clean 0.03% distortion.

Not only is this circuit simpler than the analog lowpass scheme, it provides highly predictable output amplitude[24], it provides tunability – that's because the filter tracks the input frequency, keeping its lowpass breakpoint at $1.28 f_{IN}$ as the input frequency is varied.[25] In this circuit we varied the 555's frequency by factors of ten, and measured a maximum of 0.1% distortion for output sinewave frequencies over the range of 100 Hz to 10 kHz.

B. Wien bridge oscillator

At low to moderate frequencies the Wien bridge oscillator (Figure 7.20) is a useful source of low-distortion sinusoidal signals. The idea is to make a feedback amplifier with 0° phase shift at the desired output frequency, then adjust the loop gain so that a self-sustaining oscillation just barely takes place. For equal-value R's and C's as shown, the voltage gain from the noninverting input to op-amp output should be exactly +3.00. With less gain the oscillation will cease, and with more gain the output will saturate. The distortion is low if the amplitude of oscillation remains within the linear region of the amplifier, i.e., it must not be allowed to go into a full-swing oscillation. Without some trick to control the gain, that is exactly what

[22] Depending on the waveform's symmetry, the Fourier series may consist of odd harmonics only, in which case the first higher harmonic is at $3f_0$; this is the case for a square wave of 50% symmetry.

[23] Putting the breakpoint at 1.2 kHz lowered the distortion to a mere 0.1%, or −60 dBc (dB relative to the carrier).

[24] Given by the first term of a square wave's Fourier series, $A_{PP}=(4/\pi)V_{cc}$, thus 2.25 Vrms.

[25] An elegant variation is illustrated in the LTC1799 datasheet, in which the tracking switched-capacitor lowpass filter is further configured with a stopband *notch* at the input signal's $3f_0$, thus providing additional attenuation at the square wave's strongest harmonic.

will happen, with the amplifier's output increasing until the effective gain is reduced to 3.0 because of saturation. The tricks involve some sort of long-time-constant gain-setting feedback, as we will see.

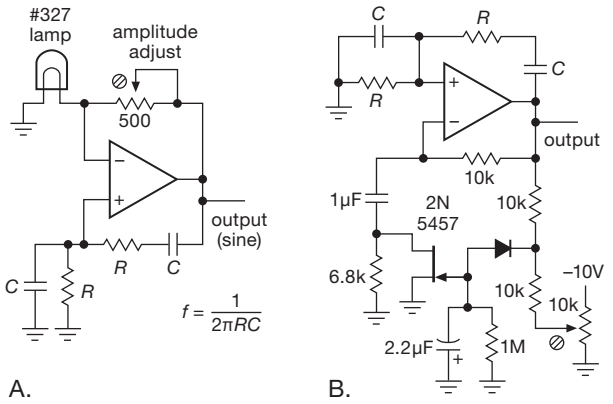


Figure 7.20. Wien-bridge low-distortion oscillators. A. Amplitude control with incandescent lamp. B. Amplitude control by a JFET's variable resistance.

In the first circuit, an incandescent lamp is used as a variable-resistance feedback element. As the output level rises, the lamp heats slightly, increasing its resistance and therefore reducing the noninverting gain. The circuit shown has less than 0.003% harmonic distortion for audio frequencies above 1 kHz; see LTC App. Note 5 (12/84) and App. Note 43 (6/90) for more details.[26]

In the second circuit, an amplitude discriminator consisting of the biased divider and diode charges a long-time-constant *RC*; this voltage adjusts the ac gain by varying the resistance of the FET, which behaves like a voltage-variable resistance for small applied voltages (see §3.2.7). Note the long time constant used (2 seconds); this is essential to avoid distortion, because fast feedback will distort the wave by attempting to control the amplitude within the time of one cycle.

Another interesting technique for amplitude control is shown in Figure 7.21, where a photoresistive optocoupler is used for gain feedback. These devices consist of an LED illuminating a resistive element, with the output terminals providing a resistance of good linearity (<0.1% distortion

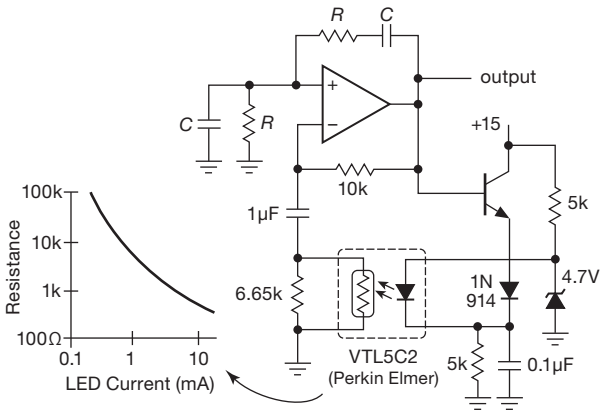


Figure 7.21. A photoconductive opto-isolator provides another method of amplitude control in the Wien-bridge oscillator. (Courtesy Steve Cerwin)

for applied voltages <1 Vrms), varied over several decades of resistance according to the LED current. Unlike silicon devices, they are intrinsically slow (tens of milliseconds for the device shown), which is helpful in this application.

It has been claimed that, with careful design, Wien-bridge oscillators can be built with distortion "well into the parts-per-billion range." Tricks to do this, which include the use of cascaded op-amps (for high loop gain, therefore low distortion), and cancellation of remaining harmonic distortion, are described in *Linear Technology Magazine*, Feb. 1994, pp. 26–28.

An ultralow-distortion design

We are skeptical of this claim.[27] However, with some attention to detail the distortion can be brought down to levels of a few parts per million (0.0001%) using quite conventional components and techniques. Figure 7.22 shows such a circuit, which we designed and tested to find out just how difficult this might be.

We began with a two-op-amp circuit variant, which has the advantage of using the inverting configuration; this reduces distortion by eliminating the common-mode signals present in the simpler non-inverting amplifier. The OPA627 is a fast (f_T=16 MHz), low-noise (e_n=4.5 nV/$\sqrt{\text{Hz}}$), low input-current ($I_B \approx 1$ pA) op-amp, with the particular advantages here of intrinsically low distortion (0.00003% as a unity-gain follower with a 10 V signal at 1 kHz) and the ability to operate with ±15 V supplies. (We want to operate with a large signal swing to minimize the effect of noise on the sinewave purity.)

[26] The use of a lamp to stabilize the Wien bridge oscillator was invented and patented (#2,268,872) by William Hewlett in 1942. The resulting model 200A audio oscillator was the first commercial product sold by Hewlett, with his partner David Packard; it cost $54.40 (but not because of cost calculations: they liked the slogan "54-40 or Fight!", presidential candidate Polk's aspirations for the northwestern border of the United States). The rest is history.

[27] We counsel a cautious skepticism when confronting any claims of parts-per-billion linearity in analog circuits.

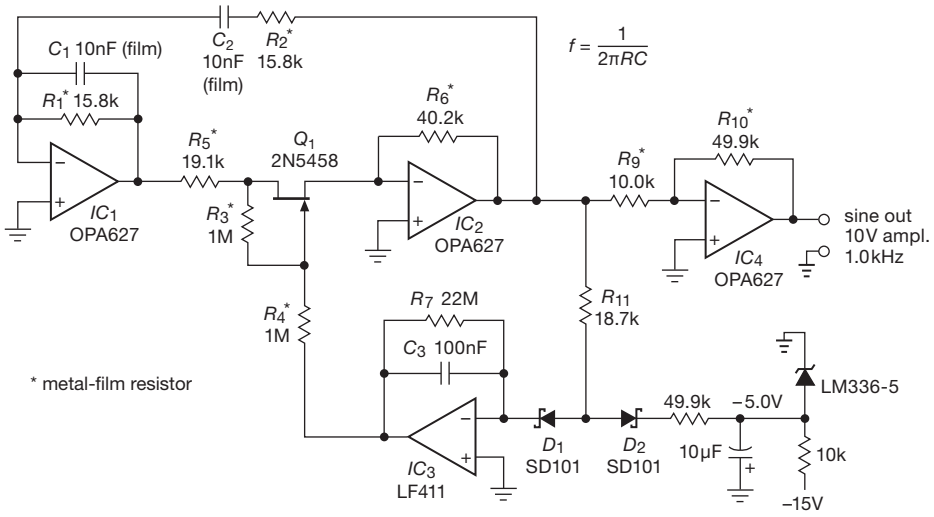


Figure 7.22. Wien-bridge (1 kHz) oscillator with unusually low distortion (<0.001%). To obtain best performance, include a trimmer in R_5, adjusted to optimize the controlled value of the JFET's resistance.

For this configuration, sustained oscillation occurs when IC_2's voltage gain is -2.00. We chose R_5 to be 5% smaller than critical, with the series JFET providing the adjustable 1k (nominal) additional resistance. That puts a 100 mV (peak-to-peak) sinewave across the JFET, which we judged small enough for good linearity, especially with the linearizing divider R_3R_4 (see §3.2.7A). Amplitude control is provided by integrator IC_3, which receives pulses of input current (via the divider to a stable -5 V reference) when the sinewave output of IC_2 reaches 2 V amplitude: its negative-going output back-biases the JFET's gate, relative to the source at virtual ground, raising the JFET's resistance and thus lowering the gain of IC_2 to maintain this output amplitude.[28] For the values shown, the JFET's minimum R_{ON} (i.e., at V_{GS}=0) must be less than 1k, which requires a minimum g_m of 1 mS (see §3.2.7); the 2N5458 specifies a minimum g_m of 1.5 mS, so the circuit is guaranteed to start up. We tacked on an inverting gain-of-5 stage to produce a healthy 10 V amplitude output.

The circuit worked "out of the box" – correct frequency and amplitude (1 kHz, 10 V) and a good-looking sinewave. The measured total harmonic distortion (THD) was an admirable 0.002%.[29] Before celebrating, though, we tried some variations: (a) Replacing the film capacitors with ceramic ("X7R" type) raised the distortion[30] a hundredfold, to 0.22%! (b) Dropping the swing across the JFET to 50 mVpp (by raising R_5 to 19.6k) halved the distortion, to 0.001%; from here on we stuck with this smaller JFET swing. (c) Next we trimmed the ratio of R_3/R_4 slightly, to minimize the (dominantly second harmonic) distortion, achieving a final THD figure of 0.0002%; that's -114 dB down from the signal, a mere 2 parts per million! (d) Finally, to see the effect of the linearizing gate divider, we omitted R_4, which raised the distortion 50-fold, to 0.01%.

Some important lessons from this exercise, if you want the lowest distortion, are (a) avoid inexpensive ceramic capacitors, (b) use the gate-linearizing trick (subtracting $V_{DS}/2$ from V_{GS}), and (c) keep the swing small across JFETs being used as resistors, preferably less than 100 mV (which, however, causes a rather lengthy amplitude settling time). Because JFET nonlinearity dominated the distortion, even when trimmed, we could have reduced the distortion still further by running the oscillator at lower amplitude, say 0.5 V, at the expense of added broadband noise produced by the fixed noise contribution of the op-amps.[31]

C. *RC* phase-shift oscillator

Unlike the relaxation oscillator (where an *RC* time constant is combined with voltage thresholds to make an

[28] We chose the integrator gain to put the unity-gain frequency of the control loop roughly at 50 Hz.

[29] Almost entirely second harmonic.

[30] Now dominated by third harmonic.

[31] Better yet, use a photoresistive gain control, as in Figure 7.21. Jim Williams did this, and he also added a lowpass filter between IC_3 and R_4 to attenuate the integrator's small cycle-by-cycle error-correcting waveform, achieving a measured distortion below 3 ppm; see LTC appnote AN132.

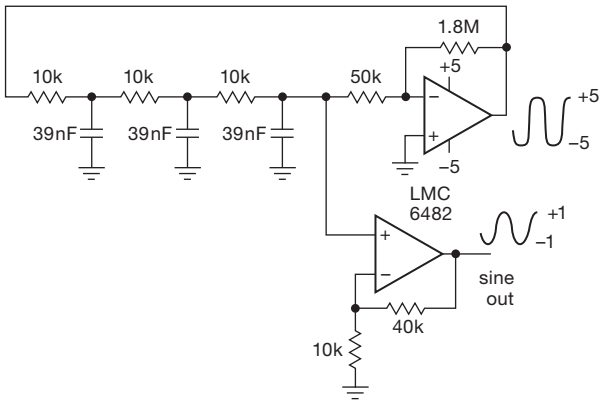


Figure 7.23. Phase-shift oscillator. Three *RC* sections produce a 180° phase shift, converted to positive feedback by the inverting amplifier.

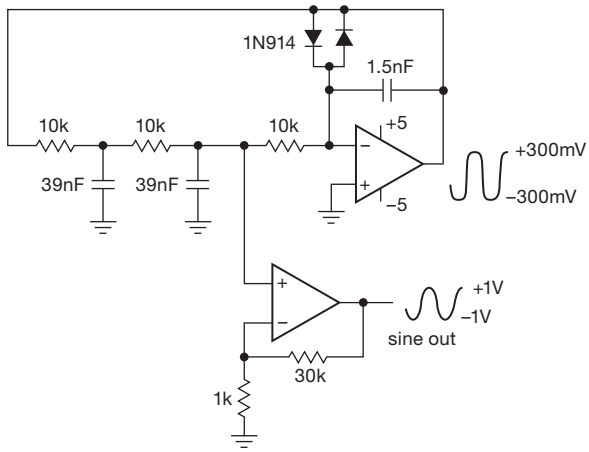


Figure 7.24. A variation on the phase-shift oscillator. An integrator adds 90° lagging phase shift (and inversion) to a 2-section *RC*. (Courtesy Tony Williams)

oscillation), the preceding Wien-bridge oscillator exploits the phase-shifting characteristics of an *RC* network, in a circuit with positive feedback, to select its operating frequency. This same idea is used in *phase-shift* oscillators: gain and feedback are applied around a network of several *R*'s and *C*'s, arranged so that the resulting loop oscillates at a frequency set by the network. Figure 7.23 shows a classic example.

The three-section *RC* produces a lagging phase shift that increases with frequency, reaching 180° at approximately ω=2.4/*RC*, where the loss through the network is a factor of 26.[32] The inverting amplifier provides the remaining 180° phase shift, and also the needed voltage gain (here a conservative $G_V=-36$). The circuit oscillates at 1 kHz, with a (rather distorted) clipped sinewave that swings rail-to-rail (i.e., ±5 V). However, the waveform at the last *RC* section is nicely sinusoidal, and, after an ×5 gain stage, emerges as a 1 V amplitude sinewave with just 0.9% distortion.

For phase-shift oscillator devotees, there is a world of possible variations: discrete transistor configurations, amplitude-limiting feedback schemes, and so on. Although we tried mightily, we are unable to resist the temptation to show another phase-shift oscillator (Figure 7.24). Here an (inverting) integrator provides 270° of lagging phase shift, so only two *RC* sections are needed to close the loop in-phase. This circuit illustrates also the use of back-to-back diode clamps for amplitude limiting. As with Figure 7.23, we've derived the output from the last *RC* section, where its distortion is minimum; the circuit's output is a 1 kHz sinewave of 1 V amplitude and 1% distortion. If you were instead to put the diode limiters across the first 39 nF phase-shifting capacitor, the integrator output would be another sinusoidal wave of low distortion – in fact, it would be a 90° lagging wave (cosine, inverted), thus creating a "quadrature pair" (§7.1.9, though here of unequal amplitudes).

[32] The effects of loading cause those values to deviate from the ideal (fully isolated stages) values of $\omega=\sqrt{3}/RC$ and loss factor of 8.

D. *LC* oscillators

At high frequencies (say above a megahertz) a favorite method of sinewave generation is to use a *resonator* of some sort to establish the frequency of oscillation. The resonator itself may be electrical (e.g., an *LC* circuit), or electromechanical (e.g., a piezoelectric quartz crystal), or even atomic or molecular (e.g., a hydrogen maser). Some resonators are easily tuned (e.g., *LC*), whereas others are quite stably fixed (e.g., a quartz crystal). Resonator-based oscillators are fundamentally different from the preceding *RC*-based oscillators, because they exploit a system that has an intrinsic resonant frequency (like a crystal resonator), compared with an *RC* circuit's nonresonant time constant (or phase shift). Because these resonances can be both narrow in frequency and stable over time, they are well suited for conscription into the noble service of oscillation.

We begin with *LC*-controlled oscillators, which play an important role in communications, and in which a tuned *LC* circuit is connected in an amplifier-like circuit to provide gain at its resonant frequency. Overall positive feedback is then used to cause a sustained oscillation to build up at the *LC*'s resonant frequency; such circuits are self-starting.

Figure 7.25 shows two popular configurations. The first

circuit is the trusty Colpitts oscillator,[33] a parallel tuned *LC* at the input, with positive feedback from the output (inverted in phase, because the JFET is inverting). For this circuit, operating at 20 MHz, the distortion is typically less than −60 dB. The second circuit is a Hartley oscillator, built with an *npn* transistor; the variable capacitor is for frequency adjustment. Because a common-base amplifier is noninverting, the feedback signal is not inverted. Both circuits use output *link coupling*, just a few turns of wire acting as a step-down transformer.

A third oscillator configuration – the emitter-coupled oscillator (or Peltz oscillator) – is shown in Figure 7.26, which is used in the 1648 "voltage-controlled oscillator" chip of the ECL III digital logic family.[34] You can think of this as a feedback noninverting differential amplifier, with a parallel *LC* to set the frequency of oscillation. The 1648 will operate to 200 MHz, with the operating frequency set, as usual, by the resonant frequency of the parallel *LC*: $f_0 = 1/2\pi\sqrt{LC}$. Although the datasheet claims "high spectral purity," we found it to be mediocre at best when compared with a Clapp oscillator using a single JFET (see Figure 7.30 later in this section).

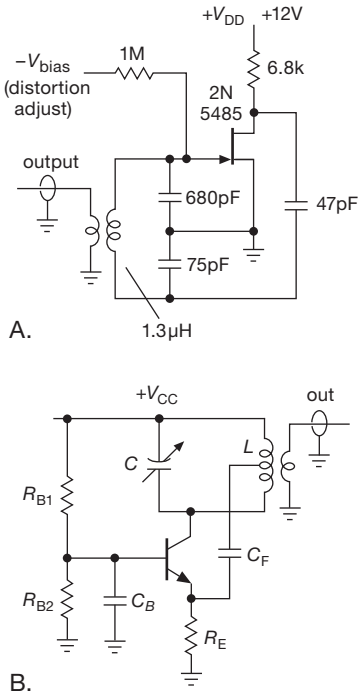


Figure 7.25. Popular *LC* oscillator configurations. A. Colpitts with center-tapped resonating capacitor. B. Hartley with a center-tapped resonating inductor.

Electrical tunability

LC oscillators can be made *electrically* tunable over a modest range of frequency. The trick is to use a voltage-variable capacitor ("varactor") in the frequency-determining *LC* circuit. The physics of diode junctions provides the solution, in the form of a simple reverse-biased diode: the capacitance of a *pn* junction decreases with increasing reverse voltage (§1.9.5B). Although any diode acts as a varactor, you can get special varactor diodes designed for the purpose; Figure 7.27 shows the tuning characteristics of some representative types. And Figure 7.28 shows how to use a varactor to achieve ±1% electrical tunability, in this case with simple Armstrong-type JFET oscillator (with transformer-coupled feedback from the source). In this circuit the tuning range has been made deliberately small to achieve good stability, by using a relatively large fixed capacitor (100 pF) shunted by a small tunable capacitor (maximum value of 15 pF). Note the large biasing resistor (so the diode bias circuit doesn't load the oscillation) and the dc blocking capacitor.

Varactors typically provide a maximum capacitance of a few picofarads to a few hundred picofarads, with a tuning range of about 3:1 (although there are wide range varactors with ratios as high as 15:1). Because the resonant frequency of an *LC* circuit is inversely proportional to the square root of capacitance, it is possible to achieve tuning ranges of up to 4:1 in frequency, though more typically you're talking about a tuning range of ±25% or so.

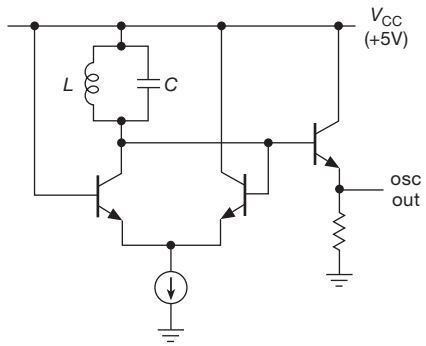


Figure 7.26. Emitter-coupled oscillator, a simplified version of that used in the MC1648 ECL-family IC.

In varactor-tuned circuits, the oscillation itself (as well as the externally applied dc tuning bias) appears across the varactor, causing its capacitance to vary at the signal

[33] Edwin H. Colpitts, US patent 1624537, filed in 1918, but not granted until 1927.

[34] It is an *oscillator*; but to get voltage control you've got to use a tuning varactor, as explained below.

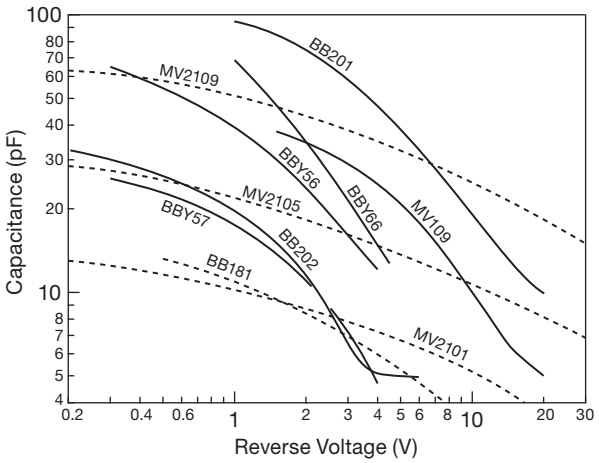


Figure 7.27. A reverse-biased diode exhibits capacitance that varies with applied voltage, shown here for several typical "varactor" tuning diodes. Those with steeper solid curves have "hyperabrupt" diode junctions.

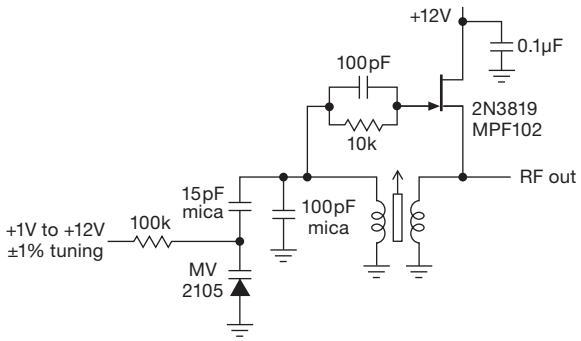


Figure 7.28. Varactor-tuned *LC* oscillator.

frequency. This produces oscillator waveform distortion, and, more important, it causes the oscillator frequency to depend somewhat on the amplitude of oscillation. To minimize these effects, you should limit the amplitude of the oscillation (amplify in following stages, if you need more output); also, it's best to keep the dc varactor bias voltage above a volt or so, in order to make the oscillating voltage small by comparison.

An additional technique that helps mitigate this signal-biasing effect is to use a series pair of back-to-back varactors, so that the oscillating voltage seen by the two varactors acts to change their capacitances in opposite directions. This is illustrated in the low-noise oscillator of Figure 7.29 (see also Figure 6.8), used within a phase-locked loop (§13.13) to produce a clean 60 MHz "local oscillator" for a radio-astronomy receiver. This particular configuration is known as a Clapp oscillator, for which the frequency is ordinarily set by the series resonant L_1C_1. Here we've added a parallel capacitance across C_1, consisting of the series pair of varactors. The tuning voltage is applied via R_2, which puts them both in equal back-bias (relative to their anodes at zero volts). The varactors, of equal capacitance, each take half the oscillation voltage, producing capacitance changes of opposite sign and (if the signal is not too large) of approximately equal magnitude. The net effect is a greatly reduced change in the capacitance of the series pair, and therefore lower distortion and frequency pulling. The measured signal purity was approximately 10 dB better than a good commercial frequency synthesizer (HP 3325A). A frequency spectrum of its output is shown in Figure 7.30, where its signal purity is compared with that of an MC1648 emitter-coupled oscillator using similar *LC* components and operating at approximately the same frequency.[35]

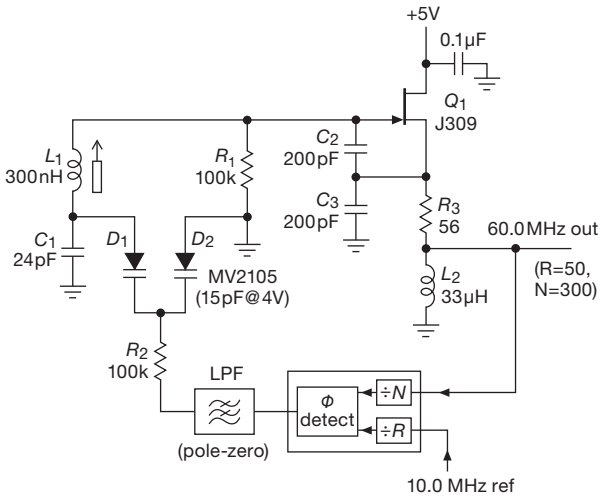


Figure 7.29. Low-noise JFET *LC* oscillator, used within a phase-locked loop (see §13.13). This design has unusually low sideband noise, as shown in the spectra of Figure 7.30.

Electrically tunable oscillators are used extensively to generate frequency modulation, as well as in RF phase-locked loops like this one. PLLs are treated in Chapter 13.

For historical reasons we should mention a close cousin of the *LC* oscillator, namely the tuning-fork oscillator. It used the high-Q oscillations of a metallic tuning fork as the frequency-determining element of an oscillator, and it found use in low-frequency standards (stability of a few parts per million, if run in a constant-temperature oven) as well as in wristwatches. These objects have been super-

[35] To preserve the oscillator's free-running spectral purity when embedded in the PLL, we included an *LC* trap to suppress spurs at the phase-detector's 200 kHz reference frequency, see Figure 6.8.

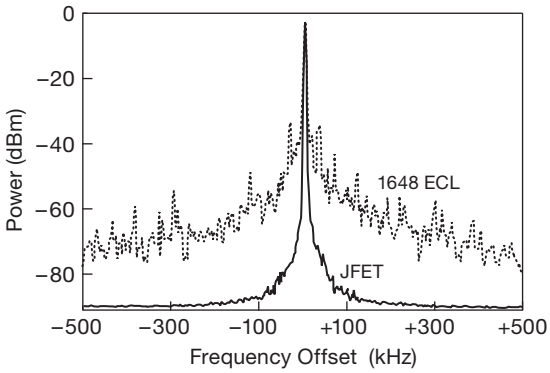


Figure 7.30. Frequency spectrum of the 60 MHz JFET *LC* oscillator of Figure 7.29, compared with that of a bipolar ECL oscillator (MC1648). Vertical: 10 dB/div. Horizontal: 200 kHz/div.

seded by quartz ("crystal") oscillators, which are discussed in §7.1.6. Interestingly, however, quartz crystals made for *low-frequency* operation (e.g., the 32.768 kHz used in wristwatches) oscillate in a mechanical tuning-fork mode.

E. Parasitic oscillations

Suppose you have just made a nice amplifier and are testing it out with a sinewave input. You switch the input function generator to a square wave, but the output remains a sinewave! You don't have an amplifier; you've got trouble.

Parasitic oscillations aren't ordinarily as blatant as this. They are usually observed as fuzziness on part of a waveform, erratic current-source operation, unexplained op-amp offsets, or circuits that behave normally with the oscilloscope probe applied, but go wild when the 'scope isn't looking. These are bizarre manifestations of untamed high-frequency parasitic oscillations caused by unintended Hartley or Colpitts oscillators exploiting lead inductances and interelectrode capacitances.

The circuit in Figure 7.31 shows an oscillating current source, born in an electronics lab course where an analog volt-ohm-milliammeter (VOM) was used to measure the output compliance of a standard transistor current source. The current seemed to vary excessively (5% to 10%) with load voltage variations within its expected compliance range, a symptom that could be "cured" by sticking a finger on the collector lead! The combined meter capacitance and the collector–base capacitance of the transistor resonated with the meter inductance in a classic Hartley oscillator circuit, with feedback provided by collector–emitter capacitance. Adding a small resistor in series with the base suppressed the oscillation by reducing the high-frequency common-base gain. This is one trick that often helps.

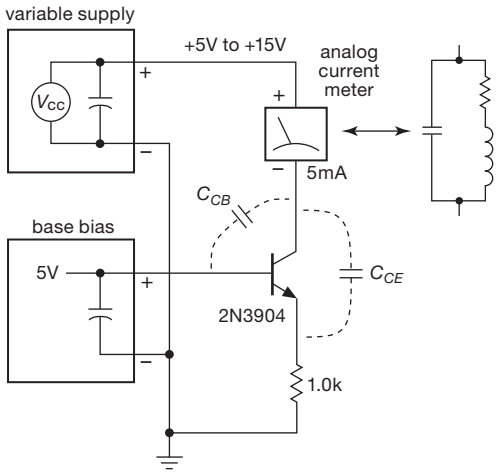


Figure 7.31. Unintended Hartley oscillator causes current-source mischief.

There are opportunities for parasitic oscillations in any active circuit that has gain. You've just got to be alert, and pay attention to any unexpected or weird circuit behavior. Sometimes you'll see the telltale signs of fuzz on only a portion of the waveform. With experience you'll come to recognize the symptoms of oscillations in op-amps (generally in the neighborhood of f_T, say, a megahertz) or in discrete small-signal transistors (generally in the 10s to 100s of megahertz).

"Pickup"

It's easy[36] to mistake various forms of signal *pickup* for an oscillation, because they too can cause fuzz on the displayed signal. If you suspect pickup, check to see if you've got 60 Hz[37] (or perhaps 120 Hz), a clear indication of powerline coupling. This can originate in capacitive coupling to a high-impedance point in the circuit. Or it can come from inductive coupling into a portion of your circuit that has some geometrical area linked by the alternating magnetic field. A third possibility is through a *ground loop* (portions of the circuit referenced to local grounds that are not at the same potential). Even in a well-layed-out circuit this latter problem can be severe, for example when it is connected to some external instrument that is plugged into a different ac power outlet. Higher-frequency pickup is common, too: coupling from switching power supplies (treated in Chapter 9), generally in the range of 20 kHz to 1 MHz; or

[36] Easier than you might think, because of (among other things) confusion caused by "aliasing" in digital oscilloscopes.

[37] 50 Hz if you live almost anywhere other than the Americas (or one of the other globally-scattered 60 Hz enclaves).

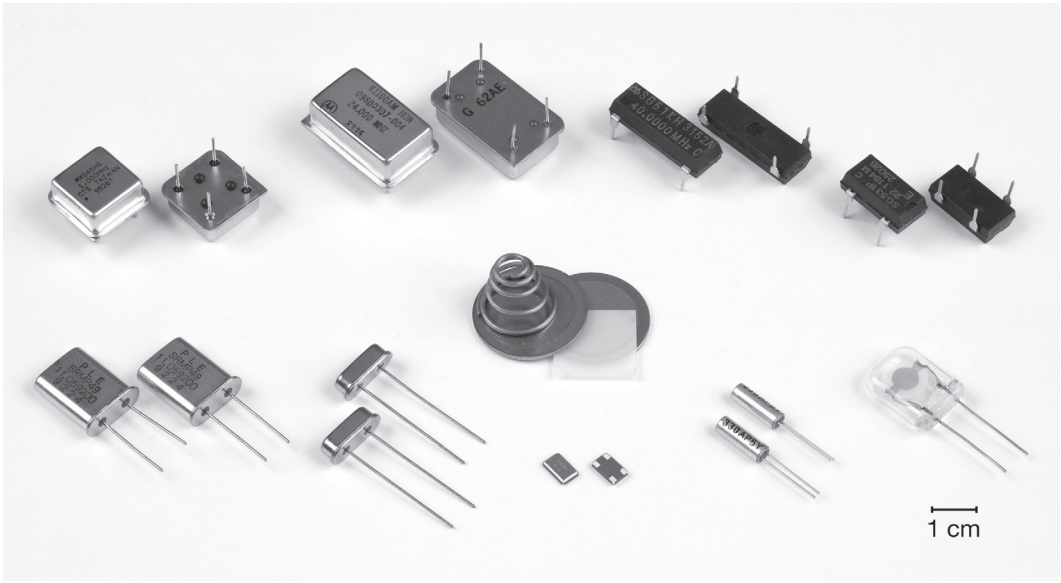


Figure 7.32. Quartz crystal packages. Across the top row are complete oscillator modules in DIP-8 and DIP-14 sizes; a much smaller alternative is the tiny 7mm×5mm surface-mount oscillator module at bottom center. The strange object in the middle is a bare crystal, shown with its spring-loaded electrode plates. You don't see those anymore; instead, crystals come in the popular sealed packages known as (left to right, bottom row) HC49/U, HC49/US, and 3mm tubular. We were lucky to find that oddball glass case at the right: inside you can see the gorgeous quartz disk with its plated electrodes.

modulated RF pickup from broadcast stations (US allocations are 0.5 MHz–1.7 MHz for AM, 88 MHz–108 MHz for FM, and anywhere from 55 MHz–700 MHz for television).

7.1.6 Quartz-crystal oscillators

RC (or capacitor plus current source) relaxation oscillators can easily attain stabilities approaching 0.1%, with initial predictability of 5% to 10%. That's good enough for many applications, such as a vacuum fluorescent display (VFD) in which the individual characters of a multicharacter display are driven sequentially in rapid succession (a 100 Hz overall rate is typical); this is called a *multiplexed display* (see the example circuit in §10.6.2). Only one character is lit at any time, but, if the entire display is "refreshed" rapidly enough, your eye sees the whole display without obvious flicker. In such an application the precise rate is quite irrelevant – you just want something in the ballpark. As stable sources of frequency, *LC* oscillators can do a bit better, with stabilities of 0.01% over reasonable periods of time. That's good enough for nondemanding applications like an inexpensive radio. Both kinds of oscillators are easily tunable – with a variable *R* or current source (for the relaxation oscillator) and with a mechanically or electrically tunable capacitor, or a *slug-tuned* inductor (for the *LC* oscillator).

But for real stability there's no substitute for a crystal oscillator. This uses a piece of quartz (silicon dioxide, the primary ingredient of glass) that is cut and polished to vibrate mechanically at a certain frequency. Quartz is *piezoelectric* (a strain generates a voltage, and vice versa), so acoustic waves in the crystal can be driven by an applied electric field and in turn can generate a voltage at the surface of the crystal. By plating some contacts on the surface, you wind up with an honest electrical circuit element that can be modeled by a sharply resonant *RLC* circuit, pretuned to some frequency (which is the mechanical resonant frequency of the little slab of single-crystal quartz). Quartz crystals come packaged as bare components or as complete oscillator modules; some examples are shown in Figure 7.32.

The quartz crystal's high Q (typically around 10^4–10^5) and good stability make it a natural for oscillator control as well as for high-performance filters. As with *LC* oscillators, the crystal's equivalent circuit provides positive feedback and gain at the resonant frequency, leading to sustained oscillations.

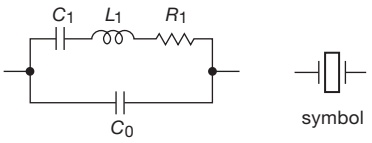


Figure 7.33. Quartz-crystal equivalent circuit. C_0 is the actual electrode and lead capacitance, whereas the series *RLC* models the electrically coupled *mechanical* resonance. Typical values for a 1 MHz crystal might be C_0=4 pF, and, for the series *RLC*, 1 H, 0.02 pF, and 75 Ω ($Q \sim 10^5$).

A. Series and parallel modes

The crystal's resonant behavior, as modeled by its equivalent circuit, deserves some additional explanation. The equivalent circuit contains two capacitors, giving a pair of closely spaced (within 0.1%) series- and parallel-resonant frequencies (Figure 7.33). The effect is to produce a rapidly changing reactance with frequency (Figure 7.34).

The label "resonant frequency" f_R is given to the series-resonant frequency of L_1 and C_1 (it's also called the series-resonant frequency f_S), so the net series reactance goes from being capacitive (below f_R) to inductive (above f_R). At f_R the net reactance of the series pair (L_1 and C_1) is zero, and the magnitude of the impedance is a minimum (and equals R_1).[38] Slightly above this in frequency (typically $\sim$0.1% higher) is the "antiresonant frequency" f_a, where the series combination of C_0 and C_1 (which is slightly less than C_1 alone) resonates with L_1. (Alternatively you can think of this as the parallel resonance of C_0 with the net series reactance of L_1 and C_1, which becomes increasingly inductive above f_R.) This is also called the parallel resonant frequency f_p, although that term should properly be reserved for the actual circuit situation in which an external load capacitance C_L is intentionally added in parallel (more on this soon). At this frequency (f_a or f_p) the net reactance again goes through zero, but this time with a peak in the magnitude of the impedance. When a crystal is operated in parallel resonance, the additional parallel capacitance added by the external circuit adds to the crystal's C_0 and lowers the resonant frequency somewhat. Crystals intended for operation in parallel resonant mode will specify a value of external shunt capacitance (typically in the range of 10–35 pF) for oscillation at the specified frequency stamped on the crystal.

This business of parallel and series is important, and any device that uses an external crystal will specify which mode it's using, along with some guidance on crystal parameters (maximum allowable R_S, value of parallel capacitance). Better yet, you will see a listing of crystal manufacturers and part numbers that are known to work properly.

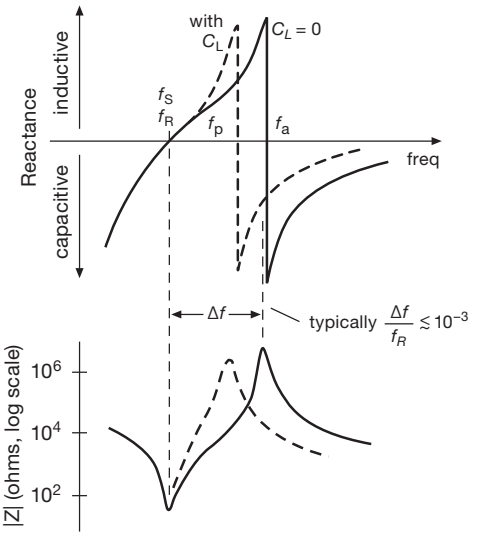


Figure 7.34. Reactance and magnitude of the impedance of a quartz crystal near its resonances, on a greatly expanded frequency scale. f_S and f_a are the series- and parallel-resonant frequencies [more precisely, the *resonant* (f_R) and *antiresonant* (f_a) frequencies, respectively]. An additional external capacitance C_L lowers the parallel-resonant frequency to f_p, when operated in that mode.

B. Exploring a quartz crystal

You can find plenty of sketches like Figure 7.34. But do you believe they accurately represent what real crystals do?

We weren't sure, so to find out we took a sample crystal (a CTS type MP100, specified as 10.0 MHz±45ppm, in series-resonant mode), and measured its impedance with a high-resolution vector impedance test instrument (an HP4192A). The latter can measure from 0.01 Ω to 200 kΩ, with a resolution of 1 Hz, and at frequencies to 13 MHz – perfect for this job. Our particular crystal had a measured series-resonant frequency of f_S=10.000086 MHz (that's a frequency error of +8.6ppm), an impedance (resistive) at resonance of R_1=4.736 Ω, and a parallel capacitance of C_0=5.5 pF. We know the product L_1C_1 (from the resonant frequency), but not their individual values.[39] But we can get at them indirectly, by measuring the (unspecified) parallel-(anti)resonant frequency, and its variation

[38] Recall that the impedance of an ideal parallel *LC* goes to infinity at resonance, whereas that of a series *LC* goes to zero; §1.7.14.

[39] Recall there's no actual inductor or capacitor in there. These represent the electrical equivalent of the sharply resonant mechanical crystal, as piezoelectrically coupled by the attached electrodes. They're sometimes called a "motional" inductor and capacitor.

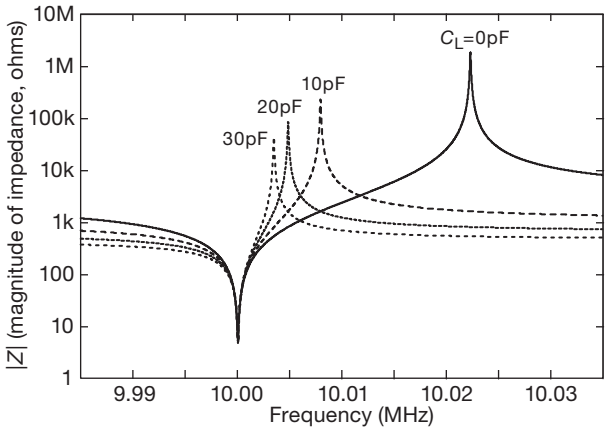


Figure 7.35. Impedance versus frequency for a sample 10.0 MHz series-resonant crystal, as modeled by SPICE from measured values of its *RLC* electrical model. Curves for four values of external parallel capacitance C_L are plotted.

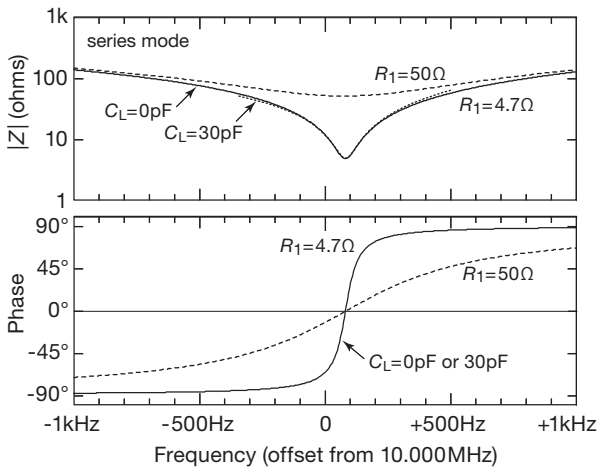


Figure 7.36. Impedance and phase in the neighborhood of the series resonance, for the 10.0 MHz crystal of Figure 7.35. Note that the graphs of impedance and phase are unaffected by external capacitance. The high-Q resonance of our sample, with its R_1=4.7 Ω, is considerably degraded for a crystal with worst-case specified R_1=50 Ω.

with added external parallel capacitance C_L. These we measured as f_a=10.02245 MHz (no C_L: C_0 only), and f_p=10.00355 MHz (with C_L=30 pF external parallel capacitance).

From these we can back out the values of L_1 and C_1, namely 10.3324 mH and 0.024515 pF.[40] And with those values we can enjoy many happy hours running the SPICE simulator, to learn how graphs of the crystal's impedance, phase shifts, and Q-value really look. Figures 7.35, 7.36, and 7.37 show such results.

These show the expected minimum at the series resonant frequency f_R (where $|Z|$=4.7 Ω), and that it varies very little with external capacitance (not visible, in fact, even in the expanded plot of Figure 7.36; that is, much less than 1 ppm going from C_L=0 pF to 30 pF). In contrast, the *parallel*-resonant frequency (impedance maxima) depends relatively strongly on external capacitance, which effectively "pulls" its resonance downward by ∼2000ppm when 30 pF is added.

The fact that the parallel resonance is higher than the 10.0 MHz stamped on the case doesn't mean there's anything "wrong" with this crystal. Its frequency is simply specified for series-resonant circuit operation. If it were instead specified for parallel-resonant operation, this particular sample would be stamped "10.00355 MHz," and would specify "C_L=30 pF."[41] (Of course, you'd buy instead a 10.0 MHz standard frequency crystal specified for parallel mode; for this manufacturer that would have the part number MP101.)

From the relatively large frequency shifts with load capacitance, it's clear that you have to be careful to use the specified load capacitance (taking into account wiring and chip capacitances) when operating a crystal in parallel-resonant mode. Looking at the positive side, that means you can use an external variable capacitor to trim the operating frequency (or lock it, over a narrow range, using an electrically tunable varactor capacitor). On the negative side, it means that even small drifts in the external circuit's capacitance will cause frequency shifts. For example, to achieve 0.1 ppm frequency stability (assuming the crystal is that good over temperature or time) the external capacitance must not change by more than 0.002 pF; this is likely a difficult constraint for the external amplifier that closes the oscillation loop.

Our crystal measured an impressively low value of R_1=4.7Ω, compared with the manufacturer's worst-case maximum of 50 Ω. To see how that changes things, we included that worst-case value in the expanded plots of the series and parallel resonances (Figures 7.36 and 7.37). For the series resonance the impedance minimum is shallower, and the phase shift versus frequency is more gentle. The

[40] Formulas, for those who want them: $C_1 = 2(1 - f_B/f_A)/(1/C_A - 1/C_B)$, then $L_1 = 1/C_1(2\pi f_S)^2$.

[41] More formulas: the capacitive loading produces a parallel resonant frequency $f_P = f_S\{1 + C_1/2(C_0 + C_L)\}$. And, knowing the crystal's parameters, the load capacitance required for getting a parallel-resonant frequency f_P is $C_L = \{f_S C_1/2(f_P - f_S)\} - C_0$.

shallower phase change ($\sim$1.3°/ppm, versus $\sim$13°/ppm) means that the external oscillator circuit must keep changes in its phase shift an order of magnitude smaller to maintain the same stability (here 0.13° versus 1.3°, for stability of 0.1ppm). In an oscillator circuit such a crystal would be less stable against other variations in circuit parameters as well (amplifier input impedance, gain, etc.), and in fact it may refuse to oscillate entirely. Worse yet, the circuit may oscillate at an unrelated frequency, an unhappy situation that we've encountered more than once.

Finally, the expanded plot around the parallel resonance, for a single value of load capacitance (C_L=30 pF; the others are off-scale), shows the same sort of Q degradation with maximum (worst-case) specified R_1. It is interesting to note that the sharpness of the phase change (and width of the impedance maximum) is similar to that of the series-resonant case, contrary to statements that you'll sometimes hear.

To complete the oscillator circuit, the crystal is connected within a positive feedback loop. Some common oscillator circuits are shown in §7.1.6D, where you'll see external loading capacitances, a large-value resistor to complete the bias path, and (sometimes) a smaller series resistor. You can simulate the full circuit configuration, if you like, by using the crystal model above.[42] We've done so, but we will show restraint and conclude this discussion, declaring victory in revealing the nature of those often misunderstood crystal resonances.

C. Standard crystal frequencies

Quartz crystals are available from about 10 kHz to about 30 MHz, with overtone-mode crystals going to about 250 MHz. Although crystals have to be ordered for a given frequency, most of the commonly used frequencies are available off the shelf. Frequencies such as 100 kHz, 1.0 MHz, 2.0 MHz, 4.0 MHz, 5.0 MHz, and 10.0 MHz are always easy to get. A 3.579545 MHz crystal was used in analog TV color-burst oscillators. Digital wristwatches use 32.768 kHz (divided by 2^{15} to get a useful 1 Hz), and other powers of 2 are also common. A crystal oscillator can be adjusted slightly by varying a series or parallel capacitor, for example one of the 27 pF or 32 pF capacitors in Figures 7.38D and E. Given the low cost of crystals (much less than a dollar), it is worth considering a crystal oscillator in any application where you would have to strain the capabilities of *RC* relaxation oscillators.[43]

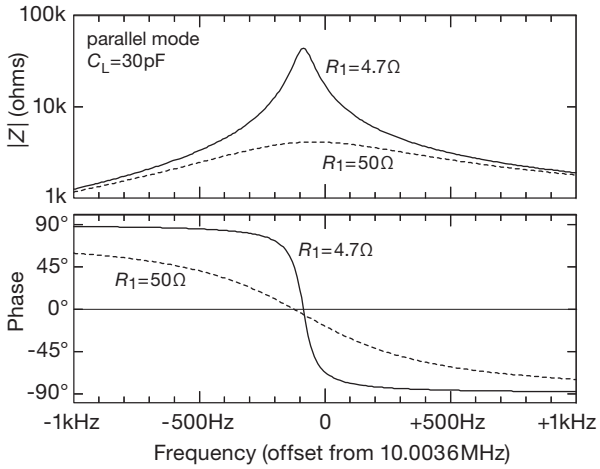


Figure 7.37. Impedance and phase in the neighborhood of the parallel resonance for the 10.0 MHz crystal of Figure 7.35. Here only one value of load capacitance (C_L=30 pF) is shown, because the other values are completely off-scale. Once again, the high-Q resonance of our sample is considerably degraded for a crystal with worst-case specified R_1=50 Ω.

Although crystal oscillators are not tunable, in the manner of relaxation oscillators or *LC*-based oscillators, you can use a varactor to vary some added external capacitance, thereby "pulling" the natural frequency of a parallel-mode quartz-crystal oscillator. The resulting circuit is called a "VCXO" (voltage-controlled crystal oscillator), and it augments the good-to-excellent stability of crystal oscillators with a small degree of tunability. The best approach is probably to buy a commercial VCXO, rather than attempt to design your own. Typically they produce maximum deviations of $\pm$10 ppm to $\pm$100 ppm from center frequency, though wide-deviation units (up to $\pm$1000 ppm) are also available.

An alternative, and a popular one at that, is to synthesize (with either a phase-locked loop, §13.13, or by direct digital synthesis, §7.1.8) any desired output frequency, using a crystal oscillator's fixed-frequency output as a "reference." The synthesized frequency can be changed easily under digital control, and it is as stable as the crystal oscillator itself. As a consequence, most contemporary communications equipment (radios, televisions, transmitters, cellphones, etc.) uses DDS or PLL synthesis to generate the needed internal frequencies.

[42] Readers unfamiliar with SPICE may benefit from Appendix J.

[43] A recent challenger in the low-cost resonator-based oscillator category is MEMS (microelectromechanical systems), in which the resonator is made as a tiny etched silicon structure. Although these oscillators are not as stable as quartz, they can be *very* small (SiTime makes a series that measures 2 mm×2.5 mm×0.8 mm), and they can incorporate temperature compensation and frequency synthesis circuitry naturally within the same silicon technology.

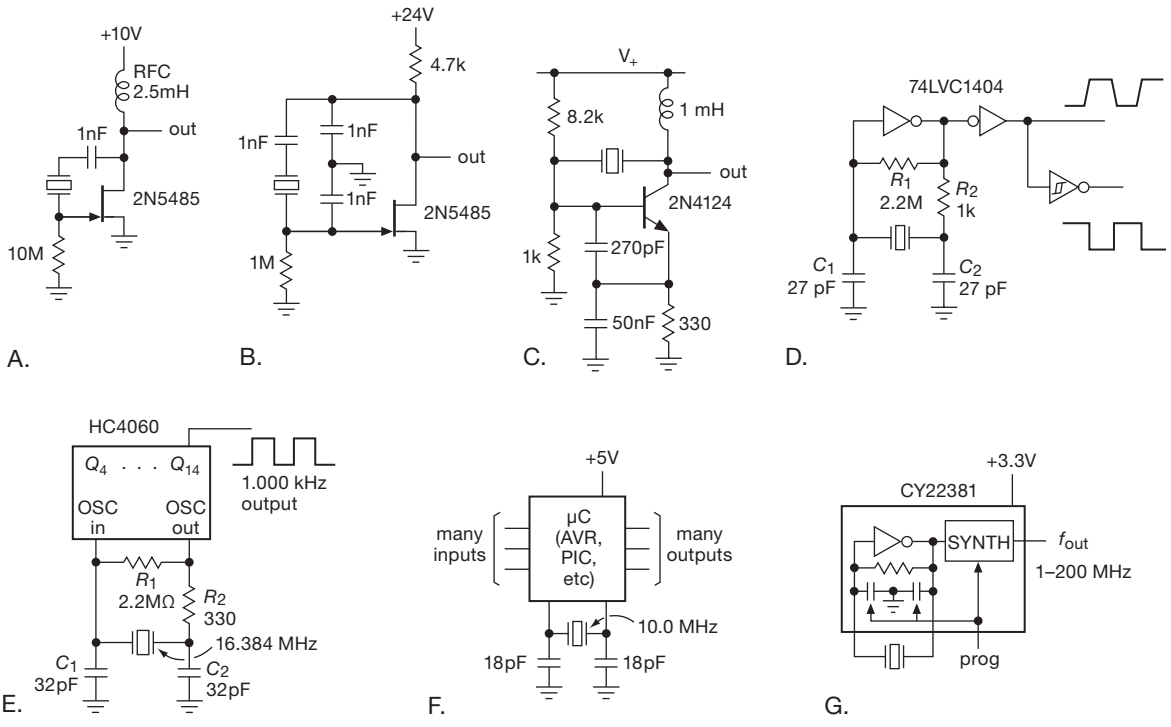


Figure 7.38. Various crystal oscillators. Circuits D–G exploit portions of digital logic circuits: an inverter, a 14-stage binary counter, a microcontroller, and a frequency synthesizer, respectively.

D. Crystal oscillator circuits

Figure 7.38 shows some crystal oscillator circuits. In Circuit A the classic Pierce oscillator implemented with the versatile FET (see Chapter 3). The Colpitts oscillator, with a crystal instead of an *LC*, is shown in circuit B. An *npn* bipolar transistor with the crystal as feedback element is used for circuit C. The remaining circuits generate logic-level outputs by using digital logic functions (circuits D through G). It's common to see an unbuffered logic inverter (i.e., a single CMOS transistor pair, as in Figure 3.90) used as a crystal oscillator (Figure 7.38D); in that application the inverter is biased into the linear region with a high-value feedback resistor, with the crystal providing (parallel-mode) resonant feedback. The LVC1404 is designed particularly for this application, with its pair of unbuffered inverters plus an optional Schmitt-trigger inverter (for generating abrupt output transitions); for low voltages (down to 0.8 V) the 'AUP1GU04[44] works well. In this circuit (and in Figure 7.38E) the series resistor R_2 should be chosen comparable to the reactance of C_2 at the oscillator frequency.[45]

Let us pause for a moment to ask how these latter parallel-mode circuits can oscillate, given that the crystal has a 0° phase shift at its resonance (whether series or parallel mode; see Figures 7.36 and 7.37). What happens is that the load capacitance C_L is actually wired as a pair of capacitors (C_1 and C_2) in series, with the midpoint grounded. So when there's an oscillating voltage across the crystal, the two ends are seesawing 180° out of phase with each other; it's just like a center-tapped transformer winding. The inverting amplifier completes the 360° phase shift needed for a sustained oscillation.

Turning back to the remaining circuits in Figure 7.38, it's quite common to see a pair of "XTAL" terminals on more complex digital ICs (microprocessors, waveform synthesizers, serial communications chips, etc.), an invitation to use the chip's internal oscillator circuit (usually a prebiased unbuffered inverter). In Figures 7.38E–G we've shown three such examples – a 14-stage binary frequency

[44] Our practice is to strip off unimportant prefixes (hence the apostrophe) when talking about standard digital logic, as we'll explain in Chapter 10.

[45] For example, ∼330 kΩ for a 32 kHz oscillator, and ∼1 kΩ for a 5 MHz oscillator.



Figure 7.39. The coveted *Art of Electronics* logo wristwatch runs 3 years on a 1.5 V, 28 mAh cell – that's just 1 *micro*amp!

divider generating a precise 1.000 kHz square-wave output, a microcontroller whose serial port timing is set by the external 10.0 MHz crystal, and a frequency synthesizer chip that generates accurate frequencies needed for applications such as multimedia, communications, and data conversion (through a programmable PLL, f_{out} from 1 MHz to 200 MHz).

Figure 7.39 shows an interesting niche application for crystal oscillators, the "quartz wristwatch." You need quartz stability here (there are 86,400 seconds in a day, so stability of "only" 1 part in 10^4 would cause a drift of a minute per week); and, you need *really* low power. These inexpensive mass-produced products run their oscillators, frequency divider electronics, and drive power for a tiny stepping motor on a power budget of about a microwatt.

A micropower design

Challenged by the astonishingly low power of highly customized quartz-oscillator wristwatch circuits, we looked further into what can be done using only standard components. We chose the low-voltage 74AUP logic family (specified for 0.8–3.3 V operation), and we tested the standard parallel-resonant Pierce configuration, using an unbuffered inverter for the oscillator followed by a Schmitt inverter second stage to generate a clean switching waveform (Figure 7.40A).

The measured total supply current versus supply voltage is plotted in Figure 7.41 for both a 32.768 kHz (wristwatch) crystal and a 2.5 MHz crystal; these curves (marked "R_3=R_4=0") show a rapid increase in supply current with increasing supply voltage, caused by the "class-A" oscillator current (the overlapping conduction of the nMOS and pMOS inverter pair for mid-supply input voltages; see Figure 10.101) during the transitions of the input waveform.

A nice trick to reduce greatly this effect is to add a pair

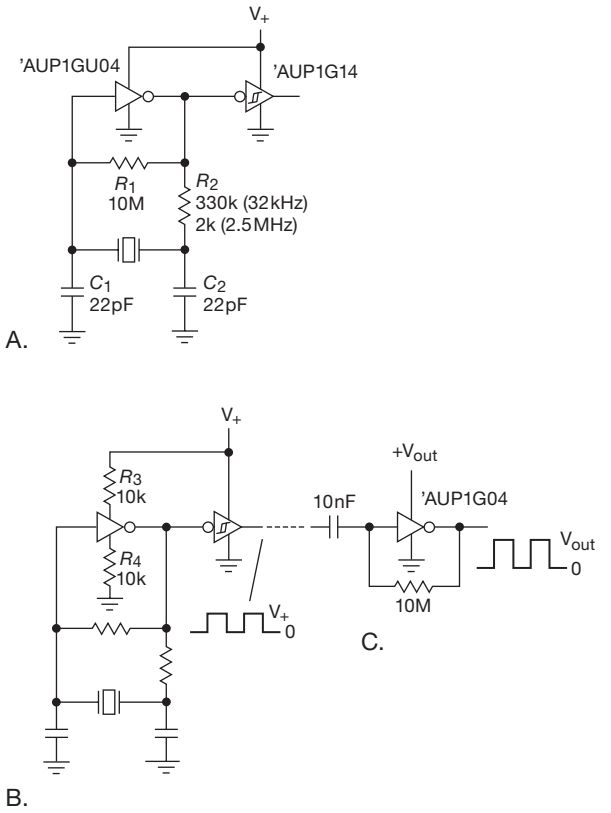


Figure 7.40. Micropower crystal oscillator. A. Unbuffered inverter oscillator with Schmitt second stage. B. Reducing linear "shoot-through" current. C. Low-voltage oscillator drives full-voltage output inverter.

of resistors in the supply leads (Figure 7.40B); this produced the supply-current curve marked "R_3=R_4=10k," a factor of 20–50 reduction for the 32.768 kHz oscillator. So we have a sub-microamp 32 kHz oscillator – but only for output voltages of less than a volt, which is lower than just about any logic device you might want the oscillator to drive. The final trick is to use that low-voltage oscillator to drive a full-voltage output stage (Figure 7.40C), with a blocking capacitor and large feedback resistor to bias the output stage in its linear region. With the oscillator and second stage running at 1.0 V, feeding a biased output inverter third stage running at 1.8 V, we measured 2.4 μA total supply current, compared with 12.8 μA for the 2-stage oscillator alone (Figure 7.40B) running at 1.8 V; this is a five-fold improvement.[46]

These modest experiments lead to the obvious question:

[46] The analogous figure for a 2.5 V output stage was 13.8 μA, a factor of 2.4 better than the two-stage circuit alone.

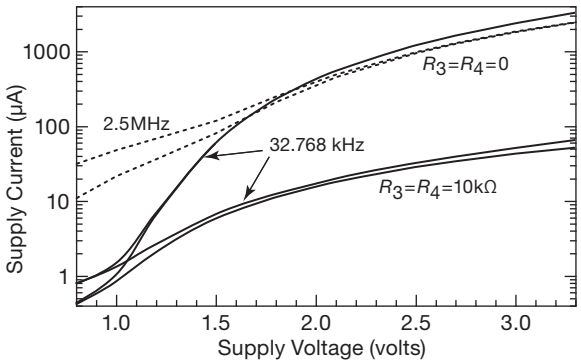


Figure 7.41. Measured power-supply current for the oscillators of Figure 7.40. Each curve pair plots first-stage (lower) and two-stage (upper) supply current.

how do the watchmakers do it? If you poke around a bit, you can find some very interesting datasheets. For example, a company called "EM Microelectronic" offers a tiny IC, the EM7604, which it describes as a "Low Power Crystal Oscillator Circuit 32.768 kHz." And they mean *low*: this puppy can run from 1.2 V to 5.5 V, and has a typical operating current of 0.3 μA or less at supply voltages between 3 V and 5 V. Perhaps a revealing hint as to their customer base is provided by the notation "Swatch Group Electronic Systems."

E. A caution

The proper design of crystal oscillators is not at all trivial – it is essential to ensure that the product of circuit gain A and crystal loss B (i.e., the loop gain AB) is greater than unity, and that the overall phase shift going around the loop is an integral multiple of 360°, at the desired frequency of oscillation.[47] The crystal loss (caused by R_S in the equivalent circuit) may prevent proper oscillation, with the shunt capacitance C_P alone providing an adequate path for oscillation at a frequency unrelated to that stamped on the crystal. You must be careful, also, to select a crystal intended for operation at its series or parallel resonance, as required by the external oscillator circuit.[48] Oscillator chips and other ICs that accept an external crystal for their internal clocking (such as microcontrollers, see for example Figure 15.4) will state it clearly on the datasheet; here's an example, from the datasheet for the MPC9230 PLL synthesizer:

> . . . As the oscillator is somewhat sensitive to loading on its inputs, the user is advised to mount the crystal as close to the MCP9230 as possible to avoid any board level parasitics. . . . Because the series resonant design is affected by capacitive loading on the XTAL terminals, loading variation introduced by crystals from different vendors could be a potential issue. For crystals with a higher shunt capacitance, it may be required to place a resistance across the terminals to suppress the third harmonic. . . . The oscillator circuit is a series resonant circuit and thus for optimum performance a series resonant crystal should be used. Unfortunately, most crystals are characterized in a parallel resonant mode.

And here's an example from the 174-page (!) datasheet for the PIC16F7x microcontroller, for which you connect the crystal between two pins, and a capacitor from each to ground:

> The PIC16F7x oscillator design requires the use of a parallel cut crystal.
>
> ⋮
>
> **Capacitor selection for crystal oscillator (for design guidance only):**
> These capacitors were tested with the crystals listed below for basic start-up and operation. These values were not optimized. Different capacitor values may be required to produce acceptable oscillator operation. The user should test the performance of the oscillator over the expected V_{DD} and temperature range for the application. See the notes following this table for additional information.
> [*a table of selected manufacturers' crystals follows, then more warnings.*]

These are not idle warnings (though the severity of their disclaimers may have been encouraged by the company's legal department). In various situations we've found that one company's crystals work fine, and another's, with similar specifications, do not. This is most likely due to poorly specified properties such as effective series resistance and mount capacitance. Our experience with discrete crystal-oscillator circuits has been, well, checkered.

F. Crystal oscillator modules

For reasons like these we favor the use of complete oscillator *modules*, to get bulletproof reliability. These cost more than bare crystals,[49] but they include oscillator circuitry

[47] These are the so-called *Barkhausen criteria* for oscillation.

[48] See, for example, RCA App. Note ICAN-6539.

[49] Typically $1.50 versus $0.30, in 100-piece quantities; twice that in single-piece quantities.

guaranteed to work, and they provide logic-level square-wave output. You can use them as the oscillator for any IC to which you could attach a bare crystal, because all such ICs will also accept a clocking square-wave input.

Oscillator modules come in IC-style packages, such as standard DIP and smaller 4-pin surface-mountable styles. They come off-the-shelf in lots of standard frequencies (e.g., 1, 2, 4, 5, 6, 8, 10, 16, and 20 MHz, up to 100 MHz or so), as well as some weird frequencies used in microprocessor systems (e.g., 14.31818 MHz, used for video boards; or 22.118 MHz, a favorite for 8051-type microcontrollers because of serial port timing). These "crystal clock modules" typically provide accuracies (over temperature, power-supply voltage, and time) of a modest 0.01% (100 ppm), but you get that performance inexpensively, reliably, and you don't have to wire up any circuitry.

If you need a non-standard frequency, you can get "programmable oscillator modules" that let you select the frequency, typically in the range of 1 MHz to 125 MHz or more; these cost roughly double that of standard modules (about $5 each in small quantities), and are "one-time" programmable (you select the frequency when you order; or you can buy a programmer for ∼$500 and program the "blank" oscillator modules in the field). These use PLL techniques to synthesize the desired output frequency from the standard-frequency internal oscillator (§13.13).[50]

Some manufacturers of crystal (and ceramic) oscillator modules are Cardinal Components, Citizen, Connor Winfield, Crystek, CTS, Ecliptek, ECS, Epson, Fox, Seiko, and Vishay.

G. Ceramic resonators

Before going on to discuss oscillators of *higher* stability (§7.1.7), we must mention *ceramic resonators*. Like quartz crystals, these are piezoelectric mechanical resonators, with electrical properties similar to those of quartz. They come in a limited selection of frequencies from about 200 kHz to 50 MHz. However, they are less precise (typically ±0.3%), with correspondingly poorer stability (typically 0.2%–1% over temperature and time). The good news is that they are small, dirt cheap ($0.15–$0.25 in small quantities), available with built-in capacitors (for about $0.25–$0.50), generally interchangeable in any quartz crystal oscillator circuit, and can be "pulled" over a frequency range of a few parts per thousand (owing to their lower Q value). They occupy a useful niche between *LC* resonators and quartz crystals. They are available from companies like Abracon, AVX, ECS, Murata, Panasonic, and TDK.

H. SAW-based oscillators

If you need a stable oscillator at frequencies higher than are supported by crystal (or ceramic) resonators, there is the technology of surface acoustic waves (SAWs), used both for filters and for oscillators. You can get SAW oscillator modules in the frequency range from 100 MHz to 1 GHz. They are tiny (just like crystal-oscillator modules) and they have comparable stabilities (50 ppm over temperature). The down side is that only a sparse set of standard frequencies is available, and these things tend to be somewhat pricey (upward of $50, in small quantities).

On the other hand, if you want to roll your own, you can get bare SAW *resonators* in the frequencies that are popular for garage-door openers, key fobs, and the like (433 MHz is widely used for these things) for about $1; add an inexpensive bipolar transistor and a few passive components and you've got an oscillator. Attach a short piece of wire, and you've got a transmitter!

7.1.7 Higher stability: TCXO, OCXO, and beyond

Without great care you can obtain frequency stabilities of a few parts per million over normal temperature ranges with crystal oscillators. By using temperature-compensation schemes you can make a TCXO (temperature-compensated crystal oscillator) with somewhat better performance. Both TCXOs and uncompensated oscillators are available as complete modules from many manufacturers, e.g., Bliley, Cardinal Components, CTS Knights, Motorola, Reeves Hoffman, Statek, and Vectron. They come in various sizes, ranging from modules down to SMT and DIP packages. TCXOs deliver stabilities of 1 ppm over the range 0°C to 50°C (inexpensive) down to 0.1 ppm over the same range (expensive).

A. Temperature-stabilized oscillators

For the utmost in stability, you may need a crystal oscillator in a constant-temperature oven ("OCXO"). A crystal with a zero temperature coefficient at some elevated temperature (80°C to 90°C) is used, with the thermostat set to maintain that temperature. Such oscillators are available as small modules for inclusion into an instrument or as complete frequency standards ready for rack mounting. The 1000B from Symmetricom is typical of high-performance

[50] Some manufacturers provide modules that offer a pin-selectable choice of several frequencies. An example is the ECS-300C series from ECS, which come in 8-pin packages, with 3 pins setting the binary division ratio (from 1/2 to 1/256 of the base frequency).

modular oscillators, delivering 10 MHz with stabilities of a few parts in 10^{11} over periods of seconds to hours.

When thermal instabilities have been reduced to this level, the dominant effects become crystal "aging" (the frequency tends to increase continuously with time), power-supply variations, and environmental influences such as shock and vibration (the latter are the most serious problems in quartz wristwatch design). To give an idea of the aging problem, the 1000B oscillator has a specified aging rate (after a month of break-in) of 1 part in 10^{10} per day, maximum. Aging effects are due in part to the gradual relief of strains, and they tend to settle down after a few months, particularly in a well-manufactured crystal.[51]

Oven-controlled crystal oscillators can be miniaturized, if need be, for portable applications that demand excellent oscillator stability. Valpey Fisher, for example, puts a resistive plating directly on the quartz crystal, creating a small OCXO ($\sim$1.3 cm^3) requiring just 0.15 W of heater power.

B. Atomic standards

Atomic frequency standards are used where the stability of ovenized-crystal standards is insufficient. These use a microwave absorption line in a rubidium (Rb) gas cell, or atomic transitions in an atomic cesium (Cs) beam, as the reference to which a quartz crystal is stabilized. Commercially available Rb and Cs frequency standards achieve accuracy and stability of a few parts in 10^{11} and 10^{13}, respectively. Cesium-beam standards are the official timekeepers in the United States, with timing transmissions from the National Institute of Standards and Technology (NIST) and the Naval Observatory.

Atomic hydrogen masers are yet another highly stable standard. Unlike Rb and Cs standards, the hydrogen maser is an actual oscillator (rather than a passive reference), with claimed stabilities approaching a few parts in 10^{14}. Recent research in stable clocks has centered on techniques using "cooled" trapped ions or atoms, or "atomic fountains," to achieve even better stability. These schemes are being used to create precise *optical* standards, which are then bridged to a radiofrequency reference via an "optical comb." Many physicists believe that ultimate stabilities of parts in 10^{17}–10^{18} are achievable.

Finally, you don't have to spend vast amounts of money to get an accurate frequency standard. Instead you can derive a precise 10 MHz clocking signal, along with 1 pps pulses, by receiving navigation signals from the Global Positioning System (GPS).[52] This is a constellation of 24 satellites in 12-hour orbits, covering all but the arctic latitudes, designed for precise navigation and timing. The satellites carry stable atomic clocks, and they transmit "navigation messages" using sophisticated dual-frequency spread-spectrum methods, at 1.575 GHz and 1.228 GHz. A GPS receiver on the ground, collecting signals from four satellites, can triangulate its position, and time as well. The inexpensive handheld GPS receivers you can buy (from companies like Garmin and Magellan, or included in cell phones) are intended for navigation, and they do not recover or regenerate a reference frequency. However, you can spend more and get a carefully engineered laboratory standard. An example is the Symmetricom 58503B "GPS Time and Frequency Reference Receiver," which delivers a 10 MHz output (stable to 1 part in 10^{12}, averaged over a day), and also 1 pps timing pulses (accurate to 20 nanoseconds); it costs $4,500. All you need, to get this kind of precision, is a place to put the doorknob-size antenna; your tax dollars do the rest.

In Table 7.2 we've gathered together various oscillator technologies, and their characteristics. We hope you like it.

7.1.8 Frequency synthesis: DDS and PLL

A stable reference is *stable*, but not tunable. But, as we hinted earlier, there are two nice techniques that let you create an output frequency of your choosing, with the stability of the reference: direct digital synthesis (DDS), and phase-locked loop (PLL) synthesis. These are digital "mixed-signal" techniques, which we treat in detail in Chapters 12 and 13. But they are closely related to oscillators and frequency generation, and so we will describe them here at a basic level.

A. Direct digital synthesis

The idea here is to program a digital memory with the numerical values of sine and cosine for a large set of equally spaced angle arguments (say for every 1°). You then make sinewaves by rapidly generating the sequential addresses, reading the memory values for each address (i.e., each sequential angle), and applying the digital values to a digital-to-analog converter. (DAC)

Figure 7.42 shows the scheme, in both its simplest notional form (a counter increments addresses to a sine-lookup ROM), and in the (far better) method used in

[51] The Symmetricom 1000B datasheet suggests (but does not guarantee) that a well-aged specimen will typically mellow out at "parts in 10^{11} per day."

[52] There are two analogous navigation systems: the Russian GLONASS, and the European Galileo. The latter is slated to be fully operational by 2020.

Table 7.2 Oscillator Types[a]

Class	Type	Stability	Tunability	Agility	Freq Range	Cost	Notes
Relaxation and delay	RC relaxation	10^{-2} to 10^{-3}	wide (>10:1)	high	Hz to 10's MHz	low (<$10)	555, 1799, etc
	delay	10^{-2} to 10^{-3}	modest (~5:1)[b]	high	10 MHz to 100's MHz	low (<$10)	within an IC
	Wien bridge	10^{-3}	modest (<10:1)	slow-slewing	Hz to MHz	low (<$10)	sinewave output
Resonator	LC	10^{-3} to 10^{-5}	modest	high	kHz to 100's MHz	low (<$10)	
	ceramic	10^{-2} to 10^{-3}	small (<10^{-3})	high	100's kHz to 10's MHz	low (<$10)	
	crystal ("xtal")	10^{-5}	very small (10^{-4})	high	10's kHz to 100 MHz	low (<$10)	ubiquitous
	xtal - TCXO	10^{-6} to 10^{-7}	very small (10^{-4})	high	10's kHz to 100 MHz	medium ($10-100)	
	xtal - OCXO	10^{-8} to 10^{-9}	very small (10^{-4})	high	10's kHz to 100 MHz	high ($100-1000)	
	SAW	10^{-4}	very small (10^{-4})		100's MHz	low (<$10)	small, low jitter
	cavity[c]	10^{-5}	modest[d]	low to high[d]	10's MHz to 10's GHz	low to medium	
Atomic	Rb vapor	10^{-10}	N.A.	N.A.	10 MHz derived ref[e]	high ($1k)	
	Cs beam	10^{-13f}	N.A.	N.A.	10 MHz derived ref[g]	very high ($10k)	the def'n of the sec!
	H maser	10^{-14}	N.A.	N.A.	10 MHz derived ref[h]	even higher (>$100k)	
	GPS	10^{-13k}	N.A.	N.A.	10 MHz derived ref	high ($1k)	ref'd to NIST Cs stds
Reference-derived	PLL synthesis	equal to ref	wide	t_s= 0-100 ms	Hz to GHz	low-med ($10-100)	
	"direct" synth	equal to ref	wide	t_s= 5-10 ms	Hz to GHz	high (>$1k)	
	DDS	equal to ref	wide	immediate	Hz to GHz	low-high ($10-1000)	

Notes: (a) somewhat subjective evaluations. (b) tuning via operating current. (c) cylindrical cavity, waveguide, or dielectric "pill" resonator. (d) tuning via mechanical plunger, varactor, or YIG. (e) from 8.634488275 GHz resonance. (f) long-term. (g) from 9.192631770 GHz resonance. (h) from 1.420405751767 GHz oscillation. (k) long-term.

practice. In the latter a ROM with an n-bit address space (thus values of the sine at 2^n phases within a single 360° cycle) is driven by a phase register that accumulates steps of phase according to the value of the "frequency tuning word" (FTW). At each clock the phase advances by $\Delta\phi = (360°/2^n) \cdot \mathrm{FTW}$, generating an output frequency $f_{\mathrm{out}} = \mathrm{FTW} \cdot f_{\mathrm{clk}}/2^n$, with any extra phase being carried forward from cycle to cycle.

This method has some drawbacks. The output is actually a staircase wave, since it is constructed from a set of discrete voltages, one for each table entry. You can, of course, use a lowpass filter to smooth the output; but having done so, you cannot span a wide range of frequencies, because the lowpass filter must be chosen to pass the sinewave itself while blocking the (higher) angle step frequency (the same problem applies to the switched-capacitor resonator; §7.1.9B). Decreasing the angular step size helps, but reduces the maximum output frequency.

Contemporary DDS chips include the sine lookup table, DACs, and everything else you need except for the stable fixed-frequency clocking input (usually provided by a simple crystal oscillator; §7.1.6). They are remarkably fast, and inexpensive. For example, the AD9850 series includes the AD9852, which goes to 150 MHz output frequency with 48-bit frequency resolution (that's a *micro*hertz!!); it costs $15 in quantity. If that isn't fast enough for you, perhaps the AD9912 ($37), with its 1 GHz clocking speed ($f_{\mathrm{out(max)}} = 400$ MHz) and 14-bit DAC will do the job. DDS chips let you program frequency sweeps (a ramping frequency versus time); and also amplitude, frequency, and phase *modulation* (periodic variation over time). You can send frequency-changing commands at very high rates (100 million new frequencies per second, for the AD9852) to make an *agile oscillator*.

Various members of the family permit phase synchronization and precise phase offsetting, so you can make a precision programmable quadrature oscillator (i.e., simultaneous sine and cosine outputs) with stunning performance with just a few chips (and for a few dollars more).

B. Phase-locked loops

This mixed-signal technique synthesizes an output frequency f_{out} that is related to the reference oscillator frequency f_{osc} by a rational fraction; i.e., $f_{\mathrm{out}} = (n/r) \times f_{\mathrm{osc}}$, where n and r are integers. You can think of this as a generalization of the simple divide-by-N frequency divider (where $f_{\mathrm{out}} = f_{\mathrm{osc}}/N$). Because PLLs blend analog and digital techniques, we'll discuss them in detail only later, in §13.13.

What's important to know now is that this gives you wonderful flexibility in oscillator frequency generation. For example, if we use a fixed-frequency 16 MHz crystal oscillator to drive a PLL chip and set $r = 16$, then the output frequency will be exactly n MHz, thus creating a high-frequency single-chip oscillator with output frequency settable in 1 MHz steps (with a typical output-frequency range

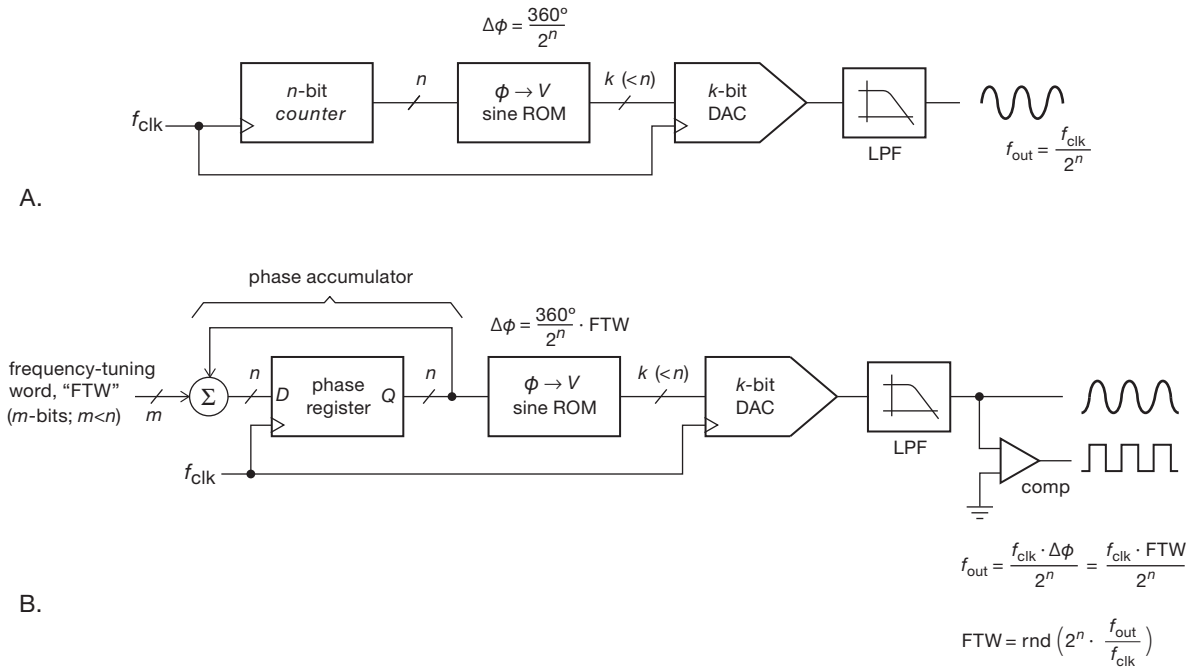


Figure 7.42. Direct digital synthesis creates its sinewave output from precomputed values of a sinusoid stored in ROM. The simplest configuration (A) increments a counter to address successive values from ROM. Far better is a configuration with a *phase accumulator* (B), which delivers n bits of output frequency resolution.

of 25–500 MHz or so). PLL synthesis is used in telecommunications devices (such as radios, televisions, and cellular telephones) to set the operating frequency for each channel.

As this example suggests, the somewhat complex PLL clock-synthesis technique is available in easy-to-use chips, with someone else having done all the hard work (specifically, making a phase detector and a voltage-controllable oscillator, and stably closing the loop). There are, in addition, the inexpensive "programmable oscillator modules" we mentioned in §7.1.6F that incorporate a crystal oscillator and a PLL, so you can stock a single module type, and program its frequency when you're ready to use it. You can get these from the companies that make fixed-frequency modules, for example Epson (SG8002 series), ECS (ECS-P series), Citizen (CSX-750P series), CTS (CP7 series), and Cardinal (CPP series). They come in the same DIP and SMT packages as conventional factory-set oscillators, with a slight price premium.

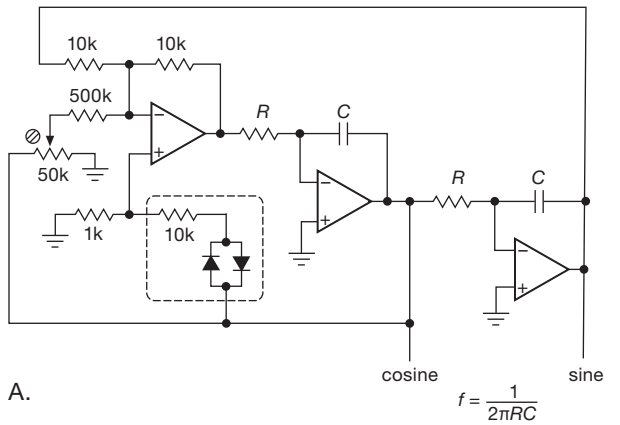
7.1.9 Quadrature oscillators

There are times when you need an oscillator that generates a simultaneous *pair* of equal-amplitude sine waves, *90° out of phase*. You can think of the pair as sine and cosine (or I and Q, for in-phase and quadrature). This is referred to as a *quadrature pair* (the signals are "in quadrature"). One important application is in radio communications circuits (quadrature mixers, single-sideband generation). Of great utility, as we explain below, a quadrature pair is all you need to generate any arbitrary phase.

The first idea you might invent is to apply a sinewave signal to an integrator (or differentiator), thus generating a 90°-shifted cosine wave. The phase shift is right, but the amplitude is wrong (figure out why). Here are some methods that do work.

A. Paired integrators

Figure 7.43 is a variation on a circuit that has been bouncing around now for several decades. It uses a pair of cascaded integrators (−90° phase shift each) inside a feedback loop, closed by a unity-gain inverting amplifier (180° phase shift). Oscillation occurs at a frequency at which each integrator has unity voltage gain; that is, at a frequency at which the capacitive reactance $1/2\pi fC$ equals R. The diodes limit the amplitude to ∼300 mV. We tested this circuit with LMC6482 op-amps running from ±5 V, and with R=15.8k and C=10 nF (f_{osc}=1 kHz), producing



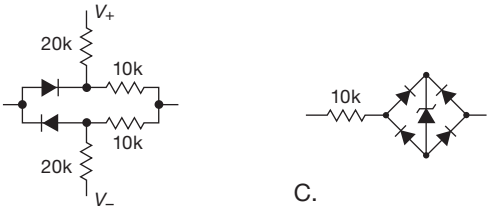


Figure 7.43. Quadrature sinewave oscillator (adapted from a circuit by Tony Williams): A. Basic circuit, with diode limiter (dashed box). B. Biased limiter. C. Zener limiter.

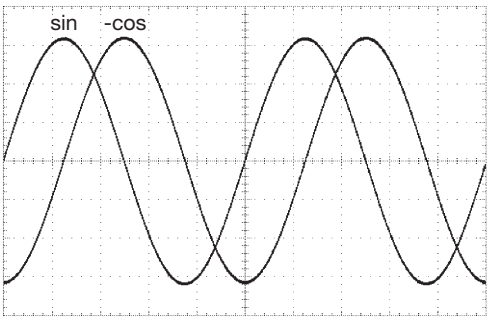


Figure 7.44. Measured output from the circuit of Figure 7.43A. Horizontal: 200 μs/div. Vertical: 100 mV/div.

B. Switched-capacitor resonator

This is a clever method: Figure 7.45 shows how to use a switched-capacitor filter IC as a self-excited bandpass filter to generate a quadrature sinewave pair. The easiest way to understand it is to assume there is already a sinewave output present. Op-amp U_{2a}, wired as a comparator, converts this to a ± 5 V square wave, which is fed back as the filter's input. The filter has a narrow bandpass (Q=10), so it converts the input square wave to a sinewave output, sustaining the oscillation. A square-wave clock input (CLK) determines the bandpass center frequency and hence the frequency of oscillation, in this case $f_{CLK}/100$. The circuit

the waveforms in Figure 7.44. The measured frequency was 997 Hz, with distortion of 0.006% and 0.02% for the sine and cosine outputs, respectively.

The simple diode limiter does not provide particularly good amplitude control, and it also limits the amplitude to ~300 mV. An improved limiter is shown in Figure 7.43B, in which the output is biased toward the supply rails by a pair of dividers, so that the diodes conduct only at a larger amplitude (set by the divider ratio). With the component values shown, the measured amplitude was 3.3 V. Figure 7.43C shows another limiter that some designers favor, using a zener wrapped in a diode bridge (therefore bidirectional) to set the amplitude. You can use regular zener diodes for larger amplitudes (5 V and above), but low-voltage zeners perform poorly, with their "soft knees" (see Figure 1.17). But you can use instead a low-voltage 2-terminal reference, which behaves like a nearly perfect zener; some examples are the LM385-1.2 and -2.5 (1.24 V and 2.50 V), the AD1580 (1.22 V), and the ADR510 (1.0 V). With any of these schemes it's a good idea to put a capacitor across the zener (or reference) to hold its voltage during zero crossings of the waveform.

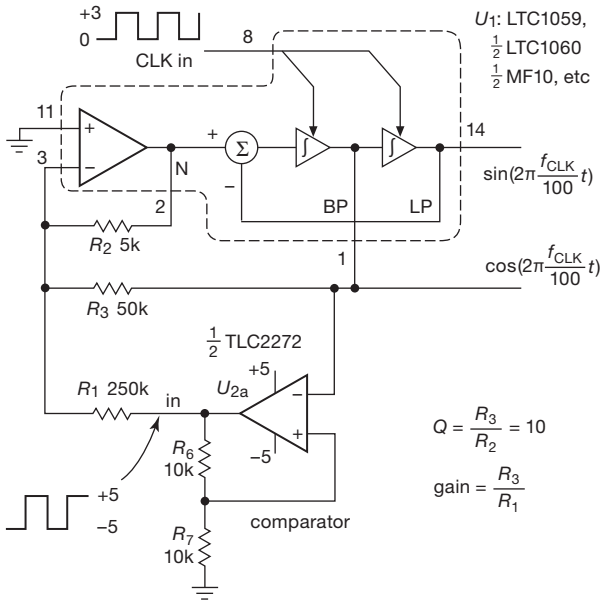


Figure 7.45. You can generate a pair of quadrature sinewaves by feeding back the squared-up output of a narrow bandpass filter, implemented in a switched-capacitor filter IC.

generates a quadrature pair of sinewaves of equal amplitude, and is usable over a frequency range of a few hertz to more than 10 kHz. Note that the output is actually a "staircase" approximation to the desired sinewave, owing to the quantized output steps of the switched filter – see Figure 7.46.

An interesting feature of this circuit is its ability to "calculate pi": the filter is configured with an accurate gain of R_3/R_1=0.2, and its input is an accurate $\pm$5 V square wave, so one might expect an output amplitude of $\pm$1 V. But no – the filter retains only the fundamental frequency component of the square wave, which has an amplitude equal to $4/\pi$ times that of the square wave. So, voilà, we get an output amplitude of $4/\pi$ volts (about 1.27 V), as seen in the 'scope traces.

C. Direct digital synthesis

We saw this popular method earlier (§7.1.8) as a general way to synthesize a sinewave (or any "arbitrary" wave, if you want) of precise frequency, relative to an accurate reference frequency input. Among its many virtues, this method lends itself well to the generation of a pair of signals in quadrature (or any other phase relationship, if you want). Figure 7.47 shows a nice chip from Analog Devices that is intended for quadrature sinewave generation; it permits frequency adjustment in steps of a *micro*hertz, along with a large bag of tricks that includes digital mixers (for quadrature amplitude modulation), and timers and accumulators (for frequency or phase modulation, nonlinear chirp, etc). The datasheet makes for some good reading!

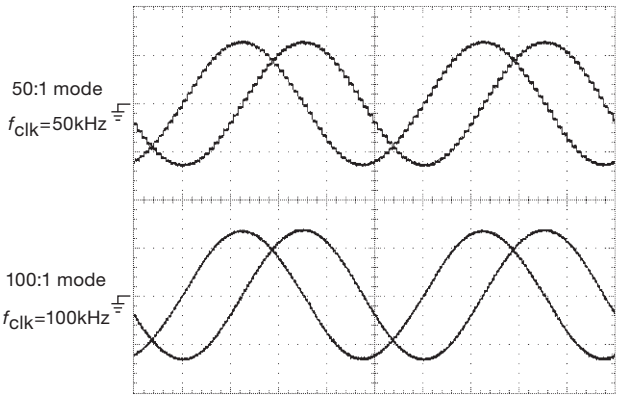


Figure 7.46. Observed waveforms from the circuit of Figure 7.45. The switched capacitor filter generates a staircase approximation to the ideal quadrature sinewaves, more evident when the filter is set to the coarser 50:1 f_{CLK}/f_{out} ratio. This filter knows the value of π: the output amplitude is $4/\pi$ volts! Horizontal: 200 μs/div; Vertical: 1 V/div.

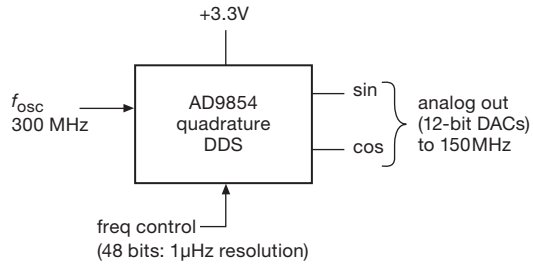


Figure 7.47. Quadrature generation by DDS. The AD9854 has many additional tricks up its sleeve (see text).

D. Phase-sequence filters

There are tricky *RC* filter circuits that have the property of accepting an input sinewave and producing as output a pair of sinewave outputs whose phase *difference* is approximately 90°. The radio hams know this as the "phasing" method of single-sideband generation (due to Weaver), in which the input signal consists of the speech waveform that you want to transmit. Unfortunately, this method works satisfactorily only over a rather limited range of frequencies and requires precision resistors and capacitors.

A better method for wideband quadrature generation uses "phase-sequence networks," consisting of a cyclic repetitive structure of equal resistors and geometrically decreasing capacitors, as in Figure 7.48. You drive the network with a signal and its 180°-shifted cousin (that's easy, since all you need is a unity-gain inverter). The output is a fourfold set of quadrature signals, with a 6-section network giving $\pm$0.7° error over a 40:1 frequency range; an 8-section network extends the range to 150:1. This is demonstrated in Figure 7.49, a SPICE simulation of an 8-section phase-sequence network, showing the behavior of a pair of quadrature outputs with good performance from 400 Hz to 50 kHz.

E. Quadrature square waves

For the special case of square waves, generating quadrature signals is a lead-pipe cinch. The basic idea is to generate twice the frequency you need, then divide by 2 with digital *flip-flops*. Figure 7.50 shows a simple circuit that does this, using type-D flip-flops (Chapter 10). This technique is essentially perfect from dc to at least 100 MHz.

F. Radiofrequency quadrature

At radio frequencies (upward of a few megahertz) the generation of quadrature sinewave pairs again becomes easy, using devices known as *quadrature hybrids* (or quadrature *splitter–combiners*). At the low-frequency end of the radio spectrum (from a few megahertz to perhaps 1 GHz) these

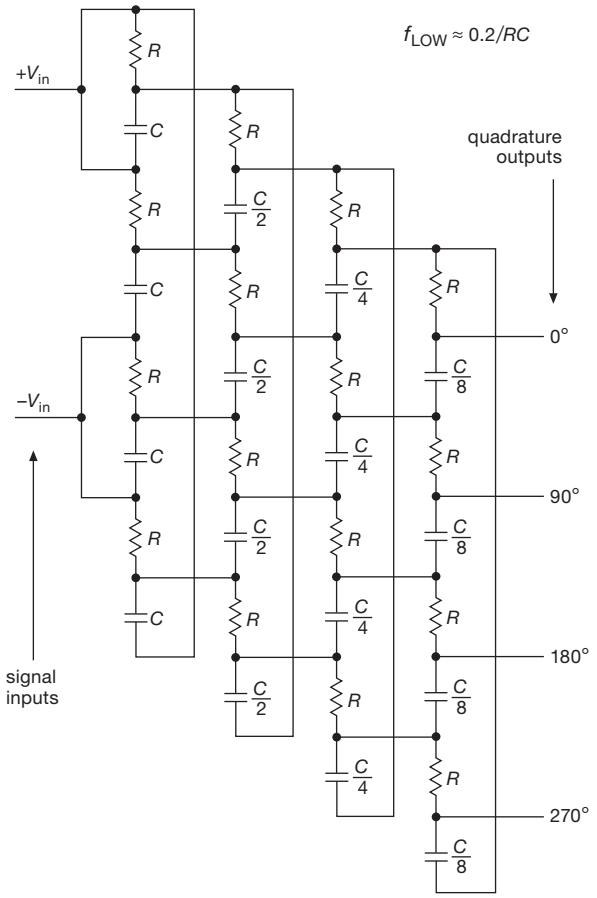


Figure 7.48. Four-section phase-sequence network. The network is effective for frequencies extending from a low frequency $f_{LOW} \approx 0.2/RC$ to a high end that depends on the ratio of capacitances of the first and last section, as $f_{HIGH}/f_{LOW} \approx C_{first}/C_{last}$.

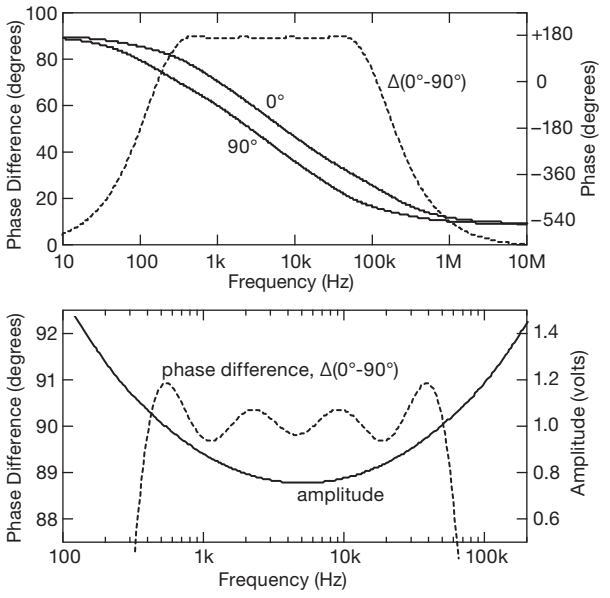


Figure 7.49. Phase and amplitude behavior of an 8-section phase-sequence network (with R=10k, C=40 nF), as modeled in SPICE. The lower plot expands the region of accurate quadrature. Note that it is the *difference* of phases (dashed curves) that are in quadrature.

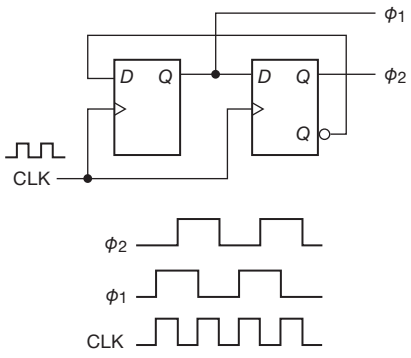


Figure 7.50. Square waves in quadrature, courtesy of the versatile type-D flip-flop.

take the form of small core-wound transformers, while at higher frequencies you find incarnations in the form of stripline (strips of foil insulated from an underlying ground plane; see Chapter *1x*) or waveguide (hollow rectangular tubing). These techniques tend to be fairly narrowband, with typical operating bandwidths of an octave (i.e., ratio of 2:1).

G. Generating a sinewave of arbitrary phase

Once you have a quadrature pair, it's easy to make a sinewave of *arbitrary* phase. You simply combine the in-phase (I) and quadrature (Q) signals in a resistive combiner, made most easily with a potentiometer going between the I and Q signals. As you rotate the pot, you combine the I and Q in different proportions, taking you smoothly from 0° to 90° phase. If you think in terms of phasors, you'll see that the resulting phase is completely independent of frequency; however, the amplitude varies somewhat as you adjust the phase, dropping 3 dB at 45°. You can extend this simple method to the full 360° by simply generating the inverted (180°-shifted) signals, I' and Q', with an inverting amplifier of gain $G_V = -1$.

The generation of sinusoidal bursts of predetermined phase (and amplitude) is of great importance in digital communications. It is used in particular in the method of *quadrature amplitude modulation* (QAM, pronounced

"quahm"), in which several bits are encoded in each QAM "symbol." For example, most digital cable television is encoded as 256-QAM, with each symbol (out of a constellation of 256) carrying 8-bits of information; you can think of the individual symbols as little bursts of several sinusoidal cycles, with a particular phase and amplitude characterizing each symbol.[53]

With DDS it is even easier to set the output phase (relative to a synchronizing pulse, or to a second DDS synthesizer that is synchronized to the first), because DDS chips let you add a user-specified offset to the internal accumulating phase that is used to generate the sinewave. For example, the AD9951 has a 14-bit *phase-offset word*, giving you a phase-settable step size of $360°/2^{14}$, or 0.02°. You can change the phase on-the-fly to produce phase modulation.

7.1.10 Oscillator "jitter"

In addition to the primary oscillator parameters – frequency, amplitude, and waveform – there is *stability*: if an oscillator drifts with time, temperature, or supply voltage, we talk about stability (or lack thereof), and we may assign corresponding coefficients. For example, a crystal oscillator might specify a tempco of 1 ppm/°C. That's important for applications such as timekeeping, communications, or spectroscopy.

But it's not the whole story. You could, for example, have an oscillator that maintains a precise 10.0 MHz average frequency, independent of time and temperature, but that exhibits variations in the timing of its zero crossings from cycle to cycle. (If you like, you can think of this as an instability on a short time scale, as compared with the long time scales of effects like drift with time and temperature.[54]) This undesirable property has several names, depending on the context: if it's a sinewave, and you're using it in a communications application, you call it *phase noise*, or *spectral purity*; if instead it's a square wave or digital pulse train, and you're using it for waveform sampling and reconstruction, or for clocking fast digital data links, you call it *jitter*.

A bit more on this business of jitter. Timing jitter arises whenever an output transition is created by a periodic signal crossing a decision threshold, as for example in an *RC* relaxation oscillator or a crystal oscillator. Figure 7.51 shows the situation. A signal of finite slew rate S (volts/second) is crossing a voltage threshold, and both the signal and the voltage threshold are imperfect, each with some additive noise voltage v_n; we'll call these $v_n(\text{sig})$ and $v_n(\text{th})$. So there's a timing uncertainty, such that the time at which the (noisy) signal crosses the (noisy) threshold can (and will) vary by

$$\Delta t = \frac{\sum v_n}{\text{slew rate}} = \frac{v_n(\text{sig}) + v_n(\text{th})}{S}$$

Jitter is greater with slowly slewing signals, for a given amount of signal or threshold noise. A sinewave of frequency f has a maximum slew rate of $S{=}2\pi f V_0$, where V_0 is the peak signal amplitude (though less if the threshold is not at the amplitude midpoint); the slew rate of a pulse or step can be approximated by $S{=}V_{\text{step}}/t_r$, where t_r is the rise time (or, more generally, transition time).

Oscillator stability and jitter are important subjects, and we'll have more to say about them in Chapter 13, in connection with digital–analog conversion, serial digital communication, and phase-locked loops.

7.2 Timers

Oscillators generate a periodic signal, characterized by its waveform (sine, square, pulse), its frequency, and its amplitude. Closely related to oscillators are *timers*: circuits that generate a delayed pulse, or a pulse of a given width, following a triggering event. Here you talk about delay *times* and pulse *widths* rather than about *frequency*. The techniques, however, are quite similar, consisting mostly of *RC*-style analog waveforms (driving comparator–trigger circuits), or oscillator-driven digital counters or dividers.

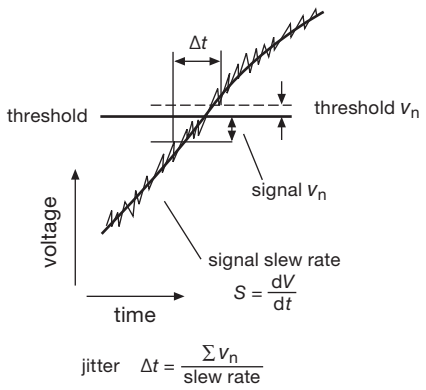


Figure 7.51. Timing jitter basics: a signal of finite slew rate passes through a decision threshold.

[53] The constellation of phases and amplitudes are chosen to minimize error, given the characteristics of the cable.

[54] Those souls (you know who you are!) who derive pleasure from the misfortunes of other oscillators have more sophisticated measures of stability, notably the *Allan variance*, which is basically a plot of oscillator stability versus averaging time.

There's quite a range of techniques, depending on the time scale involved and the accuracy required. We visit most of the techniques in common use.

7.2.1 Step-triggered pulses

Sometimes you need to generate an output pulse, of some duration, from an input trigger. The input may be a short pulse, from which you want to generate a longer output pulse; alternatively, the input may be a "step" (usually a logic-level transition, longer than the desired output pulse), from which you want to generate a short pulse.

A. Short pulse from step: *RC* + discrete transistor

An *RC* differentiator turns a voltage step into a step with a trailing decay of time constant *RC* (Figure 7.52A). You can use the output directly, or you can run it through a transistor switch to generate something more like a rectangular pulse. Figure 7.52B shows the ubiquitous 2N7000 small MOSFET used to generate an output pulse of either polarity (swinging between a positive rail and ground); the pulse width is of order $\tau=RC$, but depends upon the ratio of gate threshold voltage to input step size. In Figure 7.52C we show the analogous circuits with a fast bipolar switching transistor (a PN2369: it uses gold-doping to reduce base charge storage time to 13 ns maximum, compared with 200 ns for a plain ol' jellybean like the 2N3904[55]). Note that a series resistor (R_1) is needed to limit the peak base current. The exact timing behavior with a BJT is more complicated than with a MOSFET, because of forward base-voltage clamping: for the circuit with rising-edge input, the output pulse width is determined primarily by R_1C (which sets the decay of forward base drive current); for the circuit with falling-edge input, on the other hand, the output pulse width is determined primarily by R_2C (which times the recovery from base back-bias to forward bias, assuming $R_2 \gg R_1$), as given by the formula shown. For the latter circuit the input step amplitude V_{step} should be no larger than ~6 V to prevent base-emitter reverse breakdown.

A caveat: with the exception of the falling-edge BJT in Figure 7.52C, the pulse widths generated by these circuits are somewhat unpredictable, and they should not be used where accurate timing is needed. We will see better methods shortly.

This is a situation where SPICE simulation provides a useful tool for exploring detailed circuit behavior. Figure 7.53 shows a comparison of bench measurement and simulation for the falling-edge circuit of Figure 7.52C, running from a +5 V supply, when driven by a 5 V falling step of 5 ns fall time. The rather good agreement validates the use of the simulation models and tools. Note particularly the accurate simulation of delay time (the input step began at 20 ns), transistor saturation voltage, base voltage, and output waveform (which, though not visible in the printed figure, overshoots the +5 V rail by about 40 mV, owing to capacitive coupling of the rising base voltage).

We'll be seeing more accurate and predictable ways to generate output pulses from input edges. But this simple method is fine for noncritical jobs, for example driving a hefty latching power relay, as shown in Figure 7.54. This

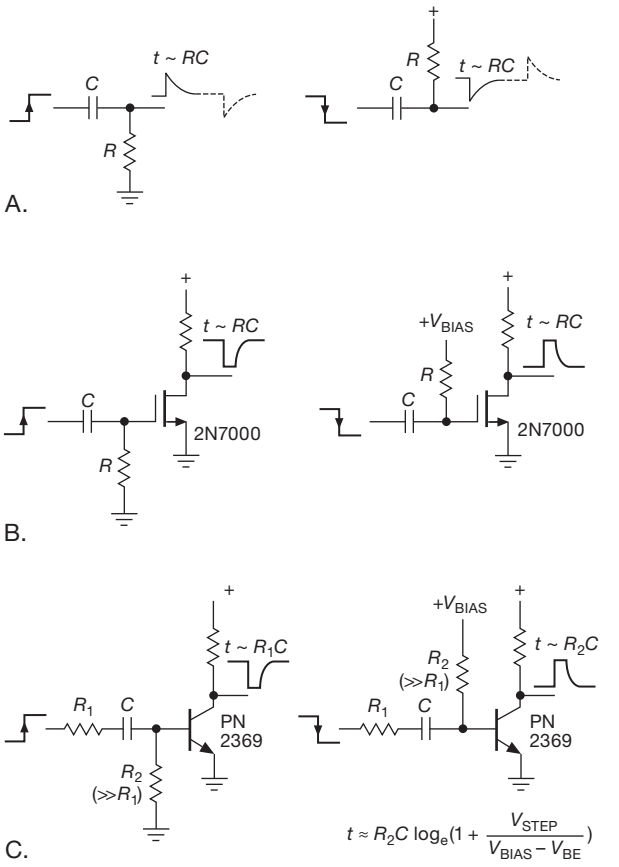


Figure 7.52. Generating a pulse from a step with *RC*'s and transistors. A. Bare *RC* differentiator. B. *RC* drives MOSFET switch. C. *RC* drives BJT switch.

[55] For a SOT-23 surface-mount package there's the MMBT2369. The 2N5771 (MMBT5771) and PN3640 (MMBT3640) are similar *pnp* types with 20 ns storage-time specs. These gold-doped transistors are all low-voltage types, rated at 12 to 20 V. Gold doping leads to low beta and higher leakage current; these transistors are well suited for fast switching, but not much else.

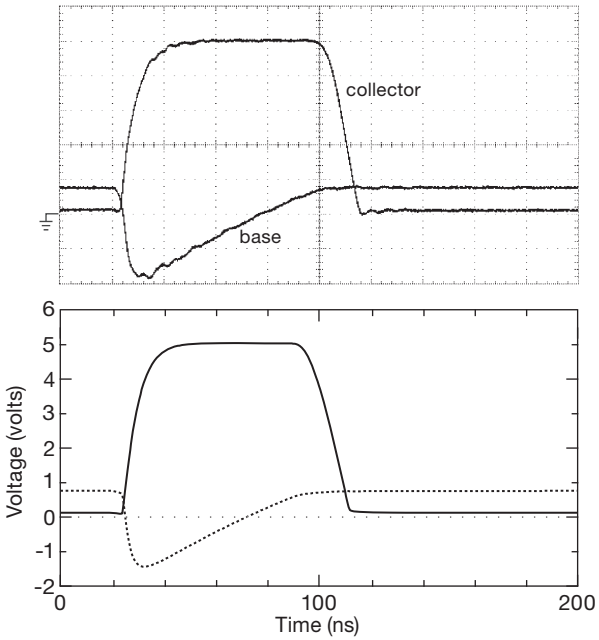


Figure 7.53. Simulated (lower) and measured (upper) waveforms of the right-hand circuit of Figure 7.52C, with R_1=220 Ω, C_1=47 pF, R_2=2.2k, R_C=470 Ω, and with 'scope probing capacitance to ground of 8 pF at base and collector. The SPICE plot and the 'scope trace have been scaled identically, at 20 ns/div and 1 V/div.

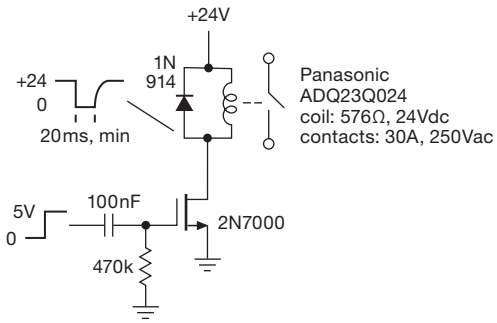


Figure 7.54. A 20 ms pulse actuates a latching power relay. This is a "2-coil" latching relay (for SET and RESET), only one of which is shown. This particular relay costs $23, and includes high-current PCB and FastOn (spade lug) contacts.

particular relay's contacts are rated at 30 A and 250 Vac; it has a pair of coils (set, reset), which you drive by applying 24 Vdc for a minimum of 20 ms. We tested the circuit in our lab – it made an impressive clatter that could be heard down the hallway.

B. Pulse from step or edge

With the addition of a couple more transistors you arrive, finally, at a pulse output circuit that is happy to trigger either on a step or on a short pulse; that is, it is *edge* triggered, and insensitive to whether the triggering edge belongs to a pulse that is shorter or longer than the desired output pulse length. We showed this classic BJT "one-shot" circuit earlier, in the transistor chapter – see Figure 2.12. You can wire up that configuration, if you want. But you're probably better off using an integrated *monostable multivibrator*: these devices take care of the details, and they provide flexible triggering options. We'll look at them next, after taking a brief look at the use of digital logic gates for generating short pulses.

C. Short pulse from step: logic gates

A closely related technique uses digital logic gates (Chapter 10) in place of discrete transistors. This is particularly useful if you want output signals to drive additional logic, because the output signals are of the correct logic voltages and speeds. In Figure 7.55 a logic inverter (§10.1.4D) with built-in Schmitt trigger (§4.3.2A) is used to create output pulses with abrupt transitions; although the edges are fast (owing to the Schmitt trigger), the timing is only approximate, because the Schmitt voltage thresholds are loosely specified (e.g., the specified hysteresis amplitude spans a range of 3:1).

In Figure 7.56A the short *propagation delay* of logic inverters (of order several nanoseconds), combined with the logic of *gates*, substitutes for *RC* delays to create the output pulse width; the three circuits shown respond to rising, falling, or both edges, respectively. The circuits in Figure 7.56B extend this gate idea to longer output pulse widths, by using an *RC* delay, sharpened up with a Schmitt inverter, to provide the delayed input to the gate.

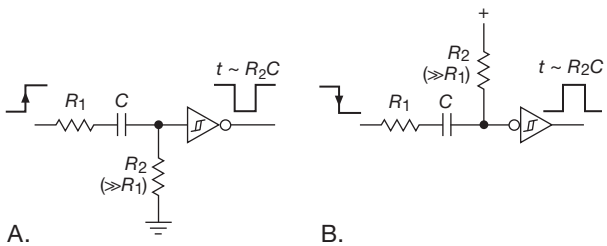


Figure 7.55. A logic inverter with Schmitt-trigger input creates a clean output pulse. R_1 limits gate input current when the input signal returns to its initial level, during which the gate input is driven below ground or above V_+, respectively.

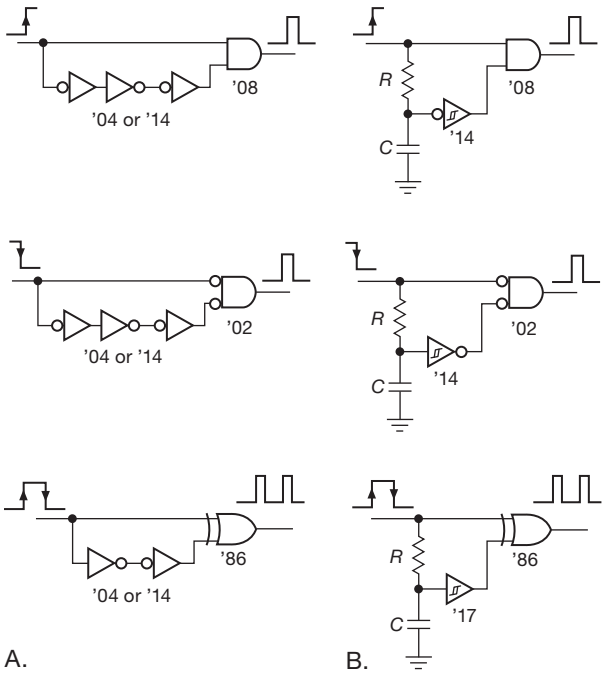


Figure 7.56. Generating a pulse from a step, with logic gates. A. Short pulses made with 2-input gates and cascaded inverter delays. B. Longer pulses made with 2-input gates and *RC* delays.

D. Short pulse from edge: monostable multivibrators

If these last circuits appeal to you, you're in luck: the semiconductor industry has created a class of integrated circuits that combines digital logic with an *RC* timing circuit, in the form of the *monostable multivibrator* (also known as a "one-shot," emphasis on the word "one"). These are edge triggered, with timing set with good accuracy by an external *RC* and with clean logic-level outputs. The output pulse can be short compared with the input duration (as with the circuits above); or it can be longer than the input (as with the circuits we discuss next). This topic has plenty of complexity, so we postpone a full discussion to §7.2.2.

E. Long pulse from trigger: the 555 returns!

The circuits above create an output pulse shorter than the input. That's OK, if it's what you want. But you might want the opposite: a brief input pulse triggers a longer output – think of the one-minute zapper button on a microwave oven, for example. The methods here can be grouped, roughly, into two categories: (a) *RC*-mediated analog timing circuits, and (b) oscillators followed either by dedicated digital counter–dividers, or by the full computational glory of microprocessors (computers-on-a-chip). In the first category are devices like the classic 555 (which we used earlier to make an *RC* relaxation oscillator, §7.1.3), and the monostable multivibrator (a dedicated pulse-generating chip).

Looking back in the chapter, we made an oscillator with the 555 by driving both the TH and the TR′ inputs with the voltage on the capacitor (Figures 7.9–7.14); those inputs flipped the output state (and the DIScharge transistor), causing the cyclic charging and discharging of the capacitor. To make instead a *single* pulse, connect the capacitor to the TH input only, and use the TR′ terminal as a trigger input (Figure 7.57). A negative-going input starts the cycle, turning off the DIS transistor and causing the output to go HIGH; when the capacitor voltage reaches 2/3 of V_+ the cycle ends, with the output returning to LOW and the DIS pin discharging the capacitor rapidly to ground. It is easy to calculate the pulse width, namely $t = 1.1RC$. Note that the input trigger must be removed before the end of the pulse, that is, it must be shorter than the intended output pulse. Look back at Figure 4.41 to see how to make a programmable one-shot pulse generator (with "cold switching" control of the pulse width) with a 555.

Roughly speaking, a CMOS 555 (see Table 7.1 on page 430) can generate pulse widths from about 1 μs to 100 seconds. At the long end the limit is set by residual TH input current, which is less than 10 nA, so values of R as large as 10–100 MΩ can be used. At the short end, the minimum resistor value is limited by the maximum DIS current (∼15 mA with a 5 V supply) and the intrinsic speed of the 555. Another limitation of the 555 is the awkward input trigger condition, namely a negative-going pulse that must return to its positive quiescent voltage before the end of the output. The latter problem is addressed nicely in the class of timer chips known as monostable multivibrators.

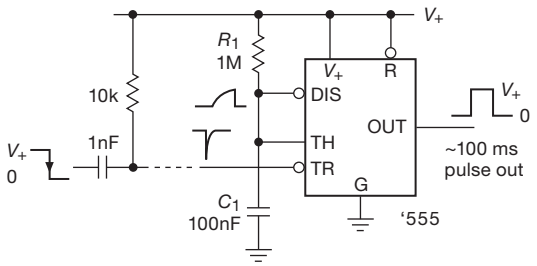


Figure 7.57. The venerable 555 generates a single positive output pulse when connected in *monostable* mode. The input *RC* differentiator converts a falling step into a trigger pulse; it can be omitted if a short trigger pulse is available to drive the TR′ input.

7.2.2 Monostable multivibrators

Within the various families of digital logic (§10.1.2B, §10.2.2) you can get *monostable multivibrators* (one-shots), which you can think of as logic-friendly versions of the 555 as it was used in Figure 7.57. They are edge triggered (by standard logic levels), and generate a logic-level output pulse Q (and its complement Q') whose width is determined by an external R and C. Monostables are very useful (some would say *too* useful!) for generating pulses of selectable width and polarity. Making one-shots with RC's combined with discrete transistors or gates (as we've just done) is tricky, and depends, for example, on the details of a gate's input circuit, since you wind up with voltage swings beyond the supply voltages. Rather than encouraging bad habits by illustrating more such circuits, we encourage you to adopt the one-shot as an available functional unit. In actual circuits it is best to use a packaged one-shot; you construct your own only if absolutely necessary, e.g., if you have a gate available and no room for an additional IC package (even then, maybe you shouldn't).

A. What's inside

Although you can happily use monostables without ever worrying about what's happening inside, it is an interesting place to visit. Figure 7.58 shows the internal circuit scheme used in most monostables. There are pins for an external C and R; the latter charges the capacitor toward V_+, which can range from +2 V to +15 V, depending on the particular logic family.[56] In the resting state the capacitor is fully charged, and the output flip-flop is reset, i.e., Q is LOW (ground) and Q' is HIGH (V_+).

Look at the waveforms in the figure, and assume for now that the three resistors R_1–R_3 are of equal resistance.[57] When the trigger condition is met (more on this below), for example by bringing input B HIGH while input A' is LOW, two things happen: (a) the output Q goes HIGH; and (b) the capacitor is rapidly discharged toward ground by the lower MOSFET.[58] When the capacitor's falling voltage reaches $\frac{1}{3}V_+$, the control circuitry removes gate drive from the lower MOSFET, allowing the capacitor to charge back up through R_{ext}; this begins the RC-mediated timing interval, which ends when the capacitor's voltage reaches $\frac{2}{3}V_+$,

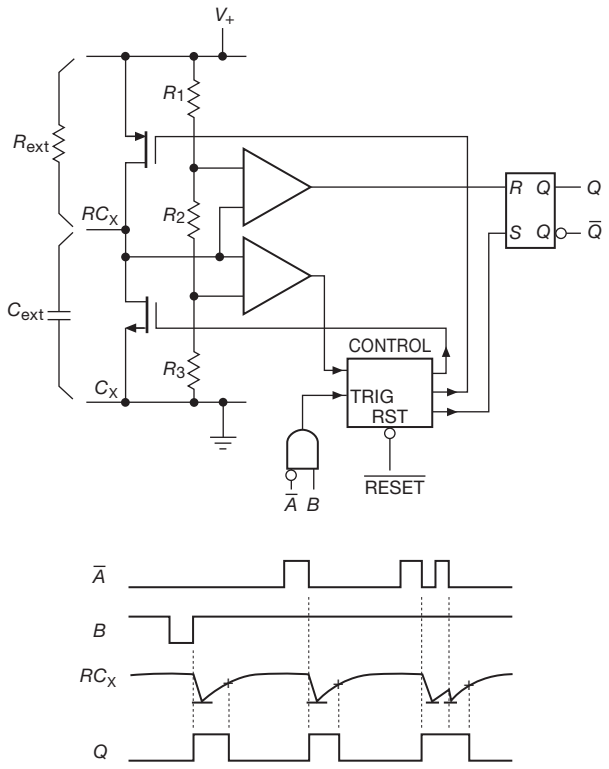


Figure 7.58. Internal circuit and waveforms for a monostable multivibrator. The waveforms show *retriggerability*.

at which time the output flip-flop is reset, ending the output pulse by bringing Q again LOW. It's a straightforward exercise in RC time constants to find that this takes a time

$$t = RC \log_e \frac{V_+ - V_L}{V_+ - V_H}, \tag{7.1}$$

where V_L and V_H are the lower and upper threshold voltages.

Exercise 7.6. Derive this formula.

B. One-shot characteristics

Inputs

One-shots are triggered by a rising or falling edge at the appropriate inputs. The only requirement on the triggering signal is that it have some minimum width, typically 25 ns to 100 ns; it can be shorter or longer than the output pulse. Two inputs are usually provided so that a signal can be connected to one or the other, to trigger the one-shot either on a rising edge or on a falling edge; alternatively, both inputs can be used, with a pair of separate triggering sources. The extra input can also be used to inhibit triggering. Figure 7.59 shows two examples.

[56] Popular logic supply voltages are +5 V, +3.3 V, and +2.5 V.

[57] They usually aren't, though it's not important here. More on this later, in §7.2.2C.

[58] Typical discharge currents are 30–80 mA. So a 0.01 μF timing capacitor may take ~600 ns to discharge from 5 V down to the lower threshold at 1.6 V. This time is folded into the manufacturer's specified "K-factor" (see later discussion).

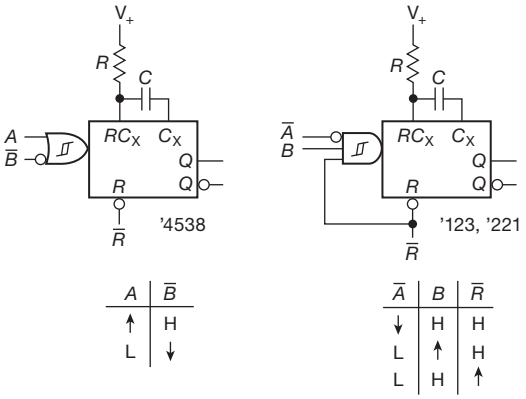


Figure 7.59. Two popular one-shots and their truth tables. Monostables trigger on input *transitions*, and usually accommodate either polarity, with internal AND or OR gating. Most monostables include Schmitt trigger inputs.

Each horizontal row of the truth tables represents a valid input triggering transition. For example, the '4538 is a dual monostable with OR gating at the input; if only one input is used, the other must be disabled, as shown. The popular '123 is a dual monostable with AND input gating; unused inputs must be enabled. Note particularly that the '123 triggers when RESET is disabled if both trigger inputs are already asserted. This is not a universal property of monostables, and it may or may not be desirable in a given application (it's usually not). The '423 is the same as the '123, but without this "feature."

When monostables are drawn in a circuit diagram, the input gating is usually omitted, saving space and creating a bit of confusion.

Retriggerability

Most monostables (e.g., the 4538, '123, and '423 mentioned earlier) will begin a new timing cycle if the input triggers again during the duration of the output pulse (as in Figure 7.58). They are known as *retriggerable* monostables. The output pulse will be longer than usual if they are retriggered during the pulse, finally terminating one pulse width after the last trigger. The '221, by contrast, is nonretriggerable: it ignores input transitions during the time that the output pulse is asserted.

Resettability

Most monostables have a jam RESET input that overrides all other functions. A momentary input to the RESET terminal terminates the output pulse. The RESET input can be used to prevent a pulse during power-up of the logic system; however, see the preceding comment about the '123.

Table 7.3 Monostable Multivibrators

Type	Retrig?	Trig	Families
'123[a]	•	!A & B & !R	AHC(T), HC(T), LS, LV, LVC, VHC
'221	-	!A & B	74C, HC(T), LS, LV, VHC
'423	•	!A & B	HC(T), LS
'4538	•	A or B	4000, HC(T)

Notes: (a) See Table 7.4.

Pulse width

Pulse widths from 40 ns up to milliseconds (or even seconds) are attainable with standard monostables, set by an external capacitor and resistor combination. A device like the 555 can be used to generate longer pulses, but its input properties are sometimes inconvenient. Very long delays are best generated digitally (see §7.2.4).

Table 7.3 lists the commonly available monostables. In addition to these traditional logic-family monostables, be sure to look at the LTC6993 "TimerBlox" one-shots (see table on page 433). It has only one trigger input, but four variants let you choose rising- or falling-edge triggering, and retriggerable or non-retriggerable. You set the pulse width with a range selection pin (via a voltage divider from V_+), and another external resistor that allows continuous tuning over a 16:1 span within the selected range. The eight digitally-divided ranges jump by successive factors of 8, thus a total range of 2^{21}:1. Output pulse widths range from 1 μs to 34 seconds, with worst-case timing accuracy of a few percent.

C. Cautionary notes about monostables

Monostables have some problems you don't see in other digital circuits. In addition, there are some general principles involved in their use. First, a rundown on monostable pathology.

Some problems with monostables

Timing One-shots involve a combination of linear and digital techniques. Because the linear circuits have the usual problems of V_{GS} (or V_{BE} and beta) variation with temperature, etc., one-shots tend to exhibit temperature and supply-voltage sensitivity of output pulse width. A typical unit like the '4538 will show pulse-width variations of a few percent over a 0–50°C temperature range and over a ±5% supply-voltage range. In addition, unit-to-unit variations give you a ±10% prediction accuracy for any given circuit. When looking at temperature and voltage sensitivity, it is

important to remember that the chip may exhibit self-heating effects and that supply-voltage variations *during the pulse* (e.g., small glitches on the V_+ line) may affect the pulse width seriously.

Brand variation Monostables with the same generic part number, but made by different manufacturers, may have somewhat different specifications, particularly involving timing components. The usual way this is specified is by a "K-value," where the output pulse width (for all but small-value capacitors) is given approximately by $t_w = KRC$ (if the datasheet does not mention K, look for the pulse width with R=10k and C=100 nF). And here's how it goes: nearly all monostables fall into three clusters of K-values, either K=0.7 (all of the '4538 monostables), or K is either ≈0.45 or ≈1.0 (most other monostable part numbers).

The '4538 monostables are boring! They all have K=0.7 (hey, not a bad reason for choosing it. . .). But there's some excitement with the other part numbers, because the K-value for a given part number may be 1.0 or 0.45, depending on who makes it. For example, the 74HC123 is available from at least five manufacturers.[59] The FSC and Toshiba parts guarantee an output pulse width of 0.9–1.1 ms (with 1.0 ms typ) for 10k and 100 nF, but the NXP part instead specifies 0.45 ms (typ, with no minimum or maximum spec). The other two brands have chosen different *RC* combinations, at which they specify typical values only: ST uses 100 nF/100 kΩ (4.4 ms typ), and TI uses 10 nF/10 kΩ (45 μs typ). Evidently these parts are not fully interchangeable![60] This caution extends generally to analog and mixed-signal devices of all kinds (consider yourself warned).

To illustrate this point, we've collected in Table 7.4 a comprehensive listing of all available '123-style monostables. The data include the timing variable K that you can use to predict the pulse width according to $t_w = KRC + t_{min}$.[61] The effective value of K varies with supply voltage, so in Figure 7.60 we've made graphs showing this dependence for the monostables listed in Table 7.4.

Another parameter that varies with manufacturer, and that is not generally specified on datasheets, is the particular choice of threshold voltages – that is, the ratios of the internal resistors R_1–R_3 in Figure 7.58. That does matter, though, because the rapid discharging of a small-value timing capacitor (say ∼1000 pF or less) overshoots the lower threshold (a similar effect bedevils the classic 555 sawtooth oscillator circuit, where the DIS pin pulls the timing capacitor rapidly toward ground). The result is an enlarged (and not terribly stable) pulse width. At least one manufacturer (TI, in their SN74HC4538) has addressed this problem by putting the lower threshold close to ground (at about 4.3% of V_+, or +0.2 V when running from a +5 V supply), so discharge overshoot has almost no effect – the capacitor charges back up from about the same voltage, overshoot or not.

Table 7.4 "Type 123" Monostable Timing

Mfg	Part Number[a]	T-variant[b]	Vsupply min (V)	Vsupply max (V)	K[c]	t_{min}[d] (ns)	K-graph[e]
multiple	**74LS123†**	•	4.5	5.5	0.37	-	-
Toshiba	**TC74HC123**	-	2	6	1.00	150	1
Renesas	**HD74HC123†**	-	2	6	1.00	390	2
Fairchild	**MM74HC123**	-	2	6	1.00	390	6
TI	**CD74HC123**	•	2	6	0.45	230	8
ST	**M74HC123**	-	2	6	0.44	230	9
NXP	**74HC123**	•	2	6	0.45	105	8
TI	**SN74AHC123**	•	2	5.5	1.00	110[f]	4
NXP	**74AHC123**	•	2	5.5	1.00	45	-
Fairchild	**74VHC123**	-	2	5.5	1.00	75	4
Toshiba	**TC74VHC123**	-	2	5.5	1.00	75	5
TI	**SN74LV123**	-	2	5.5	1.00	110	4
NXP	**74LV123**	-	1.2	5.5	0.43	70	7
TI	**SN74LVC1G123**[g]	-	1.65	5.5	0.95	95	3
Toshiba	**TC7WH123**[g]	-	2	5.5	1.00	75	-

Notes: (a) suffixes are omitted; all except † have Schmitt-trigger inputs, and all trigger on (!A)&B. (b) TTL thresholds, for example 74HCT123. (c) pulse width = $KRC + t_{min}$, spec'd at 5V for 10k and 0.1μF. (d) for RC=0, derived from the 2k and 28pF value by subtracting 56ns·K. (e) From the graph of K vs supply voltage. (f) 75ns for 'AHCT123. (g) single monostable in 8-pin package.

Long pulses For the generation of long pulses, you can use large value timing resistors (up to 10M should be safe, even if the datasheet shows values only to 200k, say, because these are CMOS designs with low leakage currents). Even so, the capacitor value may be a few microfarads or more; in that case electrolytic capacitors are generally necessary.[62] You have to worry about leakage current (which is insignificant with the smaller capacitor types), and you should include a diode across R (Figure 7.61A); the latter is

[59] FSC, NXP, ST, TI, and Toshiba.

[60] Measurements made on samples of the FSC and TI parts, with R=10k and C=100 nF, yielded pulse widths of 1.05 ms and 0.42 ms, respectively; these are consistent with their individual datasheets.

[61] Datasheets routinely omit the t_{min} term, but it should be included for reasonable accuracy when designing for short pulse widths.

[62] If you insist on using a high-capacitance ceramic capacitor, your selection may be limited to types with a "high-κ" dielectric; if so, be aware of their characteristic large temperature and voltage coefficients, which can cause capacitance variations of 50% or more. See §*1x.3*.

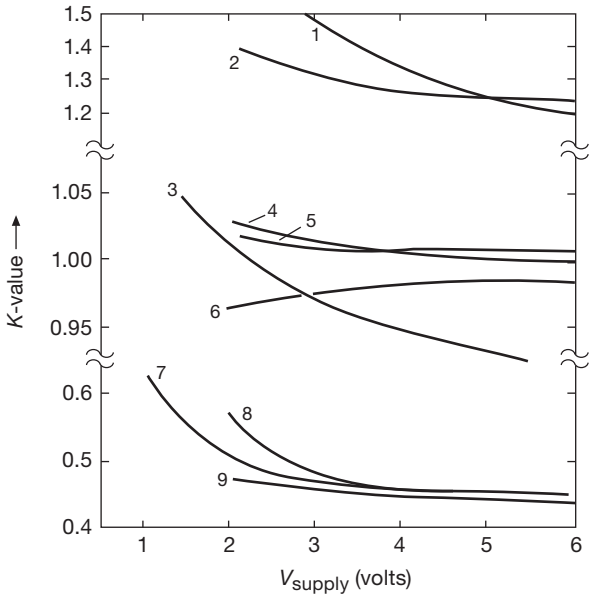


Figure 7.60. Datasheets plots of effective timing coefficient K versus supply voltage for the monostables in Table 7.4. Note the changes of scale, particularly the expanded set around K=1.0.

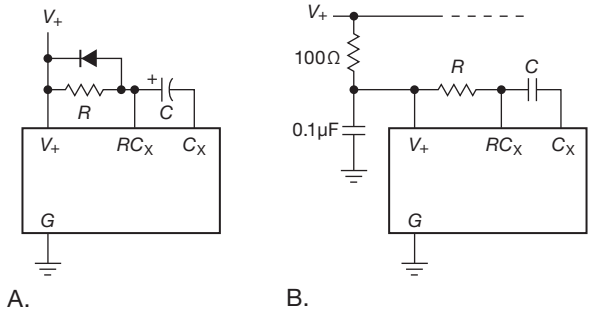


Figure 7.61. Monostable circuit variations. A. A diode prevents reverse conduction on power-down. B. A filtered "private V_+ supply" reduces supply-noise instabilities.

needed to prevent reverse conduction at the RC_X terminal by the charged timing capacitor if V_+ is turned off abruptly.

Duty cycle With some one-shots the pulse width is shortened at high duty cycle. For example, NXP's 74LV123, when powered from +3.3 V and using R=10k and C=100 nF, has constant pulse width up to 95% duty cycle, decreasing about 1.5% near 100% duty cycle. The nonretriggerable '221 monostable is considerably worse in this respect, with erratic behavior at high duty cycles. By contrast, in our tests the Fairchild MM74HC123A maintained perfect timing to 99.98% duty cycle, along with a jitter-free output pulse. (It, like the admirable TI SN74HC4538, uses a rather low V_L threshold, about 10% of V_+.)

Another thing to be aware of is the effect of capacitor size on retriggering recovery time. For example, TI's 74LVC1G123 datasheet has a plot of minimum retrigger time for various capacitances; it shows a minimum wait of 1 μs for a 10 nF timing cap, which is 1% of the 100 μs pulse width with a 10k resistor.

Triggering One-shots can produce substandard or jittery output pulses when triggered by too short an input pulse. There is a minimum trigger pulse width specified, e.g., 140 ns for the 4538 with +5 V supply, 60 ns with +15 V supply (4000-series "high-voltage" CMOS is faster and has more output drive capability when operated at higher supply voltages), 25 ns for the 'HCT423 at its specified +5 V supply, and 3 ns for the speedy 'LVC123 with +3.3 V supply.

Noise immunity Because of the linear circuits in a monostable, the noise immunity is generally poorer than in other digital circuits. One-shots are particularly susceptible to capacitive coupling near the external R and C used to set the pulse width. In addition, some one-shots are prone to false triggering from glitches on the V_+ line or ground. One way to avoid these problems is to create a "private" RC-filtered V_+ supply, as shown in Figure 7.61B; alternatively, you could supply the monostable from a separately regulated V_+, via a small linear voltage regulator, if you have available a higher-voltage supply rail.

Specsmanship Be aware that monostable performance (predictability of pulse width, temperature and voltage coefficients, etc.) may degrade considerably at the extremes of its pulse-width range. Specifications are usually given in the range of pulse widths where performance is good, which can be misleading. In addition, there can be a lot of difference from manufacturer to manufacturer in the performance of monostables of the same part number. Read the datasheets carefully!

Output isolation Finally, as with any digital device containing flip-flops, outputs should be buffered (by a gate, inverter, or perhaps an interface component like a line driver) before going through cables or to devices external to the instrument. If a device like a one-shot tries to drive a cable directly, the load capacitance and cable reflections may cause erratic operation to occur.

General considerations for using monostables

Be careful, when using one-shots, to generate a train of pulses, that an extra pulse doesn't get generated at the

"ends." That is, make sure that the signals that enable the one-shot inputs don't themselves trigger a pulse. This is easy to do by looking carefully at the one-shot truth table, if you take the time.

Don't overuse one-shots. It is tempting to put them everywhere, with pulses running all over the place. Circuits with lots of one-shots are the mark of the neophyte designer. Besides the sort of problems just mentioned, you have the added complication that a circuit full of monostables doesn't allow much adjustment of the clock rate, since all the time delays are "tuned" to make things happen in the right order. In many cases there is a way to accomplish the same job without a one-shot, and that is to be preferred. Figure 7.62 shows an example.

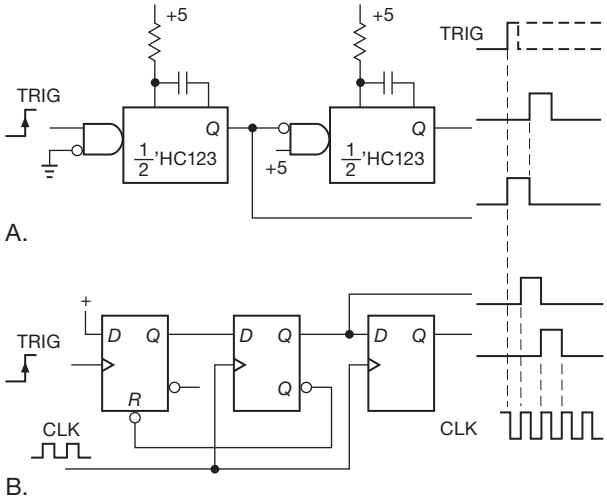


Figure 7.62. A digital delay can replace one-shot delays. Note that (unlike in circuit A, where the trigger initiates the output), in B the trigger input "arms" the circuit, whose digital output is synchronized to the next rising edge of CLK following the input trigger.

The idea is to generate a pulse and then a second delayed pulse following the rising edge of an input signal. These might be used to set up and initiate operations that require that some previous operation be completed, as signaled by the input rising edge. In the first circuit, the input triggers the first one-shot, which then triggers the second one-shot at the end of its pulse.

The second circuit does the analogous thing with type-*D* flip-flops, generating output pulses with width equal to one clock cycle. This is a synchronous circuit, as opposed to the asynchronous circuit using cascaded monostables. The use of synchronous methods is generally preferable from several standpoints, including noise immunity. If you wanted to generate shorter pulses, you could use the same kind of circuit, with the system clock divided down (via several toggling flip-flops) from a master clock of higher frequency. The master clock would then be used to clock the *D* flip-flops in this circuit. The use of several subdivided system clocks is common in synchronous circuits. Note that there is up to one clock period of jitter in the digital delay circuit, unlike the "instantaneous" response of cascaded monostables.

§7.2.4 explores further this idea of "digital timers."

7.2.3 A monostable application: limiting pulse width and duty cycle

Here's a nice monostable application, simple and effective, which has saved the day (and much grief) on numerous occasions. It's useful whenever you're driving a device with high-current short pulses (for example solenoids or LEDs), and particularly in situations where those pulses are generated from software (in a microcontroller or FPGA). The danger, of course, is that a software bug or microcontroller crash can cause a destructively long pulse.

To set the stage with a specific example, researchers in one of our labs were photographing zebrafish with a microscope fitted with an illuminator ring of sixty LEDs. The camera shutter, operating at 120 frames/sec, generated 80 μs LED-driving pulses every 8 ms. Because the duty cycle (t_{ON}/T) was only 1%, it was OK to drive the LEDs at very high current (1 A, or ten times their continuous current rating of 100 mA), and to do so without heat sinking. Fine... but a screwup in camera programming generated a long pulse that destroyed the array of LEDs; the cost was $40... and a day's labor.

The solution is a circuit like Figure 7.63. Upon triggering, the first monostable generates a pulse T_1 that limits the maximum output pulse width T_{OUT}, by disabling the AND gate after time T_1. The second monostable inhibits retriggering until it has timed out, preventing further LED-driving pulses until a total time T_2 after the last trigger. Here we chose timing components *RC* to set T_1=100 μs and T_2=5 ms. Fitted with this protection circuit, the LED illuminator has now captured many millions of portraits of zebrafish, with both fish and LED living happily.

7.2.4 Timing with digital counters

For timing jobs where you want a long delay (minutes to hours, or even days), or where you need real accuracy, stability, or predictability, these analog timing methods are inadequate. What you do instead is use digital counters in combination with a fixed oscillator (perhaps of high

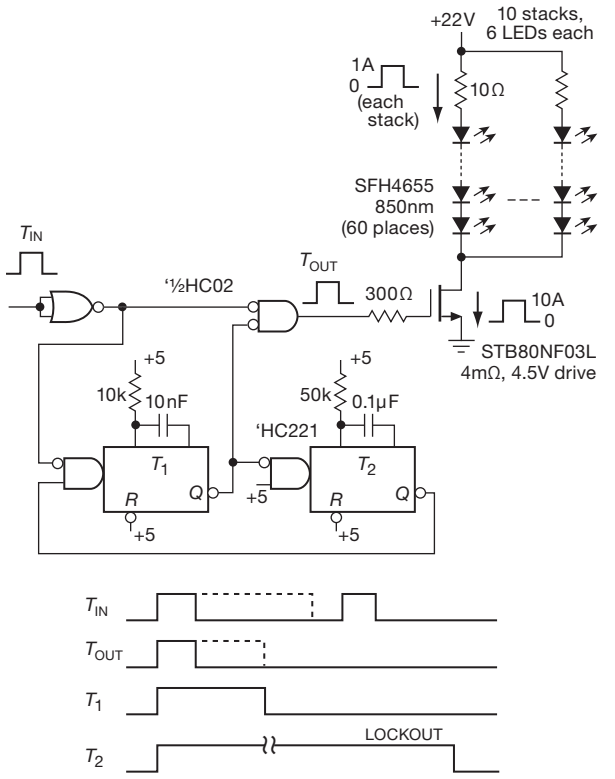


Figure 7.63. A simple protection circuit for high-current pulsed devices: limiting pulse width and duty cycle with a pair of monostables.

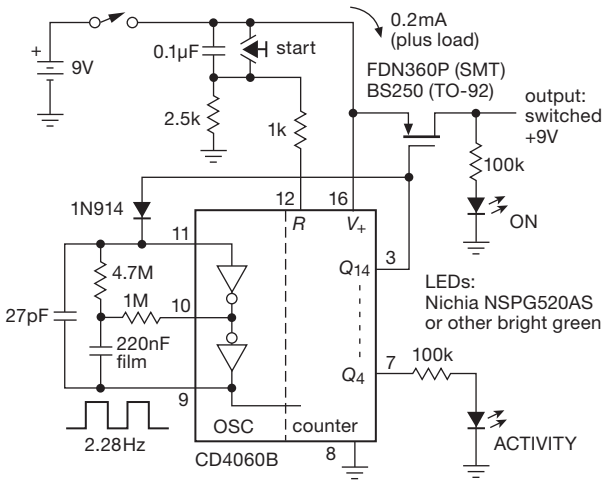


Figure 7.64. "An hour of power." The 1 MΩ (nominal) frequency-setting resistor could consist of a 750k fixed resistor in series with a 500k trimmer.

stability). Digital techniques are treated in detail in Chapters 10–13; but they form such an essential part of the business of timers that we must include them here.

A. An example: an "hour of power"

Suppose, to save battery power, you want a circuit to turn off a portable instrument after an hour. The circuit of Figure 7.64 will do the job. It exploits the CMOS 4060 binary counter chip (14 cascaded stages), which includes a pair of internal inverters intended for making an *RC* oscillator (the way we did in §7.1.2B). It belongs to the "4000B high-voltage CMOS" logic family, allowing operation from supply voltages of 3–18 V. That's handy here, because we can run it directly from the 9 V battery (whose terminal voltage begins life at about 9.4 V, and ends somewhere around 6 V).

At turn-on (or when the START button is pressed) the counter is reset to a count of zero (because the "R" reset input is brought momentarily HIGH), so all of the Q's (the counter's binary outputs) are LOW. That turns on the *p*-channel MOSFET, supplying power to the load. When the counter reaches a count of 2^{13}, the Q_{14} bit goes HIGH, both turning off power to the load and also stopping the oscillator. That's the end of the hour of power. The 4060 has negligible quiescent current ($\ll 1\mu$A), so the battery sees only MOSFET leakage current, also negligible.

A few comments. (a) Clamping the oscillator to end the cycle, as we've done here, is a trick; more typically you would use a flip-flop whose two states correspond to power-on and power-off. The popular 14541 "programmable-timer" chip offers such features internally, and is a better choice. (b) The fourth counter bit (Q_4) is used to signal activity, so you know the timer is running. We used a high-efficiency LED, running at 50 μA, to minimize current drain. (c) Tasks like this are easily accomplished with programmable *microcontrollers* (see Figure 7.69, and Chapter 15), which are flexibly reprogrammed, and which can do other tasks along the way. Our fondness for circuits like Figure 7.64 reveals, well, a certain nostalgia.

You can make an hour-of-power circuit with one of the elegant "TimerBlox" chips from LTC (introduced in §7.1.4B). The obvious choice would be the LTC6993 one-shot timer (see the table on page 433), but that part tops out at a pulse length of 34 seconds. Instead, you can press into service the LTC6991 low-frequency oscillator (maximum oscillator period of 9.5 hours!), rigged up as shown in Figure 7.65 so that it's held in the reset state after the

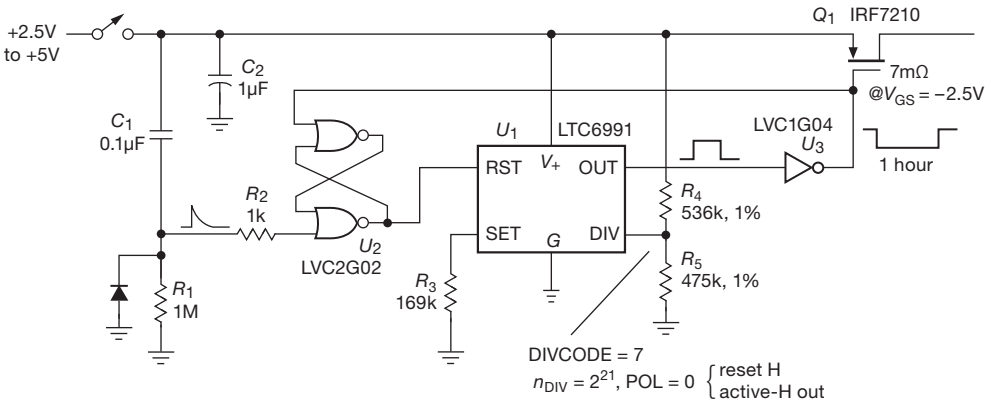


Figure 7.65. An hour of power, TimerBlox-style.

end of its first half cycle.[63] This circuit runs on the low voltages for which these chips are designed (single supply, +2.5 V to +5.5 V), which mandated a low-threshold pMOS switch for Q_1. A nice feature of this series of timers is their accuracy (±1.5% worst case): your hour-of-power will terminate within one minute of its appointed time.

B. Another example: "one second per hour"

This circuit challenge appeared on the sci.electronics.design ("sed") newsgroup: generate a 1-second pulse once per hour, given a 1 Hz input clock; and do it with a minimum number of ICs. Figure 7.66 shows one way to do this,[64] once again using an integrated binary counter. This time we've used the 4040 12-stage counter, which lacks an oscillator but which provides outputs from every stage. The AND gates (§10.1.4) provide a HIGH output when the indicated Q's are all HIGH, i.e., the count $n = 2048 + 1024 + 512 + 8 + 4 + 2 + 1 = 3599$; that sets the output flip-flop (§10.4.1), which both generates one clock cycle of HIGH output and also resets the counter. Because the count is reset to zero, the full cycle is 3600 seconds, which, by great good fortune, happens to be the number of seconds in an hour.

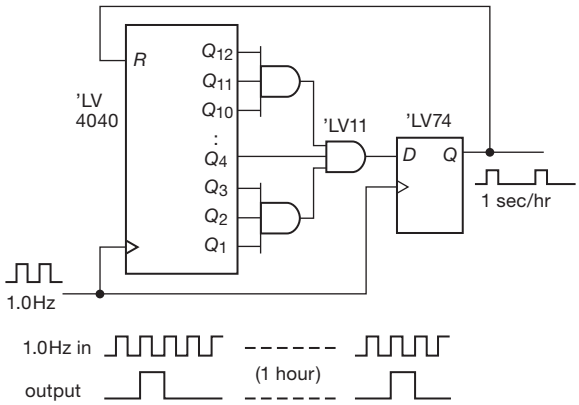


Figure 7.66. "One second per hour" timer, using digital logic.

C. A third example: remote camera control

We wanted to use a Panasonic DMC-LC1 digital camera to capture the faint light of star images as they transited an array of photomultipliers. The camera has a USB port for reading and deleting images from its memory card, but it does not provide a way to trip the shutter. However, it does have a separate connector for an electrical "shutter release," which can be activated with a switch or relay. So the idea is to use a relay to take a picture, then retrieve it via the USB connection.

Sounds easy – but there's a catch: you can't take a picture when the USB port is active. It turns out you have to deactivate the USB connection for at least 3 seconds before you take a picture, then you have to wait a second or two before reactivating it to get the image. Figure 7.67 shows a solution to this timing problem, relying again on digital counting methods. The circuit is timed by a CMOS 555, running at 1 Hz. It clocks a '4017 decimal counter, which has the interesting feature of providing

[63] A fine point, but one that can bite: if the RST pin is found to be in the asserted state when the chip finishes its initialization (here ∼1.7 ms), then (as casually stated in the datasheet) "the first pulse will be skipped." Sounds like no big deal – but that "first pulse" amounts to a full timing cycle, in other words 2 hours. That's a long time to be waiting for your hour of power! That's why we set a much longer ∼100 ms power-on jam reset to *SR* flip-flop U_2, guaranteeing that U_1 won't see its RST pin improperly asserted until its OUT pin is happily into its hour-long HIGH state.

[64] As posted there by John Fields.

individual outputs for each of its 10 states (0–9); when powered, it starts up in state 0 (because of the *RC* reset), and then progresses sequentially upward. States 0–4 all cause the "USB power" relay to actuate, because of the M^{2}L (Mickey Mouse Logic) diode array;[65] this removes USB power, which is connected through the "normally closed" relay contacts. State 3 comes along during the time the USB is deactivated, actuating the relay and shutter ("normally open" contacts, here). Then the shutter deactivates, and USB power is restored after another second.

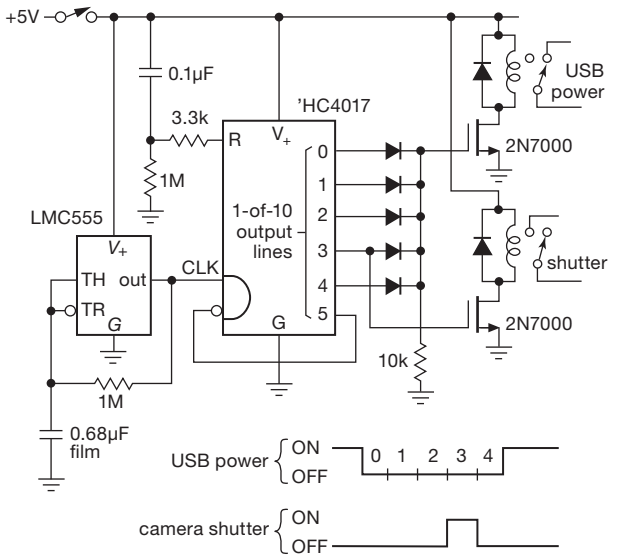


Figure 7.67. Camera control timer. The relays operate from 5 V (drawing 40 mA), and can switch up to 8 A; a suitable part is Omron G5C-2114P-US-DC5.

D. Other digital timing chips

There is a class of timer-oriented counter chips that are good for these kinds of tasks. Here are ones that we know and like (and see also the counter ICs in Table 10.5 on page 742).

ICM7240/42/50/60

This series of "Fixed and Programmable Timer/Counters" from Maxim includes an internal 555-like oscillator circuit, to which you attach an external frequency-setting *R* (to V_+) and *C* (to ground). The 8-pin 7242 has a fixed modulo-256 counter, with a pair of outputs at $f_{OSC}/2$ and $f_{OSC}/256$. The 16-pin 7240/50/60 parts allow you to set the divider modulus via programming pins: binary (1–255), decimal (1–99), and "real-time" (1–59), respectively. These low-power parts (<1 mA) run on supply voltages from 2 V to 16 V, with maximum oscillator frequency of 1–15 MHz (typical) over that range.

MC14536

This "Programmable Timer" from ON Semiconductor (formerly Motorola) includes internal inverters for making an *RC* oscillator (§7.1.2B), followed by a 24-stage binary counter chain. You can select (via 4 input pins) which of the last 16 stages you want as output; and an on-chip monostable lets you convert that output to a pulse (in the range of ∼1–100 μs wide). You can bypass the first 8 stages, for shorter timing delays, which can range from microseconds to days. This low-power part (∼1.5μA/kHz, when driven from an external oscillator) runs on supply voltages from 3 V to 18 V, with maximum clocking frequency of 1–5 MHz (typical) over the range of 5–15 V.

MC14541

This "Programmable Timer" from ON Semiconductor is similar to the MC14536, but includes only 16 stages and limits the choices of divide ratios (2^8, 2^{10}, 2^{13}, and 2^{16}). In exchange, it gives you an internal power-up reset, a choice of output polarity, a choice of single-cycle or repeating-cycle modes, and somewhat lower operating current. It includes an output flip-flop, but unfortunately does not provide inputs to set or clear it.

LTC699x "TimerBlox"

This series includes oscillator and timer functions, with single-resistor frequency (or delay) programming, and voltage-divider (2 resistors) range and mode selection; see the table on page 433. They run on a single +2.5 V to +5.5 V supply, with excellent timing accuracy (<2%, worst-case, for the oscillators; 3.4% for the timers).

In addition to the oscillators and monostables in this family, there's the LTC6994-x "Delay Block & Debouncer." The -1 variant delays one edge only, while the -2 variant delays both edges (thus pulse-width preserving). The delay range τ is 1 μs to 34 seconds, selectable (along with polarity mode) via a voltage divider; a second resistor tunes the delay within the selected range, over a 16:1 span. Input pulses shorter than τ produce no output, which is handy for debouncing, or for "pulse qualification."

TPL5000/5100

These impressive nanopower timers from Texas Instruments run from 1.8 V to 5.5 V, draw a meager 40 nA (yes, *nano*amps!), and are 3-bit programmable (via three input pins) from 1 to 16 seconds or 16 to 1024 seconds

[65] In the language of digital logic, this is a 5-input OR gate.

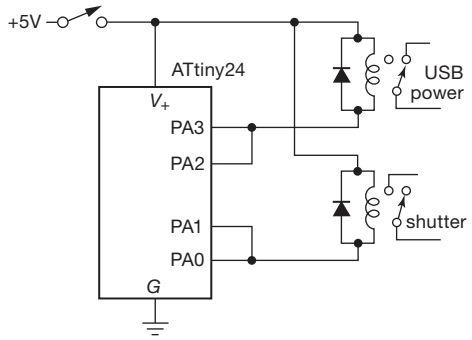


Figure 7.68. Microcontroller implementation of camera control timer.

(TPL5000 and 5100, respectively). The TPL5100 can drive a *p*-channel power MOSFET, to switch output loads.

Microcontrollers

A microcontroller is an inexpensive and flexible computer-on-a-chip, intended to be "embedded" into pretty much any kind of electronic device. We'll have much more to say about these wonderful devices in Chapter 15. But we cannot resist showing, in Figures 7.68 and 7.69, analogous microcontroller-based solutions to the camera shutter and "hour-of-power" examples whose implementation in discrete logic we showed in Figures 7.67 and 7.64, respectively.

In Figure 7.68 an Atmel "ATtiny24" microcontroller, with on-chip oscillator and timer functions (and *lots* more!) runs a program (which you have to write, of course) to do the shutter and USB power timing; its output pins can sink or source 20 mA, so we've used parallel pairs to handle the 40 mA relay current. This particular IC costs about a dollar, in small quantities. It, like all microcontrollers, is available in many variants, with additional memory, I/O, A/D converters, and so on. They are all remarkable bargains.

The microcontroller version of the "hour of power" (Figure 7.69) is a bit more complicated, because microcontrollers run only on supply voltages lower than the +9 V we chose for the earlier example in Figure 7.64; typically they require a dc supply voltage in the range +1.8 V to +5 V. The approach we've taken here is to use a low-dropout linear regulator that has a shutdown mode (in which its standby current $I_{OFF} \sim 1\mu A$), and to use a microcontroller output pin to enable the regulator during operation. The circuit is rigged up for zero battery drain (other than leakage and I_{OFF}) except when timing.

A great virtue of programmable microcontrollers is their ability to give a fine performance in any of a variety of tasks. In this circuit you could program the controller to accept other inputs and produce other outputs (for example, sense temperature and humidity, and display values on an LCD display), and omit the output power switching entirely.[66] Or more simply, as a variation on the humble hour-of-power task, you could have several "mode" inputs that would set different timings, or patterns of power, or whatever.

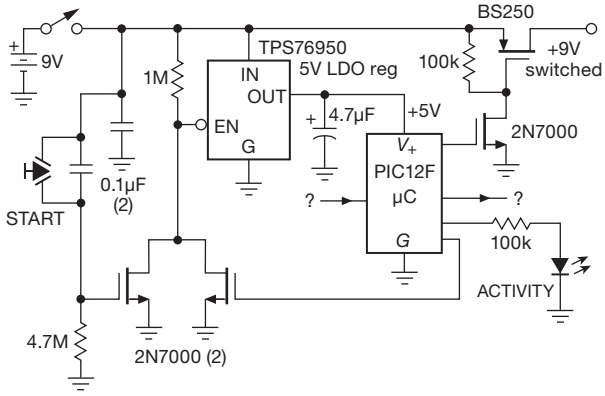


Figure 7.69. Microcontroller implementation of "an hour of power."

[66] For that kind of application you might prefer to run the whole circuit on +3 V, for example from a lithium cell, or a pair of alkaline cells.

Review of Chapter 7

An A-to-H summary of what we have learned in Chapter 7. This summary reviews basic principles and facts in Chapter 7, but it does not cover application circuit diagrams and practical engineering advice presented there.

¶A. Oscillator and Timer Overview.

Oscillators are circuits that create a periodic output waveform. The output may be as simple as a logic-level square wave (or train of pulses) for use as a "clock" in a digital system. Or it may be a highly accurate, stable, and perhaps programmable source of low-distortion sinewaves (or of other waveforms, for example a periodic ramp for use in an ADC or a PWM switchmode power converter). The signal from an oscillator is characterized broadly by its waveform, frequency, amplitude, and tunability, and, at a deeper level of scrutiny, by its phase noise, jitter, sideband suppression, distortion, temperature coefficient of frequency, and long-term stability.

Timers are circuits that generate a delayed pulse, or a pulse of a given width, following a triggering event. A timer is characterized broadly by output pulse width, delay time, and retriggerability, and, at a deeper level of scrutiny, by timing jitter, long-term stability of timing interval, and tempco.

¶B. Relaxation Oscillators.

These simple oscillators exploit an *RC* decay (or a current charging a capacitor) to generate a sustained oscillation. The *RC* can be wrapped around an op-amp or comparator, forming the classic pedagogical oscillator (§7.1.2A). More common is the use of logic inverters (§7.1.2B), or a timer IC like the ubiquitous 555 (§7.1.3 and Table 7.1) or its contemporary forebears (e.g., the LTC6900-series, §7.1.4). Such timer or oscillator ICs deliver reasonable accuracy and predictability ($\sim$1%), over frequencies ranging from a few hertz to a megahertz or more. By adding a digital counter you can lengthen the period to minutes, hours, or just about forever; examples are the 74HC4060 (*RC* oscillator plus 14-stage binary counter; §7.1.4C), and the LTC6991 (resistor-programmed oscillator, periods from 1 ms to 9 hours; §7.1.4B). At the opposite end of the timescale, logic ICs extend the *RC* oscillator frequency range to tens or even hundreds of megahertz (§7.1.4D).

¶C. Voltage-controlled Oscillators.

Sometimes you want frequency tunability, via an input control voltage. That'a a *VCO*. These can be highly linear and stable (e.g., the AD650 V-to-F converter, $\sim$0.01% nonlinearity), or just seat-of-the-pants simple (e.g., the 74LS624-629 series, accuracy and linearity not specified, but likely good to perhaps $\pm$20%). Oscillators built with resonators (*LC*, crystal, SAW, silicon) rather than *RC*'s can be tuned electrically as well; for some the tuning range is narrow (e.g., the frequency tuning range of a voltage-controlled crystal oscillator, or VCXO, is of order $\pm$100 ppm, §7.1.6C), whereas an *LC* oscillator can be varactor tuned over a 2:1 frequency range. Complete VCO modules are available for communication-band frequency ranges, for example the many modules from Crystek that together span the range from about 50 MHz to 5 GHz; the individual modules each tune over a modest range, from as little as $\pm$1% to as much as $\pm$25%.

¶D. Sinewave Oscillators.

The inherent waveforms of relaxation oscillators are segments of exponential charging and discharging (if built with an *RC*), or linear ramps (if the resistor is replaced by a current source); in either case these waveforms drive a comparator circuit to reverse the cycle. That is, the "natural" waveforms are square waves, triangle or sawtooth waves, or what might be called "shark's fin" waves. What you don't get are *sine* waves.

But sinewaves are essential for many tasks; and you *can* generate them, with some cleverness, with the frequency set by *R*'s and *C*'s only, analogous to the simple relaxation oscillator. Most famous is the *Wien-bridge oscillator* (§7.1.5B), an arrangement of two *R*'s and two *C*'s whose phase shifts cancel at a frequency $f = 1/2\pi RC$, where the attenuation is exactly a factor of 1/3. The oscillator circuit wraps this network around a noninverting op-amp with gain equal to +3. The final trick is to sense the output amplitude, and maintain it at a preset (and nonsaturated) level by controlling the gain. A pair of fellows named Hewlett and Packard founded a business with this as their first product; the rest is history.

You can make sinewaves also by lowpass filtering a nonsinusoidal waveform. This is the basis of the *RC phase-shift oscillator* (§7.1.5C), and of the digital tracking-filter technique (§7.1.5A).

An oscillator that generates a pair of sinusoidal signals with a phase difference of 90° (i.e., sine and cosine) is called a *quadrature* oscillator (§7.1.9); the signals are "in quadrature." This can be done with analog techniques (integrator pair, switched-capacitor resonator, phase-sequence filter, RF quadrature hybrid), or with digital synthesis (see ¶G below).

¶E. Resonator-based Oscillators.
Relaxation oscillators, and *RC*-based oscillators in general, can deliver stabilities no better than 1% or so. That is because their accuracy depends on an exponential decay (in the time domain), or a gradual phase shift (in the frequency domain). For better stability you need to exploit a resonant system, whose natural physical frequency can be extremely well-defined. It can have a "quality factor" Q (which is its frequency selectivity $\Delta f/f$) of a million or more.

The simple *LC* resonant circuit is widely used in the frequency range from kilohertz to hundreds of megahertz (§7.1.5D), with stabilities of order 10–100 ppm, and with tunability via a mechanical-variable capacitor or inductor, or an electrically-tunable capacitor (a varactor). For higher frequencies the *LC* tuned circuit can be replaced by a coaxial or cavity resonator.

Better stability (but less tunability) is provided by *electro-mechanical* resonators: quartz-crystal (§7.1.6), ceramic (§7.1.6G), and surface-acoustic-wave (SAW; §7.1.6H). These are inexpensive and stable ($\sim$10ppm), and quartz-crystals and crystal-oscillator modules are available in a wide selection of standard frequencies in the range of 10 kHz to 100 MHz (and crystal manufacturers are happy to provide any frequency you want as a special-order product, at quite reasonable prices). Crystal oscillators can be "pulled" up to $\pm$100 ppm with a varactor; these are sometimes called VCXOs (voltage-controlled crystal oscillators). Ceramic-resonator oscillators can be pulled over a wider range ($\pm$1000 ppm), owing to their lower Q; the downside is poorer frequency stability. They are very inexpensive, but they come in only a limited set of standard frequencies.

¶F. High-stability Oscillators.
Crystal oscillators held at constant temperature are remarkably stable, as good as 1 part in 10^9 for a well-designed OCXO (oven-controlled crystal oscillator); the residual drift is due mostly to mechanical and diffusional "aging" effects. For still better stability you need an *atomic* standard (§7.1.7B), the most accessible being a rubidium-vapor-stabilized crystal oscillator. These deliver stabilities of order 1 part in 10^{10}, and their prices (around $1k, but you can get some real deals on eBay) allow inclusion in precision test and communication equipment. There are more precise (and exotic) frequency standards – cesium beam, hydrogen maser, atomic fountain, chilled ion – but, as they say, "if you have to ask the price, you can't afford them."

Happily, you can take advantage of your tax dollars at work, namely the precise timing provided by the GPS satellite constellation. For about $1k (and a place to put a little doorknob-size antenna) you can have your own 10 MHz reference, with long-term stability of order 10^{-12}.

¶G. Frequency Synthesis.
VCOs are *tunable*, but not particularly stable, nor capable of spanning many decades of frequency; by contrast, a crystal oscillator is *stable*, but tunable at most over parts-per-million. But you can have it both ways: from a stable reference oscillator at some standard fixed frequency (typically a highly stable 10.0 MHz crystal oscillator) you can generate an output frequency of your choosing, with one of several methods of *frequency synthesis* (§7.1.8). Simplest to understand is *direct digital synthesis* (DDS), in which successive sinusoidal amplitudes are read from a table at the appropriate rate and converted to an analog output signal voltage. Complete DDS chips that include all the hardware (counters, sine lookup table, ADC) are widely available, and inexpensive. You provide the reference frequency, and you talk to these things with a microcontroller (Chapter 15) to set the output frequency, phase offset, and so on. DDS chips let you synthesize output frequencies ranging from fractions of a hertz up to a gigahertz, with frequency *agility* (the ability to change frequency rapidly and precisely).

Phase-locked-loop (PLL) synthesis is the other common technique. In its simplest form, a VCO is controlled by a phase comparator whose inputs are the rth subdivision of the reference frequency and the nth subdivision of the VCO output frequency. Then $f_{\text{out}}=(n/r)f_{\text{ref}}$. There are many subtleties here, involving loop stability, sidebands and jitter, resolution, lockup and slewing times, and so on. See §13.13 for an extensive discussion.

¶H. Timers and One-shots.
You can make simple edge-triggered pulsers with as little as an *RC*, or (better) with an *RC* assisted by a BJT or MOSFET (§7.2.1A). But you get cleaner pulses with faster edge times if you use logic gates (§7.2.1C). For better predictability it's best to use an IC designed for timer use, either a monostable multivibrator (a "one-shot"), a 555-type timer, or one of a class of specialized timers like the LTC6991/3, ICM7240/50/60, or MC14536/41. Monostables, being mixed-signal components (logic parts with a linear function), have their "issues," notably their sensitivity to supply-rail noise; they also have a somewhat limited timing range (typically tens of nanoseconds to seconds). They come in both non-retriggerable and retriggerable types; the latter extend their pulse duration if another trigger arrives during the pulse. Pulse widths are set by an

external *RC*, with a multiplying constant *K* of order unity ($\tau = KRC + t_{min}$); see Table 7.4 and Figure 7.60.

You can do much more with timer chips that incorporate a digital counter (e.g., the LTC6991 extends the timing range to ten hours), or you can configure an external binary counter. When thinking about extended timing intervals, or in fact pretty much any timing tasks, don't forget the versatile *microcontroller* (the subject of Chapter 15); there you'll find a timing example, in the form of a "suntan monitor" (§15.2).